Lecture Notes on Algebra

Groups, Rings, Modules, Vector Spaces, and Fields

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Preface

These notes offer a self-contained introduction to the central ideas of modern algebra, beginning with groups and proceeding through rings, modules, linear algebra, fields, and Galois theory. Their guiding ambition is an elementary entry point with a graduate destination: definitions are motivated by familiar examples, while the later chapters develop the structural language needed for further study in algebra, number theory, geometry, and representation theory.

The exposition is cumulative. Results are proved before they are used, and the hypotheses that make a construction meaningful are stated explicitly. In particular, the notes promote important structural results that are often assigned as exercises in the primary reference into the main narrative when later arguments depend on them. The exercises are intentionally selective: they are designed to consolidate the main ideas rather than to conceal essential theory.

The chapters may be read in order as a first course in abstract algebra with substantial linear-algebraic and homological preparation. Readers already familiar with an earlier portion may also use the later parts as a reference, subject to the stated dependencies. The author hopes that the recurring themes—symmetry, quotient constructions, universal properties, and decomposition—will make the subject feel like a connected whole rather than a collection of techniques.

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Contents

Editorial convention

Completed portions of these notes are intended to be logically complete within their chosen scope. Important results in the primary reference are incorporated and proved according to mathematical dependency, whether the source presents them in its text or its exercises. The accompanying coverage audit records each verified promotion with an exact source reference.

Chapter 1

Introduction and Guide to the Notes

1.1 Purpose and point of view

Graduate algebra is not chiefly a collection of classifications. It is a language for recognizing when apparently different problems have the same form. A quotient group, a quotient ring, and a quotient module all arise by forcing certain elements to become zero; an action describes symmetry; a polynomial in an operator turns linear algebra into module theory; and a Galois group records the symmetries of a field extension. These notes develop that common language with an eye toward the places it will be used next.

The intended reader has strong high-school or early undergraduate mathematical maturity: comfort with sets, functions, elementary proof, and familiar number systems such as $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$. No prior course in abstract algebra is assumed. Definitions begin from that elementary entry point, examples arrive before abstraction, and proofs introduce each new move before relying on it. The destination is nevertheless graduate algebra: the resulting core is designed for later commutative algebra, homological algebra, representation theory, algebraic geometry, and algebraic number theory. The last chapters give a complete first treatment of classical Galois theory.

This cumulative policy is deliberate. Early chapters do not appeal to an unseen "standard fact from algebra"; later chapters may use a result only after it has been established here. Readers who have already studied a first course can move more quickly through the examples, but they are not the prerequisite for understanding the logical development.

Readers who would like additional preparation in mathematical language, proof-writing, foundational constructions, and the number systems may consult Chapter 2 of **Lecture Notes on Mathematical Analysis** [2]. Its discussions of sets, functions, proof conventions, induction, and the construction and basic properties of $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ can help bridge to the proof-based style used here. This is useful preparation, not a prerequisite: all algebraic material required in these notes is developed within the manuscript.

Two habits are central. First, definitions are treated as constructions with universal features, rather than as names for collections of examples. Second, proofs are organized around reusable moves: pass to a quotient, use a universal property, compare kernels and images, act on a set, reduce a statement to a cyclic object, or use induction on a degree or a composition length. The same moves will recur throughout the book.

1.2 The mathematical route

Part I establishes the group-theoretic language of normal subgroups, quotients, actions, and composition factors. Sylow theory is included for its structural consequences, while lengthy classifications of groups of individual small orders are not. Part II develops ideals and polynomial rings, then the chain of implications

$$\text{Euclidean domain} \quad \Longrightarrow \quad \text{PID} \quad \Longrightarrow \quad \text{UFD}.$$

It also introduces localization, since it is the first algebraic operation that makes a global ring look local.

Part III develops four connected chapters. Chapter 6 introduces modules together with the finite-dimensional linear algebra needed later: bases, dimension, matrices, determinants, eigenvalues, diagonalization, duality, and a modest introduction to bilinear and quadratic forms. Chapter 7 proves the structure theorem over PIDs and then regards an operator as an $F[x]$-module, leading to minimal polynomials and rational and Jordan canonical forms. Chapter 8 adds inner-product geometry, adjoints, spectral theory, and singular value decomposition. Chapter 9 introduces tensor products, exterior powers, and exterior algebra: the module constructions that linearize bilinear and alternating multilinear maps and prepare the way for later commutative and homological algebra. Thus vector spaces are modules over fields, abelian groups are modules over $\mathbb{Z}$, and a linear operator on an F-vector space makes that space into an $F[x]$-module. The structure theorem for finitely generated modules over a PID therefore simultaneously explains finitely generated abelian groups and the rational canonical form of a linear map. Jordan form appears as the split-polynomial refinement of the same idea.

Part IV studies field extensions, their embeddings, separability, normality, and the correspondence between intermediate fields and subgroups of a

Galois group. Finite fields and cyclotomic fields supply recurring examples, while direct and inverse limits explain how finite Galois symmetries assemble into profinite and absolute Galois groups. Chapter 12 includes a brief local bridge through point-set topology and topological groups, solely to make the profinite and infinite-Galois language self-contained; Tychonoff's theorem is used there as an external input. It closes with a non-prerequisite survey of the later directions opened by all four parts. The later discussion of radicals illustrates how questions about polynomial formulas are controlled by group-theoretic properties.

1.3 How to read a proof

Every result used later is stated with its hypotheses and proved before its first essential use. A proof normally begins by identifying its idea: for example, the first isomorphism theorem compares an object with the quotient by the kernel of a map, whereas the orbit–stabilizer theorem compares a group action with a quotient by a stabilizer. Readers should pause at these ideas, not merely at the final calculation.

Some results are deliberately stated in a general form. This is done only when the extra generality clarifies later work. For instance, the module isomorphism theorems subsume their group and ring analogues, but the earlier versions are still proved because quotient constructions must be understood in their native settings. Conversely, the classification of finitely generated abelian groups is presented as a corollary of the PID module theorem rather than as an unrelated classification.

1.4 Exercises and completeness

Exercises have two roles: they give essential practice, and they sometimes contain theorems that deserve a place in the main narrative. We do not follow the source's typography as a measure of importance. When an exercise supplies a standard lemma, a useful converse, a structural construction, or a result needed later, the result is promoted and proved in the text. The companion coverage audit records the exact source exercise, its location here, and the reason for promotion.

At the end of each chapter, exercises are grouped as **Basic**, **Core**, **Further**, and **Looking Ahead**. The first group tests definitions; the second develops results expected of every reader; the third extends the chapter; and the last points toward later subjects without being a logical prerequisite. A routine exercise remains an exercise even when its solution is long.

1.5 Conventions and notation

Rings in these notes are associative and have a multiplicative identity; the zero ring is allowed. Ring homomorphisms preserve identity unless explicitly stated otherwise, and a subring contains the same identity as its ambient ring. Terminology concerning rings is not completely uniform: some authors allow rings without a multiplicative identity, and some do not require a subring to contain the ambient identity. The conventions used here will always be stated explicitly when a nonunital object is needed. Unless explicitly qualified otherwise, an ideal means a two-sided ideal. When one-sided ideals are relevant, we explicitly say left ideal or right ideal; in a commutative ring these notions coincide. Modules are unital left modules unless the contrary is specified. For a group G, the identity is 1, and for an additive group it is 0. The notation $H \leq G$ permits equality, while $H \trianglelefteq G$ denotes a normal subgroup. If R is a ring and M an R-module, then $\mathrm{Ann}_R(M)$ is its annihilator.

For a regular n-gon, D_n denotes its full dihedral symmetry group, so $|D_n| = 2n$. Many authors write D_{2n} for this same group; the notation in these notes records the number of polygon vertices rather than the group order. Thus D_4 is the symmetry group of a square.

Remark 1.1 (Looking Ahead). The repeated pattern "kernel, image, quotient" is the beginning of the language of exact sequences. We will use exact sequences of modules in an elementary form, but derived functors and category theory are not prerequisites for these notes.

1.6 Sources

The primary mathematical reference is [1]. The separate coverage audit records editorial provenance and verified exercise references; it is not part of the reader-facing source apparatus.

Part I

Groups

Chapter 2

Groups, Quotients, and Homomorphisms

2.1 Groups, Subgroups, and Generated Subgroups

Groups record a collection of reversible symmetries or operations. The axioms are deliberately sparse: associativity makes an unparenthesized product meaningful, the identity records "doing nothing," and inverses record the possibility of undoing an operation. The familiar examples $\mathbb{Z}$, $\mathbb{Z}/n\mathbb{Z}$, the symmetric groups S_n, and the dihedral groups D_n will recur throughout Part I.

Definition 2.1. A **group** is a set G equipped with a binary operation $G \times G \to G$, $(x, y) \mapsto xy$, satisfying the following axioms.

1. **Associativity:** for all $x, y, z \in G$,

$$(xy)z = x(yz).$$

2. **Identity:** there is an element $1 \in G$ such that, for every $x \in G$,

$$1x = x = x1.$$

3. **Inverses:** for every $x \in G$, there is an element $x^{-1} \in G$ such that

$$xx^{-1} = 1 = x^{-1}x.$$

A nonempty subset $H \subseteq G$ is a **subgroup** when the same operation makes H into a group; equivalently, it is closed under products and inverses.

Associativity permits $g_1 g_2 \cdots g_n$ without specifying parentheses.

Example 2.2 (Additive Number Systems)**.** The integers $(\mathbb{Z}, +)$ form a group: associativity is inherited from ordinary addition, the identity is 0, and the inverse of m is $-m$. The same verification gives additive groups $(\mathbb{Q}, +)$ and $(\mathbb{R}, +)$. These examples explain the word "inverse" in an additive setting: it is an additive inverse, not a reciprocal. The natural numbers $(\mathbb{N}, +)$ are not a group, because a positive integer has no additive inverse in $\mathbb{N}$.

Example 2.3 (Multiplicative Number Systems)**.** The nonzero rational numbers $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$ form a group under multiplication. The identity is 1, and the inverse of a/b is b/a. Likewise $\mathbb{R}^\times$ is a group. The zero rational or real number must be removed: it has no multiplicative inverse. This small change illustrates a common algebraic move—restrict to the elements for which an operation is reversible.

Example 2.4 (Matrices)**.** For the familiar number systems $F = \mathbb{Q}$ or $\mathbb{R}$, the invertible $n \times n$ matrices form the group $\mathrm{GL}_n(F)$. Matrix multiplication is associative, the identity is I_n, and membership means precisely that a matrix inverse exists. (The general notion of a field will be developed later.) For $n \geq 2$ this group is generally nonabelian: two reversible linear transformations need not commute. This is the first indication that groups capture more than arithmetic.

Remark 2.5. This example uses the familiar elementary notions of matrices and matrix multiplication. Their systematic definition, operations, and relation to linear transformations appear in Section 6.11.

Proposition 2.6 (Elementary Group Identities)**.** The identity and inverses are unique. Cancellation holds, and $(xy)^{-1} = y^{-1}x^{-1}$.

Proof. If e, e' are identities, use first the right-identity property of e' and then the left-identity property of e: $e = ee' = e'$. If y, z both invert x, then associativity lets us insert parentheses as follows:

$$y = y1 = y(xz) = (yx)z = 1z = z.$$

For cancellation, if $ax = ay$, multiply the equality on the left by a^{-1}:

$$x = 1x = (a^{-1}a)x = a^{-1}(ax) = a^{-1}(ay) = (a^{-1}a)y = 1y = y.$$

Right cancellation is similar. Finally,

$$(xy)(y^{-1}x^{-1}) = x(yy^{-1})x^{-1} = x1x^{-1} = 1,$$

and the same calculation in the opposite order shows that $y^{-1}x^{-1}$ is the inverse of xy. □

Thus a subgroup is not merely a subset: it is a collection of operations that is itself closed under composition and reversal. For example, $n\mathbb{Z} \leq \mathbb{Z}$, while the transpositions in S_3 do not form a subgroup because their product need not be a transposition.

We will use one kind of map before studying maps systematically. A function $f : G \to H$ between groups is a **homomorphism** if $f(xy) = f(x)f(y)$ for all $x, y \in G$. Its **kernel** is the set of elements sent to the identity of H. The structural consequences of these definitions—images, normality, and quotient groups—are developed later in this chapter.

Proposition 2.7 (Subgroup Test)**.** A nonempty subset H of G is a subgroup if and only if $xy^{-1} \in H$ for all $x, y \in H$.

Proof. If H is already a subgroup, then $y^{-1} \in H$ and closure under products gives $xy^{-1} \in H$. Conversely, assume the displayed condition. Choose $h \in H$, possible because H is nonempty. Taking $x = y = h$ gives $hh^{-1} = 1 \in H$. Now take $x = 1$ and any $y \in H$; the condition gives $y^{-1} \in H$. Finally, if $x, y \in H$, then $y^{-1} \in H$, so the condition applied to x and y^{-1} gives $x(y^{-1})^{-1} = xy \in H$. Thus H contains the identity and is closed under products and inverses. $\square$

The subgroup test is useful because it replaces three separate closure checks by a single expression. It also anticipates a recurring algebraic pattern: subobjects are often recognized by closure under a difference-like operation.

Definition 2.8. For a subset $S \subseteq G$, the **subgroup generated by** S is

$$\langle S \rangle = \bigcap_{\substack{H \leq G \\ S \subseteq H}} H.$$

It is the smallest subgroup of G containing S. A group is **finitely generated** if $G = \langle S \rangle$ for some finite S.

The intersection is a subgroup, and it is contained in every subgroup that contains S, which proves the asserted minimality. Equivalently, $\langle S \rangle$ consists exactly of finite products of elements of S and their inverses (including the empty product 1). This formulation is often the practical one; the intersection formulation is the structural one. The same "smallest subobject containing a set" construction will reappear for ideals, modules, algebras, and field extensions.

Example 2.9. In the additive group $\mathbb{Z}$, $\langle 6, 10\rangle = 2\mathbb{Z}$, since $2 = 2 \cdot 6 - 10$. In S_3, (12) and (23) generate S_3.

Proposition 2.10 (Finite Products from a Generating Set)**.** For $S \subseteq G$, the subgroup $\langle S\rangle$ consists exactly of the finite products $s_1^{\varepsilon_1} \cdots s_k^{\varepsilon_k}$, where $s_i \in S$ and $\varepsilon_i \in \{1, -1\}$, together with the empty product 1.

Proof. Let W be the set of these finite products in the already given group G. Concatenation gives closure under products, and reversal with inverse exponents gives closure under inverses. Thus W is a subgroup containing S, so minimality gives $\langle S\rangle \subseteq W$. Every subgroup containing S contains all these finite products, so its intersection contains W. $\square$

This two-containment argument will recur whenever we describe an object generated by a specified set of elements.

2.2 Cyclic Groups

For $g \in G$, $\langle g\rangle = \{g^n : n \in \mathbb{Z}\}$ is the special case $S = \{g\}$, the **cyclic subgroup generated by** g. The order $|g|$ is the least positive n with $g^n = 1$, if it exists.

The cyclic group $\langle g\rangle$ is the portion of G generated by one operation. Additively, $\langle 1\rangle = \mathbb{Z}$; multiplicatively, the rotations of a regular n-gon form C_n. The order of g measures the first time repeated application returns to the identity.

Proposition 2.11. Every subgroup of a cyclic group is cyclic. If $G = \langle g\rangle$ has order n, then for each divisor $d \mid n$, G has a unique subgroup of order d, namely $\langle g^{n/d}\rangle$.

Proof. For a nontrivial subgroup H, let r be the least positive integer with $g^r \in H$. Given $g^k \in H$, divide k by r: write $k = qr + s$, $0 \le s < r$. Then $g^s = g^k(g^r)^{-q} \in H$. Minimality of r forces $s = 0$, so every element of H is a power of g^r, and $H = \langle g^r\rangle$. In a finite cyclic group, g^a has order $n/\gcd(n, a)$. Thus $g^{n/d}$ has order d. If a subgroup has order d, write it as $H = \langle g^r\rangle$. Then $\gcd(n, r) = n/d$, so $r = (n/d)u$ with $\gcd(u, d) = 1$. Since $g^{n/d}$ has order d, there are integers v, w with $uv + dw = 1$. Consequently

$$g^{n/d} = (g^r)^v (g^{n/d})^{dw} = (g^r)^v,$$

because $(g^{n/d})^d = 1$. Hence $\langle g^{n/d}\rangle \le \langle g^r\rangle$; the reverse inclusion is immediate from $g^r = (g^{n/d})^u$. Thus $H = \langle g^{n/d}\rangle$, which proves uniqueness precisely. $\square$

Example 2.12 (The Lattice of C_{12})**.**
If $C_{12} = \langle g \rangle$, its subgroups are $\langle g \rangle, \langle g^2 \rangle, \langle g^3 \rangle, \langle g^4 \rangle, \langle g^6 \rangle$, and $\{1\}$, of orders $12, 6, 4, 3, 2, 1$, respectively. The divisors of 12 predict exactly this subgroup lattice. In contrast, the analogous subgroup problem for a general finite group is substantially harder, which is why cyclic groups are a useful first laboratory.

Proposition 2.13. The homomorphism $\varphi_g : \mathbb{Z} \to G$, $n \mapsto g^n$, has image $\langle g \rangle$, and

$$\ker \varphi_g = \begin{cases} \{0\}, & \text{if } g \text{ has infinite order,} \\ n\mathbb{Z}, & \text{if } |g| = n. \end{cases}$$

Thus an infinite cyclic group is isomorphic to $\mathbb{Z}$, whereas a cyclic group of order n is isomorphic to $\mathbb{Z}/n\mathbb{Z}$.

Proof. For integers a, b, $\varphi_g(a + b) = g^{a+b} = g^a g^b = \varphi_g(a)\varphi_g(b)$, so φ_g is indeed a homomorphism. Its image is precisely the set of all integer powers of g, namely $\langle g \rangle$. The kernel consists of precisely those integers a for which $g^a = 1$. If g has infinite order, this happens only for $a = 0$. If $|g| = n$, division with remainder shows that $g^a = 1$ if and only if $n \mid a$, so the kernel is $n\mathbb{Z}$. In the infinite-order case, φ_g is one-to-one: $g^a = g^b$ implies $g^{a-b} = 1$, hence $a = b$. It is onto $\langle g \rangle$ by its definition, and thus gives $\mathbb{Z} \cong \langle g \rangle$. If $|g| = n$, the map

$$a + n\mathbb{Z} \longmapsto g^a$$

is well-defined because $g^n = 1$; it is a bijective homomorphism onto $\langle g \rangle$. The First Isomorphism Theorem will later combine these two direct arguments into the single statement $\langle g \rangle \cong \mathbb{Z}/\ker \varphi_g$. □

Proposition 2.14. If $G = \langle g \rangle$ has order n, then g^k is a generator if and only if $\gcd(k, n) = 1$.

Proof. The order of g^k is $n/\gcd(n, k)$. Thus C_{12} has generators g, g^5, g^7, g^{11}. □

2.3 Dihedral Groups

Definition 2.15. The **dihedral group** D_n is the full symmetry group of a regular n-gon. If r is rotation through one vertex and s a reflection, then

$$D_n = \langle r, s \mid r^n = s^2 = 1,\ srs^{-1} = r^{-1} \rangle.$$

Many authors call this order-$2n$ group D_{2n}; these notes use D_n.

The relation $sr = r^{-1}s$ lets us move every occurrence of s to the left in a product. Since $s^2 = 1$ and $r^n = 1$, every product of rotations and reflections therefore has the form r^i or sr^i, with $0 \leq i < n$. These $2n$ symmetries are distinct geometrically: the first family preserves orientation and the second reverses it. Hence $|D_n| = 2n$ and $\langle r \rangle \cong C_n$. Notice that multiplying $srs^{-1} = r^{-1}$ on the right by s gives the useful rewriting rule $sr = r^{-1}s$: reflecting after rotating has the same effect as rotating in the opposite direction after reflecting.

For the square, choose r to be the quarter-turn and s a reflection in an axis through opposite vertices. The complete list is

$$D_4 = \{1, r, r^2, r^3, s, sr, sr^2, sr^3\}.$$

The first four are rotations, and the last four are reflections. In particular, $sr = r^{-1}s = r^3 s$, whereas rs is a different reflection: $sr \neq rs$. This is a concrete calculation of noncommutativity, not just an abstract consequence of the presentation.

Example 2.16 (Dihedral Groups as Permutation Groups)**.** Label the vertices of a regular n-gon by $1, \ldots, n$. Every symmetry permutes these vertices, so it determines an element of S_n. The action is injective: a geometric symmetry that fixes every vertex is the identity. Thus D_n is concretely a subgroup of S_n. The later Cayley theorem will generalize this phenomenon from dihedral groups to every finite group.

2.4 Symmetric and Alternating Groups

Definition 2.17. The **symmetric group** S_n is the group of all bijections of the set $\{1, 2, \ldots, n\}$, with composition as the operation. A k-**cycle** $(a_1\ a_2\ \cdots\ a_k)$ sends a_i to a_{i+1} for $1 \leq i < k$, sends a_k to a_1, and fixes every other letter.

Throughout this section, products of permutations are evaluated from right to left: $\sigma\tau$ means "first apply τ, then apply σ." Cycle notation records only the letters that move. For example, $(1\ 3\ 2) \in S_4$ sends $1 \mapsto 3$, $3 \mapsto 2$, $2 \mapsto 1$, and fixes 4. A cycle may be cyclically rewritten without changing the permutation: $(1\ 3\ 2) = (3\ 2\ 1)$. Cycles with disjoint supports commute because they act independently on every letter. For a worked noncommuting product,

$$(1\ 2\ 3)(1\ 2) = (1\ 3).$$

Indeed, $1 \mapsto 2 \mapsto 3$, $3 \mapsto 3 \mapsto 1$, and 2 is fixed by the composite. Reading this calculation from the right is essential.

Proposition 2.18 (Cycle Decomposition)**.** Every permutation in S_n is a product of disjoint cycles. This decomposition is unique up to reordering the cycles and cyclically rewriting each individual cycle. The order of a permutation is the least common multiple of its disjoint cycle lengths.

Proof. Begin with a letter a and follow its successive images under σ:

$$a,\ \sigma(a),\ \sigma^2(a),\ \ldots.$$

Because the set is finite and σ is injective, the first repetition must be a, giving one cycle. Repeat with a letter not yet used. The resulting cycles have disjoint supports and their product agrees with σ on every letter. The orbit of each letter under repeated application of σ determines its cycle, proving uniqueness. A power fixes every cycle exactly when its exponent is divisible by that cycle length, so the least positive such exponent is the stated lcm. □

Definition 2.19. A **transposition** is a two-cycle $(a\ b)$. An **adjacent transposition** is one of the form $(i\ i+1)$.

Proposition 2.20 (Transpositions Generate S_n)**.** Every cycle is a product of transpositions, every permutation is a product of transpositions, and every transposition is a product of adjacent transpositions.

Proof. Directly checking the images of the letters gives

$$(a_1\ a_2\ \cdots\ a_k) = (a_1\ a_k)(a_1\ a_{k-1})\cdots(a_1\ a_2).$$

The cycle-decomposition theorem therefore gives a transposition expression for every permutation. If $i < j$, then

$$(i\ j) = (i\ i+1)\cdots(j-2\ j-1)(j-1\ j)(j-2\ j-1)\cdots(i\ i+1),$$

as is verified by following the images of i, j, and every other letter. Thus adjacent transpositions also generate S_n. □

The preliminary definition of homomorphism introduced above is all that is needed for the next result; its kernel and quotient theory are still deferred to the later systematic treatment.

Theorem 2.21 (Parity and the Sign Homomorphism)**.** Every expression of a fixed permutation as a product of transpositions has the same parity of length. Consequently

$$\operatorname{sgn} : S_n \longrightarrow \{1, -1\}, \qquad \operatorname{sgn}(\sigma) = (-1)^r$$

when σ is written as a product of r transpositions, is a well-defined surjective homomorphism.

Proof. For $\sigma \in S_n$, define its inversion number to be the number of pairs (i, j) with $i < j$ and $\sigma(i) > \sigma(j)$. Multiplying by an adjacent transposition on the right swaps two adjacent entries in the list $\sigma(1), \ldots, \sigma(n)$. All inversions involving another entry are unchanged in pairs, while the inversion between these two entries is reversed; therefore the inversion number changes by exactly one. The displayed adjacent-transposition expression for $(i\ j)$ has $2(j - i) - 1$ factors. Thus every transposition changes inversion parity, and a product of r transpositions has inversion number congruent to r modulo two. Two transposition expressions for the same permutation therefore have the same parity. The definition of sgn is well-defined. Concatenating a transposition expression for σ with one for τ gives an expression for $\sigma\tau$, so $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)$. A transposition has sign -1, so the map is onto. $\square$

Definition 2.22. The **alternating group** A_n is the set of even permutations, namely $\{\sigma \in S_n : \operatorname{sgn}(\sigma) = 1\}$.

The sign rule shows directly that the product and inverse of even permutations are even, so A_n is a subgroup. The displayed sign rule says that sgn is a homomorphism. The general theory of such maps follows after cosets. There we will prove that A_n, as the kernel of the sign map, is normal of index two. For S_3, $A_3 = \langle (123) \rangle$, and the three transpositions are the odd permutations. Later quotient theory will identify the two parity classes with a copy of C_2.

2.5 Cosets, Index, and Lagrange's Theorem

Every subgroup divides a group into pieces of equal size, whether or not its left and right translates agree. A coset is a translate of the subgroup: it remembers how far an element is from belonging to H, while forgetting the particular element of H used to describe that displacement.

For $H \leq G$, a **left coset** is $gH = \{gh : h \in H\}$; its number is $[G : H]$, the **index** of H, when finite.

There are also right cosets Hg. They need not equal left cosets in a nonabelian group. A later condition, called normality, will make the two notions coincide and permit a consistent multiplication of cosets.

Lemma 2.23. The left cosets of H partition G, and each has the cardinality of H.

Proof. If $g_1h_1 = g_2h_2$, then $g_2^{-1}g_1 = h_2h_1^{-1} \in H$, which implies $g_1H = g_2H$. Thus intersecting cosets are equal. Every element lies in a coset, and $h \mapsto gh$ is a bijection $H \to gH$. $\square$

Proposition 2.24 (Coset Equality Criterion)**.** For $g, h \in G$, $gH = hH$ if and only if $h^{-1}g \in H$.

Proof. If $gH = hH$, then $g = ha$ for some $a \in H$, so $h^{-1}g \in H$. Conversely, $h^{-1}g = a \in H$ gives $gH = haH = hH$. $\square$

Example 2.25. The cosets of $n\mathbb{Z}$ in $\mathbb{Z}$ are the residue classes $r + n\mathbb{Z}$. In S_3, the left cosets of $\langle(12)\rangle$ each have two elements, so there are three of them. This latter partition exists, but its left and right cosets differ; later we will identify the precise condition that permits their consistent multiplication.

Example 2.26. In D_4, $r\langle s\rangle = \{r, rs\}$, whereas $\langle s\rangle r = \{r, sr\} = \{r, r^{-1}s\}$. The two sets differ, just as a reflection and a rotation of a square do not commute.

Theorem 2.27 (Lagrange)**.** If G is finite and $H \leq G$, then $|G| = [G : H]|H|$. Consequently, $|g|$ divides $|G|$ for every $g \in G$.

Proof Idea. The preceding lemma has already done the conceptual work. Once G is a disjoint union of translates of H, counting the pieces gives the formula.

Proof. Choose one representative from each left coset. The preceding lemma says that these cosets are pairwise disjoint, cover G, and each has exactly $|H|$ elements. If there are $[G : H]$ of them, counting the elements in this disjoint union gives $|G| = [G : H]|H|$. Taking $H = \langle g\rangle$ says that the order of g divides $|G|$. $\square$

Corollary 2.28. If G is finite, then $g^{|G|} = 1$ for every $g \in G$.

Proof. The order of g divides $|G|$. $\square$

Corollary 2.29. A finite group of prime order is cyclic.

Proof. Any element other than the identity has order dividing the prime group order, so its order is the whole prime and it generates the group. $\square$

The converse to Lagrange's theorem is false: divisibility of $|H|$ by $|G|$ does not guarantee a subgroup of order $|H|$. For example, A_4 has no subgroup of order six, although six divides twelve. This limitation is one reason later existence theorems such as Cauchy's and Sylow's are valuable.

2.6 Homomorphisms, Kernels, and Images

A homomorphism is the appropriate map between groups because it preserves the operation. Its kernel measures exactly what information the map discards, and its image records exactly what it reaches. These two subgroups are the bridge between maps and quotient constructions.

Definition 2.30.
A **homomorphism** $\varphi : G \to K$ satisfies $\varphi(gh) = \varphi(g)\varphi(h)$.

Its **kernel** is $\ker\varphi = \{g : \varphi(g) = 1\}$, and its **image** is $\operatorname{im}\varphi = \{\varphi(g) : g \in G\}$.

A homomorphism preserves the basic group operations automatically: $\varphi(1) = 1$, $\varphi(g^{-1}) = \varphi(g)^{-1}$, and $\varphi(g^n) = \varphi(g)^n$. Its kernel records exactly which elements the map identifies, since $\varphi(x) = \varphi(y)$ if and only if $x^{-1}y \in \ker\varphi$. Images, by contrast, are subgroups but need not be normal: $\langle(12)\rangle$ is the image of $C_2 \to S_3$ but is not invariant under conjugation.

For example, reduction modulo n, $\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$, has kernel $n\mathbb{Z}$ and is onto. The sign map $S_n \to \{\pm 1\}$ has kernel A_n. The latter example is a useful source of normal subgroups that is not obtained by inspection. Likewise, the map $D_n \to \{\pm 1\}$ that sends rotations to 1 and reflections to -1 has kernel $\langle r\rangle$.

Proposition 2.31. The kernel and image of a homomorphism are subgroups; moreover, φ is injective if and only if $\ker\varphi = \{1\}$.

Proof. The subgroup test gives both subgroup assertions. Also, $\varphi(x) = \varphi(y)$ exactly when $\varphi(xy^{-1}) = 1$, equivalently when $xy^{-1} \in \ker\varphi$. □

Definition 2.32. A subgroup $N \leq G$ is **normal**, written $N \trianglelefteq G$, if $gNg^{-1} = N$ for every $g \in G$.

Proposition 2.33 (Equivalent Forms of Normality)**.** For $N \leq G$, the conditions $gNg^{-1} = N$ and $gN = Ng$ for every $g \in G$ are equivalent. Thus the left and right cosets of N coincide exactly when N is normal.

Proof. Multiplying $gNg^{-1} = N$ on the right by g gives $gN = Ng$, and the reverse implication follows by multiplying on the right by g^{-1}. The coset statement is merely this equality written for each element g. □

Normality says that left and right translates of N agree: $gN = Ng$ for every $g \in G$. Equivalently, membership in N is unaffected by conjugation. This condition is natural rather than cosmetic: it is exactly what allows the product of two cosets to be independent of the chosen representatives.

Proposition 2.34. $\ker\varphi \trianglelefteq G$ for every homomorphism $\varphi : G \to K$.

Proof. For $n \in \ker\varphi$, $\varphi(gng^{-1}) = \varphi(g)1\varphi(g)^{-1} = 1$. The reverse inclusion follows on replacing g by g^{-1}. □

Corollary 2.35. The alternating group A_n is a normal subgroup of S_n of index two.

Proof. The sign homomorphism has kernel A_n, so the preceding proposition gives normality. It is surjective because a transposition has sign -1. Its two fibers are the even and odd permutations; multiplication by a fixed transposition pairs these two sets bijectively. Thus each has $n!/2$ elements, and $[S_n : A_n] = 2$. □

2.7 Quotient Groups and Isomorphism Theorems

The progression $N \trianglelefteq G \Rightarrow G/N$ is fundamental. We identify two elements when they differ by an element of N, then ask whether the group operation descends to the resulting equivalence classes. The quotient criterion answers this question precisely.

Theorem 2.36 (Quotient Criterion)**.** For $N \le G$, the rule $(gN)(hN) = (gh)N$ defines a group operation on the left cosets if and only if $N \trianglelefteq G$. The group is G/N.

Proof Idea. Changing a representative of gN multiplies it by an element of N. Normality is exactly what lets that extra factor move past the representative of the other coset and be absorbed back into N.

Proof. If N is normal, $g' = gn_1$ and $h' = hn_2$ imply

$$g'h' = gh(h^{-1}n_1h)n_2 \in ghN,$$

so the rule is well-defined; associativity, identity N, and inverse $g^{-1}N$ follow from G. Conversely, well-definedness of $(gN)(nN) = (gN)N$ shows $gng^{-1} \in N$ for every $n \in N$, which is normality. □

Example 2.37. The quotient $\mathbb{Z}/n\mathbb{Z}$ identifies integers whose difference is a multiple of n. The sign homomorphism has kernel A_n, so $S_n/A_n \cong C_2$. For the dihedral group $D_n = \langle r, s \mid r^n = s^2 = 1,\ srs^{-1} = r^{-1}\rangle$, the rotation subgroup $\langle r\rangle$ is normal and the quotient has two elements: it records whether a symmetry preserves or reverses orientation.

Example 2.38 (Why Normality Is Necessary)**.** In S_3, let $H = \langle (12) \rangle$. It is not normal because $(123)(12)(123)^{-1} = (23) \notin H$. Indeed, the proposed coset product is ambiguous: $H = (12)H$, but

$$H(13)H = (13)H \qquad \text{whereas} \qquad (12)H(13)H = (132)H,$$

and these are different cosets. The coset set exists, but it is not a quotient group.

Theorem 2.39 (Universal Property of a Quotient)**.** Let $N \trianglelefteq G$, let $\pi : G \to G/N$ be the quotient map, and let $f : G \to H$ be a homomorphism with $N \subseteq \ker f$. There is a unique homomorphism $\overline{f} : G/N \to H$ satisfying $f = \overline{f} \circ \pi$.

The following diagram gives a compact visual record of this factorization:

$$\begin{tikzcd} G \arrow[r, "\pi"] \arrow[dr, "f"'] & G/N \arrow[d, "\overline{f}"] \\ & H \end{tikzcd}$$

A diagram **commutes** when any two directed paths with the same source and target define the same map. Here commutativity says exactly $f = \overline{f} \circ \pi$. This elementary language will be used whenever a canonical map and a factorization clarify a later construction.

Proof Idea. The quotient identifies g and gn. The condition $f(n) = 1$ says that f also identifies them, so its value should depend only on the coset.

Proof. Define $\overline{f}(gN) = f(g)$. If $gN = g'N$, then $g' = gn$ for some $n \in N$, and $f(g') = f(g)f(n) = f(g)$; thus the definition is well-defined. It is a homomorphism because $\overline{f}((gN)(hN)) = f(gh) = f(g)f(h)$. The equation $f = \overline{f} \circ \pi$ holds by construction. Finally every coset is $\pi(g)$, so this equation determines $\overline{f}$ on every element of G/N, proving uniqueness. $\square$

Example 2.40 (A Factorization Computed in Full)**.** Let $f : \mathbb{Z} \to C_6$ send 1 to a chosen generator u. Since $f(6k) = u^{6k} = 1$, the subgroup $6\mathbb{Z}$ is contained in $\ker f$. The universal property produces a unique map

$$\overline{f} : \mathbb{Z}/6\mathbb{Z} \longrightarrow C_6, \qquad \overline{f}(\overline{k}) = u^k.$$

Why is this formula safe? If $\overline{k} = \overline{\ell}$, then $k - \ell = 6q$, and $u^k = u^{\ell+6q} = u^\ell (u^6)^q = u^\ell$. The quotient is not merely a convenient notation for residues: it is the unique group through which every map that kills multiples of six must pass.

Remark 2.41 (A Reusable Descent Test). Whenever a proposed formula is defined on cosets, the first question is not whether it is a homomorphism but whether it is well-defined. The standard test is: replace a representative by an equivalent one and show that the output does not change. In quotient groups this is usually equivalent to showing that the subgroup being collapsed lies in a kernel. The same sequence of checks will be used for quotient rings and quotient modules.

Example 2.42 (Revisiting Cyclic Groups)**.** For $\varphi_g : \mathbb{Z} \to G$, the image is $\langle g \rangle$. The kernel calculation in the cyclic-group section therefore becomes the uniform formula $\langle g \rangle \cong \mathbb{Z}/\ker\varphi_g$. When g has infinite order this is $\mathbb{Z}/\{0\} \cong \mathbb{Z}$, and when $|g| = n$ it is $\mathbb{Z}/n\mathbb{Z}$. This is precisely the later theorem promised in the earlier direct proof.

Theorem 2.43 (First Isomorphism Theorem)**.** For a homomorphism $\varphi : G \to K$,

$$G/\ker\varphi \cong \operatorname{im}\varphi.$$

Proof Idea. Two elements have the same image under φ precisely when their quotient lies in $\ker\varphi$. Thus the fibers of φ are the cosets of its kernel, so collapsing those cosets should leave exactly the image.

Proof. The map $g\ker\varphi \mapsto \varphi(g)$ is well-defined because equal cosets differ by an element of the kernel. It is a surjective homomorphism with trivial kernel, hence an isomorphism. □

Remark 2.44 (How to Use the First Isomorphism Theorem). The theorem is not a formula to apply after the fact. It gives a workflow: construct a homomorphism that records the feature of interest, calculate what the homomorphism forgets, and then identify its image. For the sign map the feature is parity, the forgotten permutations are the even ones, and the image is the two-element group. For reduction modulo n, the feature is a residue class, the forgotten integers are the multiples of n, and the image is $\mathbb{Z}/n\mathbb{Z}$.

Example 2.45. The sign map $\operatorname{sgn} : S_n \to \{\pm 1\}$ is surjective and has kernel A_n. The First Isomorphism Theorem therefore gives $S_n/A_n \cong \{\pm 1\} \cong C_2$. This is a typical use: calculate a kernel, then recognize a quotient without multiplying cosets by hand.

Theorem 2.46 (Second Isomorphism Theorem)**.** Let $N \trianglelefteq G$ and $H \leq G$. Then $HN \leq G$, $H \cap N \trianglelefteq H$, and

$$HN/N \cong H/(H \cap N).$$

Proof. The product HN is a subgroup because normality of N gives

$$(h_1n_1)(h_2n_2) = h_1h_2(h_2^{-1}n_1h_2)n_2 \in HN.$$

The subgroup test gives the remaining closure. Conjugating an element of $H \cap N$ by an element of H keeps it in both H and N, so it is normal in H. Restrict the quotient map $G \to G/N$ to H. Its kernel is $H \cap N$, and its image is HN/N. The First Isomorphism Theorem gives the claimed isomorphism. $\square$

The theorem says that when H enters the quotient by N, precisely the elements already in N, namely $H \cap N$, are killed.

Theorem 2.47 (Third Isomorphism Theorem)**.** If $N \trianglelefteq G$ and $N \leq H \trianglelefteq G$, then $H/N \trianglelefteq G/N$ and $(G/N)/(H/N) \cong G/H$.

Proof. Define $\psi : G/N \to G/H$ by $\psi(gN) = gH$. If $gN = g'N$, then $g^{-1}g' \in N \subseteq H$, so $gH = g'H$; hence ψ is well-defined. It is a surjective homomorphism, and its kernel consists of the cosets gN with $g \in H$, which is H/N. The First Isomorphism Theorem proves the claim, including normality of the kernel. $\square$

Example 2.48 (Quotienting in Stages)**.** The chain $12\mathbb{Z} \subseteq 4\mathbb{Z} \subseteq \mathbb{Z}$ yields

$$(\mathbb{Z}/12\mathbb{Z})/(4\mathbb{Z}/12\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}.$$

The left side first remembers residues modulo twelve and then identifies residues differing by four; the right side simply remembers residues modulo four. This is precisely the content of quotienting in stages.

This is the slogan **quotienting in stages is the same as quotienting once**. It is a basic organizational principle in all later quotient theories.

Theorem 2.49 (Correspondence Theorem)**.** Let $N \trianglelefteq G$, and let $\pi : G \to G/N$ be the quotient map. Then $H \mapsto H/N$ is an inclusion-preserving bijection between subgroups of G containing N and subgroups of G/N. Its inverse is $K \mapsto \pi^{-1}(K)$. Under this bijection, normal subgroups correspond, and $H \trianglelefteq G$ implies $(G/N)/(H/N) \cong G/H$.

Proof. If $N \leq H$, then $H/N \leq G/N$. If $K \leq G/N$, then $\pi^{-1}(K) \leq G$ and contains N. The two operations are inverse because π is surjective: $\pi^{-1}(H/N) = H$ and $\pi(\pi^{-1}(K)) = K$. Conjugation commutes with π, proving the normality assertion. The final statement is the Third Isomorphism Theorem. $\square$

For instance, subgroups of $\mathbb{Z}/n\mathbb{Z}$ correspond to subgroups of $\mathbb{Z}$ that contain $n\mathbb{Z}$. The containment condition is necessary: only such subgroups are unions of cosets of $n\mathbb{Z}$, so only they have a meaningful image as a subgroup of the quotient.

Worked Quotient Examples

The following two examples gather the ideas of normality, quotient groups, universal properties, and isomorphism theorems in concrete settings.

Example 2.50 (Reduction Modulo 12)**.** Consider $\pi : \mathbb{Z} \to \mathbb{Z}/12\mathbb{Z}$. Two integers have the same image exactly when their difference is a multiple of twelve:

$$\pi(a) = \pi(b) \quad \Longleftrightarrow \quad a - b \in 12\mathbb{Z}.$$

Thus the quotient identifies exactly the elements that the map cannot distinguish. The subgroup correspondence says that subgroups of $\mathbb{Z}/12\mathbb{Z}$ correspond to the subgroups of $\mathbb{Z}$ containing $12\mathbb{Z}$. For instance, the subgroup generated by $\overline{3}$ has four elements and corresponds to $3\mathbb{Z}$, while the subgroup generated by $\overline{4}$ has three elements and corresponds to $4\mathbb{Z}$. Inclusion reverses neither upstairs nor downstairs: the containment $12\mathbb{Z} \subseteq 4\mathbb{Z} \subseteq \mathbb{Z}$ becomes $\{\overline{0}\} \subseteq \langle\overline{4}\rangle \subseteq \mathbb{Z}/12\mathbb{Z}$.

Example 2.51 (The Sign Quotient of S_3)**.** The sign map separates the six permutations of S_3 into the three even permutations $A_3 = \{1, (123), (132)\}$ and the three odd permutations. The two cosets of A_3 are therefore A_3 itself and $(12)A_3$. In the quotient, every even permutation becomes the identity coset and every odd permutation becomes the same nontrivial coset. Its square is the identity, because the product of two odd permutations is even. Hence the quotient is the two-element group. This makes the abstract statement $S_3/A_3 \cong C_2$ concrete: the quotient retains parity and forgets every other feature of a permutation.

Proposition 2.52 (A Direct Product of Cyclic Groups)**.** For positive integers m, n, the group $C_m \times C_n$ is cyclic if and only if $\gcd(m, n) = 1$. In that case (g, h), where g and h are generators of the two factors, generates the product.

Proof. The order of (g, h) is the least positive integer k for which both $g^k = 1$ and $h^k = 1$. Thus its order is $\operatorname{lcm}(m, n)$. If m and n are coprime, this least common multiple is mn, the order of the product, so (g, h) is a generator. Conversely, suppose a single element generated $C_m \times C_n$. Its order would be mn, but the order of every element divides $\operatorname{lcm}(m, n)$. Therefore $mn \mid \operatorname{lcm}(m, n)$, which forces $\gcd(m, n) = 1$. $\square$

This example contrasts direct products with quotients. A quotient imposes identifications and thereby discards distinctions; a direct product places two independent systems side by side. Their universal properties will recur throughout algebra, but their concrete group-theoretic forms already show two basic operations of algebraic construction.

A Guided Tour of the Quotient Construction

It is useful to see the quotient construction operate several times before leaving Chapter 2. The same four questions organize each instance:

1. What elements should be regarded as indistinguishable?
2. Do these equivalence classes form cosets of a normal subgroup?
3. Does the proposed operation depend only on the classes, rather than on chosen representatives?
4. What information is retained by the resulting quotient?

For $\mathbb{Z}/n\mathbb{Z}$, the answer to the first question is that two integers are indistinguishable when their difference is a multiple of n. The retained information is a remainder. For S_3/A_3, two permutations are indistinguishable when they have the same parity; the retained information is whether the permutation is even or odd. For $D_n/\langle r\rangle$, two symmetries are indistinguishable when they differ by a rotation; the quotient retains whether the symmetry preserves or reverses orientation.

Example 2.53 (The Dihedral Quotient in Detail)**.** Let $N = \langle r\rangle \leq D_n$. The relation $srs^{-1} = r^{-1}$ implies $sr^is^{-1} = r^{-i} \in N$, so conjugation by the reflection keeps the rotation subgroup inside itself. Conjugation by a rotation plainly does the same; therefore N is normal. Its two cosets are

$$N = \{1, r, \dots, r^{n-1}\} \quad \text{and} \quad sN = \{s, sr, \dots, sr^{n-1}\}.$$

The product of two elements of the second coset lies in N, because

$$(sr^i)(sr^j) = s(r^i s)r^j = r^{j-i}.$$

Thus the nontrivial coset has order two in the quotient. This gives a direct, representative-level verification that $D_n/N \cong C_2$.

Example 2.54 (A Nonnormal Subgroup Revisited)**.** Let $H = \langle (12) \rangle \leq S_3$. It has three left cosets, each with two elements. The set of cosets is therefore perfectly meaningful, but the attempted product of cosets depends on representatives. Indeed, $H = (12)H$, yet multiplying each representative by the coset $(13)H$ produces different cosets. The obstruction is not a flaw in the formula; it is the fact that conjugating (12) by (13) produces a transposition outside H. The normality condition is exactly what removes this obstruction in the quotient criterion.

Remark 2.55 (Proof Technique: Pass to the Right Quotient)*.* When a problem contains a homomorphism, a normal subgroup, or a statement that is unchanged after "forgetting" some information, try forming a quotient. The quotient may turn a complicated relation into the identity relation. The First Isomorphism Theorem then identifies the simplified object with an image, and the Correspondence Theorem translates subgroup questions upstairs and downstairs. This is one of the central reusable moves of graduate algebra.

Why These Constructions Matter Later

The material of this chapter forms a prototype for several later branches of algebra. A quotient group is the first example of imposing an equivalence relation compatible with an operation. In ring theory, ideals play the role of normal subgroups, and the same well-definedness question produces quotient rings. In module theory, submodules replace normal subgroups; the isomorphism theorems and correspondence theorem reappear with almost identical proofs.

Group actions supply a second durable pattern. When a group acts on a set of geometric objects, on roots of a polynomial, or on a vector space, the orbit records what can be moved into what, and the stabilizer records the residual symmetry at a chosen point. This is why the elementary formula $|G| = |Gx|\,|G_x|$ later becomes a tool in Galois theory, representation theory, algebraic geometry, and number theory.

Finally, the examples S_3 and D_4 should be regarded as small models of two different sources of noncommutativity. In S_3, noncommutativity comes from rearranging different objects; in D_4, it comes from composing rotations with reflections. Both groups have meaningful quotients and semidirect-product decompositions. Revisiting them when later constructions become abstract is an efficient way to keep the structural ideas grounded.

2.8 Direct Products and Extensions

The external direct product $G \times H$ combines independent symmetries: one may act in the first coordinate without affecting the second. For example, the symmetries of a rectangle with unequal side lengths form $C_2 \times C_2$. The internal criterion below recognizes this construction inside a larger group.

Proposition 2.56. Suppose $N, H \trianglelefteq G$, $G = NH$, and $N \cap H = \{1\}$. Then $G \cong N \times H$.

Proof. Every element is nh. Such an expression is unique because $n_1h_1 = n_2h_2$ implies $n_2^{-1}n_1 = h_2h_1^{-1} \in N \cap H$. Normality makes the commutator of an element of N with one of H lie in both subgroups, hence be trivial. Therefore $(n, h) \mapsto nh$ is a bijective homomorphism. □

Remark 2.57 (Looking Ahead). The quotient map $G \to G/N$ is the prototype for quotient rings and quotient modules. Kernels, images, and quotients will later be packaged into exact sequences.

Exercises

Exercise 2.1 (Basic)**.** In the multiplicative group $\mathbb{Q}^\times$, prove that

$$\langle -1, 2 \rangle = \{\pm 2^k : k \in \mathbb{Z}\}.$$

Decide whether this subgroup is cyclic.

Exercise 2.2 (Core)**.** Prove that a subgroup of index two is normal.

Exercise 2.3 (Core)**.** Let $C_{12} = \langle g \rangle$. Determine $\langle g^3 \rangle \cap \langle g^4 \rangle$ and the subgroup generated by $\langle g^3 \rangle \cup \langle g^4 \rangle$. State the corresponding gcd/lcm pattern for two subgroups of a finite cyclic group.

Exercise 2.4 (Core)**.** Let $H \leq G$ be the unique subgroup of G having its order. Prove that $H \trianglelefteq G$. Apply this criterion to the subgroup of order three in a cyclic group of order twelve.

Exercise 2.5 (Core)**.** Let $N \trianglelefteq G$, and let $gN \in G/N$. Prove that the order of gN is the least positive integer r for which $g^r \in N$, when such an integer exists. Apply this to compute the order of $r\langle r^d \rangle$ in $D_n/\langle r^d \rangle$ for $d \mid n$.

Exercise 2.6 (Further)**.** If G is cyclic of order n, prove $\mathrm{Aut}(G) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.

Exercise 2.7 (Looking Ahead)**.** Let $N \trianglelefteq G$. Use the Correspondence Theorem to show that maximal subgroups of G/N correspond to maximal subgroups of G that contain N.

Chapter 3

Group Actions and Structural Group Theory

3.1 Actions, Orbits, Stabilizers, and Permutation Representations

An action studies a group by observing the objects it moves. This makes an abstract group concrete without choosing generators or a multiplication table: an action is a systematic way to realize its elements as permutations.

Definition 3.1. An **action** of G on X is a map $G \times X \to X$, written $(g, x) \mapsto g \cdot x$, satisfying the following two requirements:

1. the group identity does nothing: $1 \cdot x = x$ for all $x \in X$;
2. successive motions combine according to multiplication: $(gh) \cdot x = g \cdot (h \cdot x)$.

The **orbit** of x is $Gx = \{g \cdot x : g \in G\}$, and its **stabilizer** is $G_x = \{g \in G : g \cdot x = x\}$.

Equivalently, an action is a homomorphism $G \to \mathrm{Sym}(X)$. This equivalence explains the two action axioms: they say exactly that the identity acts as the identity permutation and a product acts by composition. The natural action of S_n on $\{1, \ldots, n\}$, the action of D_n on the vertices of a regular polygon, and the left-translation action of G on itself are the basic examples.

It is useful to pause over these examples. The natural action of S_n is transitive, and the stabilizer of 1 consists of permutations of the other $n-1$ letters. The geometric action of D_n is also transitive; the stabilizer of a vertex has two elements, the identity and the reflection through that vertex. The left regular action of G is transitive with trivial stabilizer. Finally, the coset action on G/H is transitive and has stabilizer H at the coset H. These are the examples to test whenever an abstract action appears later.

Definition 3.2. Let $\rho : G \to \mathrm{Sym}(X)$ be the homomorphism associated to an action. Its **kernel** is

$$\ker \rho = \{g \in G : g \cdot x = x \text{ for every } x \in X\}.$$

The action is **faithful** if this kernel is trivial.

Thus an action is faithful exactly when no nonidentity group element is invisible on X, equivalently when ρ is injective. The natural action of S_n and the left regular action of any group are faithful. By contrast, the action of D_4 on the set of diagonals of a square is not faithful: distinct geometric symmetries can induce the same diagonal action.

Proposition 3.3. The stabilizer is a subgroup, the orbits partition X, and an action gives a homomorphism $G \to \mathrm{Sym}(X)$.

Proof. For the stabilizer, $1 \cdot x = x$, so $1 \in G_x$. If $g, h \in G_x$, then $(gh^{-1}) \cdot x = g \cdot (h^{-1} \cdot x) = g \cdot x = x$, so the subgroup test applies. If two orbits meet, say $g \cdot x = h \cdot y$, then applying g^{-1} yields $x = g^{-1}h \cdot y$. Hence $x \in Gy$, and applying group elements shows $Gx \subseteq Gy$; symmetry gives equality. Finally, the assignment $g \mapsto \rho(g)$, where $\rho(g)(x) = g \cdot x$, satisfies $\rho(gh) = \rho(g)\rho(h)$ by the second action axiom. $\square$

Theorem 3.4 (Cayley)**.** Every group G is isomorphic to a subgroup of $\mathrm{Sym}(G)$.

Proof. Let G act on itself by left translation, $g \cdot x = gx$. The associated homomorphism sends g to the permutation $x \mapsto gx$. If this permutation is the identity, then evaluating it at 1 gives $g = 1$. Its kernel is therefore trivial, so the homomorphism is injective. $\square$

Example 3.5. For the left action of G on itself, the orbit of 1 is all of G and its stabilizer is trivial. For the conjugation action, the orbit and stabilizer become a conjugacy class and a centralizer. These examples show that one definition organizes both the regular action and the internal symmetry of a group.

Example 3.6 (The Coset Action)**.** For $H \leq G$, left multiplication defines $g \cdot (aH) = (ga)H$ on G/H. The action is transitive and the stabilizer of H is H. Its kernel is

$$\bigcap_{g \in G} gHg^{-1}.$$

Indeed, an element fixes every coset exactly when it belongs to every conjugate of H. This intersection, called the **core** of H, is the largest normal subgroup of G contained in H. The example is an important way to construct permutation representations from subgroups.

Theorem 3.7 (Orbit–Stabilizer)**.** For a finite group action, $|Gx| = [G : G_x]$, and therefore $|G| = |Gx|\,|G_x|$.

Proof Idea. The cosets of G_x are exactly the different ways to move x. Two elements g, h send x to the same point precisely when $h^{-1}g$ fixes x, which is the equivalence relation defining the cosets.

Proof. Define $\Phi : G/G_x \to Gx$ by $\Phi(gG_x) = g \cdot x$. To check well-definedness, suppose $gG_x = hG_x$. Then $h^{-1}g \in G_x$, whence $g \cdot x = h \cdot ((h^{-1}g) \cdot x) = h \cdot x$. Conversely, if $g \cdot x = h \cdot x$, applying h^{-1} gives $(h^{-1}g) \cdot x = x$; thus $h^{-1}g \in G_x$ and $gG_x = hG_x$. This proves injectivity at the same time. Surjectivity holds because every point of the orbit has the form $g \cdot x$. Therefore $|Gx| = [G : G_x]$, and Lagrange's theorem yields $|G| = |Gx|\,|G_x|$. ☐

Example 3.8. Let D_4 act on the four vertices of a square. The orbit of any vertex is the whole vertex set, so it has size four. Its stabilizer contains the identity and the reflection across the diagonal through that vertex, so it has size two. Thus $|D_4| = 4 \cdot 2$, as orbit–stabilizer predicts.

Corollary 3.9 (Burnside's Counting Lemma)**.** The number of orbits of a finite action on X is

$$\frac{1}{|G|} \sum_{g \in G} |\{x \in X : g \cdot x = x\}|.$$

Proof. Count pairs (g, x) with $g \cdot x = x$. Counting by g gives the numerator. Counting by orbits gives $|G|$ for each orbit, by orbit–stabilizer. ☐

Example 3.10 (Counting Colourings up to Symmetry)**.** Let the rotation group C_4 act on the two-colourings of the vertices of a square. The identity fixes all 2^4 colourings, a quarter-turn fixes two, the half-turn fixes four, and the other quarter-turn fixes two. Burnside's lemma gives $(16+2+4+2)/4 = 6$ rotational symmetry types. The point is not the answer itself: the action replaces a potentially awkward case distinction by a fixed-point count.

3.2 Conjugation, Centralizers, and the Class Equation

Conjugation answers the question “which elements look the same from inside the group?” It preserves order and many other structural properties.

Definition 3.11. For a group G and an element $x \in G$, the **centralizer** of x is

$$C_G(x) = \{g \in G : gx = xg\}.$$

The **center** of G is

$$Z(G) = \{g \in G : gx = xg \text{ for every } x \in G\};$$

thus $Z(G)$ is the collection of elements that commute with all of G. For a subgroup $H \leq G$, the **normalizer** of H is

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

It consists of the elements whose conjugation preserves H as a set.

Each of these sets is a subgroup of G. For example, the subgroup test for $C_G(x)$ follows from the fact that if a, b commute with x, then $(ab^{-1})x = a(b^{-1}x) = a(xb^{-1}) = (ax)b^{-1} = x(ab^{-1})$. The same calculation with every x gives the assertion for $Z(G)$, and the identity $(ab^{-1})H(ab^{-1})^{-1} = H$ gives it for $N_G(H)$.

Conjugation, $g \cdot x = gxg^{-1}$, has conjugacy classes as its orbits and centralizer $C_G(x)$ as the stabilizer. The elements fixed by all of G are precisely the center $Z(G)$. The normalizer similarly records the largest subgroup of G in which H is normal.

Lemma 3.12 (p-Group Action Congruence)**.** If a finite p-group P acts on a finite set X, then

$$|X| \equiv |X^P| \pmod{p},$$

where $X^P = \{x \in X : g \cdot x = x \text{ for every } g \in P\}$.

Proof. Partition X into P-orbits. Orbit–stabilizer makes every orbit size a power of p. The orbits of size one are exactly the points fixed by all of P; every other orbit has size divisible by p. Adding orbit sizes gives the stated congruence. □

This elementary congruence is a reusable engine: it converts a group action into the existence of fixed points whenever $|X|$ is not divisible by p.

Example 3.13 (A Fixed Point Forced Modulo p)**.** Let a group of order p act on a set with $p+1$ elements. The congruence gives $|X^P| \equiv p+1 \equiv 1 \pmod{p}$, so there is a global fixed point. The same orbit-counting idea powers the center theorem and Sylow congruence.

Remark 3.14 (How to Design an Action Argument). An action proof begins by choosing a set whose points encode the objects one wants to study. To prove a divisibility statement, choose a set with an accessible cardinality and compare orbit sizes. To prove a containment statement, let a subgroup act on a coset space, so that a fixed coset translates into a conjugate containment. To study elements inside one group, use conjugation: its orbits are conjugacy classes and its stabilizers are centralizers. In each case the formal calculation is short only after the right action and the right set have been chosen.

Theorem 3.15 (Class Equation)**.** If $x_1, \ldots, x_r$ represent the noncentral conjugacy classes of finite G, then

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)].$$

Proof. Partition G into conjugacy classes. Central elements yield singleton classes, and orbit–stabilizer gives the size of every other class. ☐

Example 3.16. In S_3, the three transpositions form one conjugacy class, the two three-cycles form another, and the identity is central. Thus $6 = 1+3+2$. This small example already illustrates the class equation as a partition into symmetry types rather than a numerical trick.

Example 3.17 (Centralizers in S_3)**.** The preceding equality can be recovered directly from centralizers. The centralizer of (12) is $\{1,(12)\}$, so orbit–stabilizer gives a conjugacy class of size $[S_3 : C_{S_3}((12))] = 3$. The centralizer of (123) is $\langle (123) \rangle$, so its class has size two. This is a useful practical lesson: to count conjugates of an element, compute the subgroup of elements that commute with it.

Corollary 3.18. Every finite p-group has nontrivial center.

Proof Idea. For a p-group, every noncentral orbit under conjugation has size divisible by p. The class equation therefore forces the fixed-point part, the center, to have size divisible by p as well.

Proof. For a noncentral x, the centralizer $C_G(x)$ is proper. Its order is a proper divisor of the order of the p-group G, hence $[G : C_G(x)]$ is divisible by p. Therefore the class equation gives

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)] \equiv |Z(G)| \pmod{p}.$$

Since $p \mid |G|$, also $p \mid |Z(G)|$. In particular the center has more than its identity element. ☐

3.3 The Sylow Theorems and Structural Applications

Sylow theory turns the divisibility of $|G|$ into actual subgroups. Its proofs are a model use of actions: choose a finite set on which a p-group acts, observe that nontrivial orbit sizes are divisible by p, and extract a fixed point from a count not divisible by p.

Theorem 3.19 (Cauchy)**.** If a prime p divides the order of a finite group G, then G has an element of order p.

The unusual-looking tuple set is chosen so that it has a readily computed size and a natural cyclic symmetry. Once the first $p-1$ entries are selected, the last must be $(g_1 \cdots g_{p-1})^{-1}$, so there are exactly $|G|^{p-1}$ tuples. Rotation preserves the condition that the product is one because a cyclic rotation conjugates the total product by its first entry.

Proof Idea. The set of tuples has cardinality divisible by p, while cyclic rotation organizes it into orbits of size 1 or p. The fixed points are exactly the repeated tuples, which encode elements satisfying $g^p = 1$.

Proof. Let X be the set of p-tuples $(g_1, \ldots, g_p)$ with $g_1 \cdots g_p = 1$. Its first $p-1$ entries may be chosen freely, so $|X| = |G|^{p-1}$, which is divisible by p. Cyclically rotate these tuples. Every orbit has size 1 or p, and the fixed tuples are exactly $(g, \ldots, g)$ with $g^p = 1$. Thus the number of such g is divisible by p. It includes 1, so there is a nonidentity one; its order divides the prime p, hence equals p. ☐

This proof illustrates a recurring discovery principle: construct a finite set whose cardinality is divisible by p, then let a group of order p act so that its fixed points encode the desired elements.

Theorem 3.20 (Sylow I: existence)**.** Let $|G| = p^a m$, where p is prime and $p \nmid m$. Then G has a subgroup of order p^a. Such a subgroup is called a **Sylow p-subgroup**.

Proof Idea. Induct on the group order. A central element of order p lets us reduce to a quotient. If the center does not supply one, the class equation locates a proper centralizer that still contains the full p-part of $|G|$.

Proof. We use strong induction on $|G|$; the assertion is immediate when $a = 0$. First suppose that $p \mid |Z(G)|$. By Cauchy's theorem, $Z(G)$ contains a subgroup C of order p. It is normal in G, and $|G/C| = p^{a-1}m < |G|$. Let $\pi : G \to G/C$ be the quotient map. The induction hypothesis applied to G/C gives a subgroup $\overline{P} \leq G/C$ of order p^{a-1}. Define $P = \pi^{-1}(\overline{P})$. Then $P/C \cong \overline{P}$, so $|P| = |C|\,|\overline{P}| = p \cdot p^{a-1} = p^a$.

Now suppose $p \nmid |Z(G)|$. In the class equation, the noncentral terms are conjugacy-class sizes $[G : C_G(x_i)]$. Since $|G| \equiv 0 \pmod{p}$ but $|Z(G)| \not\equiv 0 \pmod{p}$, at least one of these indices is not divisible by p. For that noncentral x_i, the centralizer $C_G(x_i)$ is a proper subgroup of G, while

$$|C_G(x_i)| = \frac{|G|}{[G : C_G(x_i)]}$$

still has p^a as its full p-part. The induction hypothesis inside this strictly smaller group produces a subgroup of order p^a, which is also a subgroup of G. □

Theorem 3.21 (Sylow II: containment and conjugacy)**.** Let Q be a Sylow p-subgroup of a finite group G. Every p-subgroup $P \leq G$ has a conjugate contained in Q. In particular, all Sylow p-subgroups are conjugate.

Proof. Let P act by left multiplication on the coset set G/Q. Its size is $[G : Q] = m$, which is not divisible by p. Every P-orbit has size a power of p, by orbit–stabilizer. If there were no fixed coset, the sum of the orbit sizes would be divisible by p, contrary to $p \nmid m$. Thus some coset gQ is fixed by every element of P. The equality $pgQ = gQ$ for all $p \in P$ is equivalent to $g^{-1}pg \in Q$ for all $p \in P$, hence to $g^{-1}Pg \leq Q$. If P is itself Sylow, the two groups in this containment have the same finite order, so they are equal. This proves conjugacy. □

Theorem 3.22 (Sylow III: number of Sylow subgroups)**.** If $|G| = p^a m$ with $p \nmid m$, the number n_p of Sylow p-subgroups divides m and satisfies $n_p \equiv 1 \pmod{p}$.

Proof. Fix a Sylow subgroup Q. Sylow II makes conjugation by G transitive on the set $\mathcal{S}$ of Sylow p-subgroups, so orbit–stabilizer gives

$$n_p = |\mathcal{S}| = [G : N_G(Q)].$$

Because $Q \le N_G(Q)$, the index divides $|G|/|Q| = m$.

Now let Q act by conjugation on $\mathcal{S}$. Suppose that $R \in \mathcal{S}$ is fixed. Then $Q \le N_G(R)$; since $R \trianglelefteq N_G(R)$, the product QR is a subgroup. It is a p-subgroup containing the Sylow subgroup R, hence $QR = R$, and therefore $Q \le R$. Equal orders give $Q = R$. Thus Q is the unique fixed point of this action. Every remaining orbit has size divisible by p, so $|\mathcal{S}| \equiv 1 \pmod p$. □

Example 3.23. If $|G| = 21$, then n_7 divides 3 and is congruent to 1 modulo 7; hence $n_7 = 1$, so the Sylow 7-subgroup is normal. This is a structural conclusion, not a classification of all groups of order 21.

Corollary 3.24. A Sylow p-subgroup is normal if and only if it is the unique Sylow p-subgroup. In particular, every group of order 45 has a normal Sylow 5-subgroup.

Proof. Conjugation carries a Sylow p-subgroup to another one. Thus a normal one is fixed by all conjugations and, by Sylow II, is the only one. Conversely, a unique Sylow subgroup is fixed under conjugation, so it is normal. For $|G| = 45 = 5 \cdot 9$, Sylow III says $n_5 \mid 9$ and $n_5 \equiv 1 \pmod 5$; among $1, 3, 9$, only 1 has this congruence. □

3.4 Semidirect Products and Split Extensions

A direct product describes two components that commute and have no interaction. A semidirect product records the next possibility: the second component acts on the first by automorphisms. The homomorphism $\alpha : H \to \mathrm{Aut}(N)$ is therefore not auxiliary data; it is the rule describing that interaction.

The multiplication formula is forced by the internal situation. Suppose $N \trianglelefteq G$, $H \le G$, and every element of G has a unique form nh. Then

$$(nh)(n'h') = n(hn'h^{-1})hh'.$$

Normality puts $hn'h^{-1}$ back in N, while the element h acts on N by conjugation. The external definition simply preserves this rule when the two groups have not yet been placed inside a common ambient group.

Definition 3.25. For $\alpha : H \to \mathrm{Aut}(N)$, the **semidirect product** $N \rtimes_\alpha H$ is $N \times H$ with $(n, h)(n', h') = (n\alpha(h)(n'), hh')$.

Proposition 3.26. This operation makes $N \rtimes_\alpha H$ a group; $N \times \{1\}$ is normal, $\{1\} \times H$ is a subgroup, and they have trivial intersection.

Proof. The identity is $(1,1)$. Since every $\alpha(h)$ is an automorphism,

$$(n,h)^{-1} = (\alpha(h^{-1})(n^{-1}), h^{-1}).$$

For associativity, the first coordinates of the two possible products are

$$\begin{aligned}
&((n,h)(n',h'))(n'',h'') : n\alpha(h)(n')\alpha(hh')(n''),\\
&(n,h)((n',h')(n'',h'')) : n\alpha(h)\big(n'\alpha(h')(n'')\big).
\end{aligned}$$

They agree because $\alpha(hh') = \alpha(h)\alpha(h')$. The two displayed subsets have the claimed subgroup properties. Finally,

$$(n,h)(n',1)(n,h)^{-1} = \big(n\alpha(h)(n')n^{-1}, 1\big) \in N \times \{1\},$$

so $N \times \{1\}$ is normal. The two subgroups meet only in $(1,1)$. □

An internal semidirect product is equivalently a split short exact sequence

$$1 \longrightarrow N \longrightarrow G \longrightarrow H \longrightarrow 1.$$

Here "split" means that the quotient map $q : G \to H$ has a homomorphic section $s : H \to G$. Then q restricted to $s(H)$ is inverse to s, so $N \cap s(H) = 1$, where $N = \ker q$. Moreover, if $g \in G$ and $h = q(g)$, then

$$q\big(gs(h)^{-1}\big) = hh^{-1} = 1,$$

so $gs(h)^{-1} \in N$ and $g \in Ns(H)$. Hence $G = Ns(H)$, and Proposition 3.29 applies. Conversely, an internal semidirect product gives $G/N \cong H$, and the inclusion of the complement gives a homomorphic section. This is only a first encounter with extensions, but it explains why semidirect products are ubiquitous.

Example 3.27. Let $C_2 = \langle s \rangle$ act on $C_3 = \langle r \rangle$ by inversion, $srs^{-1} = r^{-1}$. Then $C_3 \rtimes C_2$ has presentation $\langle r, s \mid r^3 = s^2 = 1,\ srs^{-1} = r^{-1} \rangle$, and the map sending r to (123) and s to (12) identifies it with S_3. It is not a direct product because its two factors do not commute.

Example 3.28. More generally, $D_n \cong C_n \rtimes C_2$, where the nontrivial element of C_2 acts on C_n by inversion. The relation $srs^{-1} = r^{-1}$ is exactly the twisting encoded by this action. Thus the dihedral examples introduced in Chapter 2 provide a geometric model for every semidirect product in which a reflection reverses a cyclic rotation.

Proposition 3.29 (Internal Semidirect Product)**.** If $N \trianglelefteq G$, $H \leq G$, $G = NH$, and $N \cap H = 1$, then $G \cong N \rtimes H$, where H acts on N by conjugation.

Proof. Every element has an expression nh. It is unique: if $nh = n'h'$, then $n^{-1}n' = hh'^{-1}$ belongs to both N and H, hence is 1. Sending (n, h) to nh is therefore bijective. Its compatibility with multiplication is $nhn'h' = n(hn'h^{-1})hh'$, which is exactly the external semidirect-product law for conjugation. □

3.5 Composition Series, Solvability, and Nilpotence

Composition series make precise the idea of breaking a group into indivisible layers. A simple group has no nontrivial proper normal subgroup, so it cannot be decomposed further by quotients. Jordan–Hölder says that, although the intermediate subgroups may differ, the resulting simple layers are intrinsic.

Definition 3.30. A **composition series** is a finite normal series with simple factors. A group is **solvable** when it has a finite normal series with abelian factors.

Example 3.31. The chain $S_3 \trianglerighteq A_3 \trianglerighteq 1$ is a composition series, with factors C_2 and C_3. The order in which the factors are listed has no intrinsic significance; their isomorphism types do.

Proposition 3.32. Every finite group has a composition series.

Proof. Induct on the group order. If $G \neq 1$, choose a maximal proper normal subgroup N, which exists because the group is finite. Then G/N is simple, and induction gives a composition series for N. Prepending G to that series gives one for G. □

Theorem 3.33 (Jordan–Hölder)**.** Any two composition series of a finite group have the same length and the same simple factors, up to ordering and isomorphism.

Proof. We argue by induction on $|G|$. There is nothing to prove for $G = 1$. Write the two series as

$$G = G_0 \trianglerighteq G_1 = N \trianglerighteq \cdots \trianglerighteq 1, \qquad G = H_0 \trianglerighteq H_1 = M \trianglerighteq \cdots \trianglerighteq 1.$$

The quotients G/N and G/M are simple, so N and M are maximal proper normal subgroups of G.

If $N = M$, the tails are composition series of the smaller group N. The induction hypothesis says that these tails have the same length and the same factors, and adding their common first factor G/N proves the claim.

Suppose $N \neq M$. The product NM is normal in G: for $g \in G$, $g(NM)g^{-1} = (gNg^{-1})(gMg^{-1}) = NM$. It strictly contains N, since otherwise $M \leq N$, and then the maximality of M would force $N = M$. Maximality of N consequently gives $NM = G$. Put $I = N \cap M$. Since both N and M are normal in G, so is I, and in particular I is normal in both N and M. Applying the Second Isomorphism Theorem in the forms

$$M/(M \cap N) \cong MN/N = G/N, \qquad N/(N \cap M) \cong NM/M = G/M$$

shows that M/I and N/I are simple.

Choose a composition series from I down to 1. Because N/I is simple, adjoining N as a new initial term produces a composition series of N, whose new factor is N/I; similarly, adjoining M produces a composition series of M with new factor M/I. By induction, the first constructed series has the same factors as the tail of the original G-series beginning at N, and the second has the same factors as the tail beginning at M. The complete first factor list is therefore the factors below I, followed by N/I and G/N; the second is the same factors below I, followed by M/I and G/M. The displayed isomorphisms identify $N/I \cong G/M$ and $M/I \cong G/N$, so the two lists agree up to interchanging these final two factors. $\square$

For Galois theory it is useful to recognize solvability without guessing a normal series. The commutator $[x, y] = xyx^{-1}y^{-1}$ measures the failure of x and y to commute: $[x, y] = 1$ if and only if $xy = yx$. Indeed, $xyx^{-1}y^{-1} = 1$ is equivalent, after multiplying on the right by yx, to $xy = yx$. The subgroup generated by all commutators is the **derived subgroup** $[G, G]$; define $G^{(0)} = G$ and $G^{(i+1)} = [G^{(i)}, G^{(i)}]$.

For later quotient constructions, it matters that $[G, G]$ is normal. Conjugation sends a commutator to a commutator:

$$g[x, y]g^{-1} = [gxg^{-1}, gyg^{-1}].$$

Thus conjugation preserves the subgroup generated by all commutators, and $[G, G] \trianglelefteq G$.

Proposition 3.34. The quotient $G/[G, G]$ is abelian and is the largest abelian quotient of G: every homomorphism from G to an abelian group factors uniquely through it.

Proof. Every commutator becomes trivial in the quotient, so its cosets commute. If $f : G \to A$ and A is abelian, then $f([x, y]) = 1$ for all x, y, hence $[G, G] \leq \ker f$. The universal property of the quotient map now gives the unique factorization. $\square$

Example 3.35 (The Derived Subgroup of S_3)**.** The quotient $S_3/A_3 \cong C_2$ is abelian, so the universal property above gives $[S_3, S_3] \leq A_3$. On the other hand, direct calculation gives

$$[(12), (23)] = (12)(23)(12)(23) = (123)^2 = (132).$$

This nonidentity element generates A_3, so $A_3 \leq [S_3, S_3]$. Hence $[S_3, S_3] = A_3$, and $S_3/[S_3, S_3] \cong C_2$.

Theorem 3.36. A group G is solvable if and only if $G^{(r)} = 1$ for some $r \geq 0$.

Proof. If the derived series reaches 1, its successive quotients are abelian by the preceding proposition. Conversely, let $G = G_0 \trianglerighteq \cdots \trianglerighteq G_r = 1$ have abelian factors. Inductively, $G^{(i)} \leq G_i$: if $G^{(i)} \leq G_i$, then the image of $G^{(i)}$ in the abelian group G_i/G_{i+1} has trivial derived subgroup, so $G^{(i+1)} \leq G_{i+1}$. Hence $G^{(r)} = 1$. $\square$

Solvability asks for a group built by successive abelian quotients. Nilpotence is stronger: it asks that the construction proceed from the center outward. This distinction matters already for S_3, which is solvable but not nilpotent.

Definition 3.37. Set $Z_0(G) = 1$ and define the **upper central series** inductively by

$$Z_{i+1}(G)/Z_i(G) = Z\big(G/Z_i(G)\big).$$

The group G is **nilpotent** if $Z_c(G) = G$ for some c. The least such c is its nilpotence class (unless $G = 1$).

Equivalently, a nilpotent group admits a finite chain

$$1 = N_0 \leq N_1 \leq \cdots \leq N_c = G$$

with $N_{i+1}/N_i \leq Z(G/N_i)$. Indeed, inductively $N_i \leq Z_i(G)$: this is clear for $i = 0$, and centrality of N_{i+1}/N_i gives $N_{i+1} \leq Z_{i+1}(G)$. Thus $G = N_c \leq Z_c(G)$, so G is nilpotent. Conversely, the upper central series itself is such a chain. The upper central series is therefore the canonical maximal central series, obtained by repeatedly pulling back the center of the current quotient. Thus abelian groups are nilpotent of class at most one. A nontrivial nilpotent group has nontrivial center; therefore S_3, whose center is trivial, is not nilpotent.

Theorem 3.38. Every finite p-group is nilpotent.

Proof. Induct on $|G|$. The center of a nontrivial finite p-group is nontrivial, and Cauchy's theorem gives a central subgroup C of order p. By induction, G/C has an upper central series

$$1 = \overline{Z}_0 \leq \overline{Z}_1 \leq \cdots \leq \overline{Z}_c = G/C.$$

Let Z_i be the inverse image of $\overline{Z}_i$ in G. Then $Z_0 = C$, and the chain

$$1 \leq C = Z_0 \leq Z_1 \leq \cdots \leq Z_c = G$$

is central: if $z \in Z_{i+1}$ and $g \in G$, then the image of z is central modulo $\overline{Z}_i$, so $[z, g] \in Z_i$. The first step is central because $C \leq Z(G)$. Thus this is a central series from 1 to G, proving that G is nilpotent. $\square$

Proposition 3.39. Every nilpotent group is solvable.

Proof. In the upper central series, each factor $Z_{i+1}(G)/Z_i(G)$ lies in the center of $G/Z_i(G)$, so it is abelian. Thus the upper central series is a solvable series. $\square$

Proposition 3.40 (Subgroups and quotients)**.** Subgroups and quotients of solvable groups are solvable.

Proof. Let $G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_r = 1$ be a solvable series, and let $H \leq G$. Intersecting gives

$$H = H \cap G_0 \trianglerighteq H \cap G_1 \trianglerighteq \cdots \trianglerighteq H \cap G_r = 1.$$

Each term is normal in the preceding one. The map $H \cap G_i \to G_i/G_{i+1}$ has kernel $H \cap G_{i+1}$, so its quotient $(H \cap G_i)/(H \cap G_{i+1})$ is isomorphic to a subgroup of the abelian group G_i/G_{i+1}. Removing repeated terms leaves a solvable series for H.

If $q : G \to Q$ is onto, the chain $Q = q(G_0) \trianglerighteq q(G_1) \trianglerighteq \cdots \trianglerighteq q(G_r) = 1$ has normal terms. Each nontrivial factor is an image of a subgroup of some abelian factor G_i/G_{i+1}, and is therefore abelian. After repetitions are removed, this is a solvable series for Q. $\square$

Proposition 3.41 (Extensions)**.** If $N \trianglelefteq G$, and both N and G/N are solvable, then G is solvable.

Proof. Take a solvable series for N. Take also a solvable series for G/N and pull its terms back under the quotient map $G \to G/N$. Splicing the first series to these inverse images gives a normal series for G; its factors are, respectively, factors from the series for N and factors isomorphic to those from the series for G/N. They are all abelian. □

Remark 3.42 (Looking Ahead). The action of a Galois group on roots and intermediate fields will make these methods central again. Solvability of groups eventually characterizes solvability of polynomial equations by radicals.

3.6 Two Recurring Structural Laboratories

The abstract constructions of this chapter become much easier to retain when they are repeatedly tested on a few familiar groups. The following two examples summarize how the same group can illuminate several different ideas.

The Group S_3

The six-element symmetric group is the smallest nonabelian group. It acts faithfully and transitively on three letters. Orbit–stabilizer at the letter 1 gives

$$|S_3| = |S_3 \cdot 1| \, |(S_3)_1| = 3 \cdot 2.$$

The stabilizer is $\{1, (23)\}$, a copy of C_2. Thus the action turns a concrete observation about permutations into a coset calculation.

Under conjugation, its elements separate into the classes

$$\{1\}, \qquad \{(12), (13), (23)\}, \qquad \{(123), (132)\}.$$

The class equation is $6 = 1+3+2$. This is more than bookkeeping: it shows that the three transpositions are structurally indistinguishable within S_3, whereas a transposition subgroup is not normal because conjugation moves it to another such subgroup.

The subgroup $A_3 = \langle (123) \rangle$ is normal, and the quotient $S_3/A_3 \cong C_2$ records parity. The subgroup A_3 is complemented by $\langle (12) \rangle$, and conjugation by (12) inverts (123). Hence

$$S_3 \cong C_3 \rtimes C_2.$$

Finally the chain $S_3 \trianglerighteq A_3 \trianglerighteq 1$ shows that S_3 is solvable. Its derived subgroup is A_3, so its abelianization is C_2. One group has therefore illustrated actions, cosets, normality, quotients, conjugacy, semidirect products, composition factors, and solvability.

The Group D_4

The square group $D_4 = \langle r, s \mid r^4 = s^2 = 1,\ srs^{-1} = r^{-1}\rangle$ has eight elements. Its geometric action on the four vertices is transitive, and the stabilizer of a vertex has two elements. This gives $8 = 4 \cdot 2$, the orbit–stabilizer formula in a visibly geometric form.

The rotation subgroup $\langle r\rangle \cong C_4$ is normal because $sr^is^{-1} = r^{-i}$ remains a rotation. The quotient by rotations is C_2: it distinguishes orientation-preserving symmetries from reflections. The relation $srs^{-1} = r^{-1}$ also displays D_4 as $C_4 \rtimes C_2$. This is an especially useful contrast with $C_4 \times C_2$, whose two factors commute; the dihedral relation gives the nontrivial twisting.

The center of D_4 is $\{1, r^2\}$. Since the quotient by this subgroup is abelian, the chain $1 \leq \langle r^2\rangle \leq D_4$ is a central series. Thus D_4 is nilpotent. Comparing it with S_3, which is solvable but has trivial center and is not nilpotent, makes the hierarchy

$$\text{abelian} \implies \text{nilpotent} \implies \text{solvable}$$

concrete rather than merely formal.

Remark 3.43 (A Study Method). When a new group-theoretic construction is introduced, revisit S_3 and D_4. Ask which subgroups are normal, what the natural actions are, what a quotient remembers, and whether the group decomposes as a direct or semidirect product. This practice develops structural intuition more effectively than a long list of unrelated calculations.

Exercises

Exercise 3.1 (Basic)**.** Show that the kernel of a permutation representation is the intersection of all point stabilizers.

Exercise 3.2 (Core)**.** Let G act transitively on a set X, and fix $x \in X$. Prove that the stabilizers of the points of X are conjugate. Interpret this result for the left action of G on the cosets of a subgroup H.

Exercise 3.3 (Core)**.** Compute the conjugacy classes, centralizers, and class equation of S_3. Explain which features of the computation reflect the geometry of a triangle.

Exercise 3.4 (Core)**.** Let P be a Sylow p-subgroup of a finite group G. Prove that $P \trianglelefteq G$ if and only if it is the unique Sylow p-subgroup. Deduce that any Sylow subgroup whose number is forced to be one by the Sylow divisibility and congruence conditions is normal.

Exercise 3.5 (Core)**.** Let G be a finite p-group. Prove that every nontrivial normal subgroup of G meets $Z(G)$ nontrivially.

Exercise 3.6 (Further)**.** Show that $S_3 \cong C_3 \rtimes C_2$, but is not a direct product of them.

Exercise 3.7 (Looking Ahead)**.** Let $|G| = p^2$, where p is prime. Use the center theorem and the quotient by a central subgroup of order p to prove that G is abelian. Explain why this does not classify such groups.

Part II

Rings and Polynomial Rings

Chapter 4

Rings, Ideals, and Localization

The integers have addition and multiplication, tied together by the distributive laws. A ring retains exactly this structure. The familiar examples—integers, residue classes, polynomials, and matrices—will give the definitions their meaning. Ideals permit us to force selected elements to be zero; localization instead forces selected elements to be invertible.

4.1 Rings, Subrings, and Homomorphisms

Definition 4.1. A **ring** is a set R with an abelian-group structure under $+$, an associative multiplication, and a multiplicative identity 1, such that, for all $a, b, c \in R$,

$$\begin{array}{rclrcl} (a+b)+c & = & a+(b+c), & a+b & = & b+a, \\ a+0 & = & a, & a+(-a) & = & 0, \\ (ab)c & = & a(bc), & 1a & = & a1 = a, \\ a(b+c) & = & ab+ac, & (a+b)c & = & ac+bc. \end{array}$$

A ring is **commutative** if $ab = ba$ for all a, b. A subring contains the ambient identity 1_R and is closed under subtraction and multiplication.

If $1 = 0$, then R has only one element: for every $a \in R$,

$$a = 1a = 0a = 0.$$

Thus the zero ring is included in the definition.

Remark (Ring conventions). Terminology varies among authors. Some allow rings without an identity, and some call a subset a subring when it is closed under subtraction and multiplication without requiring it to contain the ambient identity. Here rings have identities, ring homomorphisms preserve them, and subrings contain the same identity as the ambient ring. For example, $2\mathbb{Z}$ is closed under subtraction and multiplication, so it is a subring of $\mathbb{Z}$ under the looser convention, but it is not a subring in these notes because $1 \notin 2\mathbb{Z}$.

The rings $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$, and a field F, are commutative. So are $\mathbb{Z}/n\mathbb{Z}$, $F[x]$, and $R \times S$, where operations are coordinatewise. Matrix multiplication makes $M_n(F)$ a ring, usually not a commutative one. Thus commutativity must be an additional hypothesis, not a consequence of the axioms.

Proposition 4.2 (Basic identities)**.** For $a, b \in R$, one has

$$0a = a0 = 0, \qquad (-a)b = a(-b) = -(ab), \qquad (-a)(-b) = ab.$$

Proof. Since $0a = (0+0)a = 0a + 0a$, additive cancellation gives $0a = 0$; the proof of $a0 = 0$ is analogous. Now $ab + (-a)b = (a + (-a))b = 0$, giving the next identities. Applying this twice gives $(-a)(-b) = -\big(a(-b)\big) = ab$. □

Definition 4.3. A **unit** is an element u with $uv = vu = 1$ for some v. A nonzero a is a **zero divisor** if $ab = 0$ or $ba = 0$ for some nonzero b. A **domain** is a nonzero commutative ring with identity and no zero divisors; a **field** is a nonzero commutative ring in which every nonzero element is a unit.

Every field is a domain: if $ab = 0$ and $a \neq 0$, multiply by a^{-1}. The converse fails, since $\mathbb{Z}$ is a domain but 2 has no inverse. In $\mathbb{Z}/12\mathbb{Z}$, 5 is a unit, while $2 \cdot 6 = 0$ shows that 2 and 6 are zero divisors. The class of a modulo n is a unit exactly when $\gcd(a, n) = 1$, by Bézout's identity.

Example 4.4 (A useful warning)**.** Over a domain, every unit of $R[x]$ is constant. This is false in a ring with nilpotents: in $(\mathbb{Z}/4\mathbb{Z})[x]$,

$$(1 + 2x)^2 = 1 + 4x + 4x^2 = 1.$$

Thus $1 + 2x$ is a nonconstant unit. Domain hypotheses in polynomial arguments are therefore essential.

Definition 4.5. A ring homomorphism $f : R \to S$ preserves addition, multiplication, and identity. Its kernel is $\ker f = \{a : f(a) = 0\}$, and its image is $\operatorname{im} f = \{f(a) : a \in R\}$.

For example, reduction $\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ is onto and has kernel $n\mathbb{Z}$. As for group homomorphisms, $f(0) = 0$ and $f(-a) = -f(a)$. The determinant is not a ring homomorphism on matrices, because it does not in general preserve addition.

4.2 Ideals and Quotient Rings

An additive subgroup is not automatically suitable for quotienting: it must remain stable when multiplied by arbitrary ring elements. This is precisely the ring analogue of the normal-subgroup condition.

Definition 4.6. A **left ideal** of R is an additive subgroup I such that $ra \in I$ for every $r \in R$, $a \in I$. A **right ideal** is defined by the condition $ar \in I$. A **two-sided ideal** is both a left and a right ideal. Throughout these notes, unless explicitly qualified, an **ideal** means a two-sided ideal, and we write $I \trianglelefteq R$.

For $A \subseteq R$, (A) is the smallest two-sided ideal containing A. If $A = \{a_1, \ldots, a_m\}$, then

$$(a_1, \ldots, a_m) = \left\{ \sum_{\ell=1}^{N} r_\ell a_{i_\ell} s_\ell : N \geq 0,\ 1 \leq i_\ell \leq m,\ r_\ell, s_\ell \in R \right\}.$$

When R is commutative, left, right, and two-sided ideals coincide, and this simplifies to the familiar sums

$$\sum_k r_k a_k.$$

The ideals (0) and $(1) = R$ are trivial. In $\mathbb{Z}$, $(n) = n\mathbb{Z}$, while in $F[x]$, (x) is the set of polynomials with zero constant term. In $F[x, y]$, (x^2, xy) consists of sums $Ax^2 + Bxy$. The odd integers are not an ideal of $\mathbb{Z}$, since $2 \cdot 1$ is not odd. In $M_n(F)$, the matrices whose last row is zero form a right ideal but generally not a left ideal. Quotient rings therefore require two-sided, not merely one-sided, ideals.

The notation (A) is deliberately parallel to the subgroup generated by a subset: it records the smallest ideal forced by the elements of A. In a commutative ring, the principal ideal generated by a is simply $(a) = \{ra : r \in R\}$. Principal ideals reverse divisibility:

$$(a) \subseteq (b) \quad \Longleftrightarrow \quad b \mid a.$$

Indeed, $a \in (b)$ is exactly the assertion $a = br$. Thus, in $\mathbb{Z}$, $(12) \subseteq (6) \subseteq (2)$. This reversal will underlie the use of principal ideals, gcds, ascending chains, and factorization in Chapter 5.

Proposition 4.7. Every ideal of $\mathbb{Z}$ is $n\mathbb{Z}$ for a unique $n \geq 0$.

Proof. For a nonzero ideal I, choose its least positive member n. For $a \in I$, write $a = qn + r$, $0 \le r < n$. Since $r = a - qn \in I$, minimality gives $r = 0$; hence $I = n\mathbb{Z}$. The reverse containment is ideal closure, and $n \ge 0$ makes the generator unique. $\square$

For ideals I, J, the sum, intersection, and product are

$$I + J = \{i + j : i \in I, j \in J\}, \quad I \cap J, \quad IJ = \left\{\sum i_k j_k : i_k \in I, j_k \in J\right\}.$$

They are ideals. The sum is the smallest ideal containing both; the intersection is their largest common subideal. Always $IJ \subseteq I \cap J$. For example, in $\mathbb{Z}$,

$$(6) + (10) = (2), \qquad (6) \cap (10) = (30), \qquad (6)(10) = (60).$$

Thus intersection and product need not agree when the ideals are not comaximal.

Theorem 4.8 (Quotient Ring Criterion)**.** For an additive subgroup $I \le R$, the rule

$$(a + I)(b + I) = ab + I$$

is well-defined exactly when I is a two-sided ideal. In that case R/I is a ring and $\pi(a) = a + I$ is a surjective homomorphism.

Proof. If $a' = a + i$, $b' = b + j$ with $i, j \in I$, then

$$(a + i)(b + j) - ab = aj + ib + ij \in I,$$

using closure under multiplication on both sides. Thus the product is independent of representatives. All ring axioms descend from R. Conversely, $i + I = 0 + I$ implies $(r + I)(i + I) = (r + I)(0 + I)$, hence $ri \in I$, while $(i + I)(r + I) = (0 + I)(r + I)$ gives $ir \in I$. Thus I is two-sided. $\square$

Theorem 4.9 (Universal Property of Quotients)**.** If $f : R \to T$ is a homomorphism and $I \subseteq \ker f$, then there is a unique homomorphism $\bar{f} : R/I \to T$ such that $f = \bar{f} \circ \pi$.

$$\begin{tikzcd}
R \arrow[r, "\pi"] \arrow[dr, "f"'] & R/I \arrow[d, "\bar{f}"] \\
& T
\end{tikzcd}$$

Proof. Put $\bar{f}(a+I) = f(a)$. If $a+I = b+I$, then $a-b \in I$, so $f(a) = f(b)$; this proves well-definedness. The homomorphism laws follow from those for f, and every coset is $\pi(a)$, proving uniqueness. □

Thus R/I is universal among maps that force elements of I to zero, just as quotient groups were in Part I. The structural parallel is

$$\text{normal subgroup } N \trianglelefteq G \;\longleftrightarrow\; G/N, \qquad \text{two-sided ideal } I \trianglelefteq R \;\longleftrightarrow\; R/I.$$

In particular, taking $I = R$ gives R/R, the zero ring.

Theorem 4.10 (Isomorphism and correspondence)**.** For $f : R \to S$, $\ker f$ is an ideal and

$$R/\ker f \cong \operatorname{im} f.$$

Ideals of R/I correspond to ideals of R containing I, by $J \mapsto \pi^{-1}(J)$ and $K \mapsto K/I$; in particular, $(R/I)/(K/I) \cong R/K$ if $I \subseteq K$.

Proof. The kernel is additive and $f(ra) = f(r)f(a) = 0 = f(ar)$ for $a \in \ker f$. The map $a + \ker f \mapsto f(a)$ is well-defined, onto the image, and has zero kernel. Directly checking the two displayed assignments shows they are inverse ideal-preserving maps. Finally, $(r + I) + (K/I) \mapsto r + K$ is well-defined and has zero kernel. □

4.3 Prime Ideals, Maximal Ideals, Domains, and Fields

From this point through localization, rings are assumed to be commutative and unital; the zero ring remains allowed. Thus the unqualified word **ideal** continues to mean a two-sided ideal, but sidedness is automatic.

Definition 4.11. A proper ideal $\mathfrak{p}$ is **prime** if $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. A proper ideal $\mathfrak{m}$ is **maximal** if no ideal is strictly between $\mathfrak{m}$ and R.

Theorem 4.12. $\mathfrak{p}$ is prime iff $R/\mathfrak{p}$ is a domain, and $\mathfrak{m}$ is maximal iff $R/\mathfrak{m}$ is a field. Hence every maximal ideal is prime.

Proof. $(a + \mathfrak{p})(b + \mathfrak{p}) = 0$ iff $ab \in \mathfrak{p}$, giving the first claim. If $\mathfrak{m}$ is maximal and $a \notin \mathfrak{m}$, then $(a) + \mathfrak{m} = R$, so $ra + m = 1$ for suitable r, m, and $r + \mathfrak{m}$ inverts $a + \mathfrak{m}$. Conversely, a field has only the ideals 0 and itself; apply the correspondence theorem. □

In $\mathbb{Z}$, (p) is prime and maximal for prime p, while (6) is neither: $2 \cdot 3 \in (6)$, but $2, 3 \notin (6)$. In $\mathbb{Z}/12\mathbb{Z}$, the ideal (2) is maximal because its quotient is $\mathbb{Z}/2\mathbb{Z}$. Later, the division algorithm will show that $(f) \subset F[x]$ is maximal exactly when f is irreducible.

Theorem (Maximal ideals above proper ideals)**.** Let R be a commutative ring with identity. Every proper ideal I of R is contained in a maximal ideal.

Proof. Let $\mathcal{P}$ be the set of proper ideals of R that contain I, ordered by inclusion. It is nonempty because $I \in \mathcal{P}$. If $\mathcal{C}$ is a chain in $\mathcal{P}$, then

$$J = \bigcup_{K \in \mathcal{C}} K$$

is an ideal containing I: the chain condition puts any two elements of J in a common member of $\mathcal{C}$, where the ideal operations may be performed. Moreover J is proper. Indeed, if $1 \in J$, then $1 \in K$ for some $K \in \mathcal{C}$, contradicting the propriety of K. Thus every chain has an upper bound in $\mathcal{P}$. Zorn's Lemma now gives a maximal member $\mathfrak{m}$ of $\mathcal{P}$, which is maximal among all proper ideals of R, hence is a maximal ideal. □

Remark (A set-theoretic qualification). The preceding proof invokes Zorn's Lemma. For Noetherian rings, the corresponding maximality conclusion follows from the usual ascending-chain maximality argument, so no appeal to Zorn's Lemma is needed in the standard treatment. We do not pursue the precise foundational strength of the general maximal-ideal theorem here.

4.4 The Chinese Remainder Theorem

Lemma 4.13. If $I + J = R$, then $I \cap J = IJ$.

Proof. Certainly $IJ \subseteq I \cap J$. Choose $u \in I, v \in J$ with $u + v = 1$. If $x \in I \cap J$, then $x = xu + xv$, and both summands lie in IJ. □

Theorem 4.14 (Chinese Remainder Theorem)**.** If $I + J = R$, then $R/(I \cap J) \cong R/I \times R/J$.

Proof. The homomorphism $\Phi(r) = (r + I, r + J)$ has kernel $I \cap J$. With $u + v = 1$, $u \in I, v \in J$, the element $av + bu$ maps to $(a + I, b + J)$; hence Φ is onto. Apply the First Isomorphism Theorem. □

Example 4.15. To solve $x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 5$, use $6 + (-5) = 1$, with $6 \in (3)$, $-5 \in (5)$. The proof gives $x = 2(-5) + 3(6) = 8$. All solutions are $8 \pmod{15}$.

The corresponding statement for finitely many pairwise comaximal ideals follows by induction.

4.5 Localization and Local Rings

The inclusion $\mathbb{Z} \subseteq \mathbb{Q}$ permits division by every nonzero integer. Often we want only selected denominators. For instance,

$$\mathbb{Z}[1/2] = \{a/2^n : a \in \mathbb{Z},\ n \geq 0\}$$

allows division by powers of 2, without requiring arbitrary rational denominators. Localization is the controlled construction that introduces chosen denominators: it makes selected elements units while otherwise changing the ring as little as the required universal property permits. The canonical map need not be injective when zero divisors are present.

If $s \in R$, the basic case is

$$R[1/s] = \{1, s, s^2, \ldots\}^{-1}R.$$

Its elements are represented by a/s^n, and s becomes a unit with inverse $1/s$. Equivalently, $R[1/s]$ is universal among rings receiving a homomorphism from R in which the image of s is invertible. $\mathbb{Z}[1/2]$ is the first example. Inverting 6 also makes both 2 and 3 invertible, so one should not expect the only new units to be powers of the chosen element.

Fraction fields and localizations at primes are two further instances. If R is a domain, then

$$\mathrm{Frac}(R) \cong (R \setminus \{0\})^{-1}R;$$

in particular $\mathbb{Q} \cong (\mathbb{Z}\setminus\{0\})^{-1}\mathbb{Z}$. Thus the ordinary fraction-field construction inverts every nonzero element. At the other extreme, localization at a prime ideal will invert only the elements outside that prime.

Definition 4.16. A set $S \subseteq R$ is **multiplicatively closed** if $1 \in S$ and $s, t \in S$ imply $st \in S$. Define $(a, s) \sim (b, t)$ in $R \times S$ when $u(at - bs) = 0$ for some $u \in S$. The class of (a, s) is a/s, and $S^{-1}R$ has operations

$$\frac{a}{s} + \frac{b}{t} = \frac{at + bs}{st}, \qquad \frac{a}{s}\frac{b}{t} = \frac{ab}{st}.$$

The conditions in the definition are forced by fraction arithmetic: if s, t are allowed denominators, then $(a/s)(b/t) = ab/(st)$ requires st to be allowed as well; 1 is needed to represent each original element as $a/1$.

Proposition 4.17. The relation is an equivalence relation, and these operations are well-defined.

Proof. Reflexivity uses $1(as - as) = 0$, and symmetry changes the sign. If $u(at - bs) = 0$, $v(bq - ct) = 0$, then multiplying the first equality by vq, the second by us, and adding gives $uv(atq - cst) = 0$. This proves transitivity. If representatives are changed, multiplying the two witnessing equalities by the other numerator and denominator proves $ab/(st) = a'b'/(s't')$. For sums the needed cross-difference is $tt'(as' - a's) + ss'(bt' - b't)$, after multiplication by the two witnesses; it is zero. The remaining ring laws follow from common denominators. □

Example 4.18. In $R = \mathbb{Z}/6\mathbb{Z}$, take $S = \{1, 3\}$. Then $0/1 = 2/1$, since $3(0 - 2) = 0$, although $0 \neq 2$. The witness u is therefore essential in rings with zero divisors. It also shows directly that the canonical map $R \to S^{-1}R$ need not be injective.

Proposition 4.19. The canonical map $\iota : R \to S^{-1}R$, $a \mapsto a/1$, has kernel

$$\{a \in R : sa = 0 \text{ for some } s \in S\}.$$

It is injective exactly when every $s \in S$ is **regular**, meaning $sa = 0 \Rightarrow a = 0$. In the present commutative setting, this says precisely that S contains no zero divisors. Moreover, $s/1$ is a unit with inverse $1/s$.

Proof. $a/1 = 0/1$ is exactly the displayed condition. The fraction formulas prove the homomorphism statement and $(s/1)(1/s) = 1$. □

Theorem 4.20 (Universal Property of Localization)**.** If $f : R \to T$ maps each $s \in S$ to a unit, there is a unique $\widetilde{f} : S^{-1}R \to T$ such that $f = \widetilde{f} \circ \iota$.

$$\begin{tikzcd} R \arrow[r, "\iota"] \arrow[dr, "f"'] & S^{-1}R \arrow[d, "\widetilde{f}"] \\ & T \end{tikzcd}$$

Proof. Set $\widetilde{f}(a/s) = f(a)f(s)^{-1}$. A witnessing equality $u(at - bs) = 0$, after applying f and cancelling the unit $f(u)$, proves this is well-defined. The fraction formulas prove the homomorphism laws. Finally $a/s = (a/1)(s/1)^{-1}$, which forces uniqueness. □

Quotients force elements of I to be zero; localizations force elements of S to be units. These are complementary universal constructions. In particular, $R/(s)$ forces s to be zero, whereas $R[1/s]$ forces s to be invertible. For example, $\mathbb{Z}/(2) = \mathbb{F}_2$, while $\mathbb{Z}[1/2]$ still has characteristic zero.

Definition 4.21. A commutative ring is **local** if it has a unique maximal ideal.

Proposition 4.22. A commutative ring with identity is local if and only if its nonunits form an ideal.

Proof. If $\mathfrak{m}$ is the unique maximal ideal, every element outside it is a unit: otherwise its principal ideal is proper and, by the maximal-ideal theorem, lies in a maximal ideal, which must be $\mathfrak{m}$. Hence the nonunits are exactly $\mathfrak{m}$. Conversely, suppose that the nonunits form an ideal N. It is proper because 1 is a unit. Every proper ideal is contained in N, since an ideal containing a unit is all of R. Thus N itself is maximal, and every maximal ideal, being proper, equals N. This proves both existence and uniqueness. □

Remark. Under this convention the zero ring is not local: it has no maximal ideals.

For a prime ideal $\mathfrak{p}$, $1 \notin \mathfrak{p}$, and its complement $S = R \setminus \mathfrak{p}$ is multiplicatively closed. Indeed, if $a, b \notin \mathfrak{p}$ but $ab \in \mathfrak{p}$, primality would put a or b in $\mathfrak{p}$, a contradiction. Hence

$$R_{\mathfrak{p}} = (R \setminus \mathfrak{p})^{-1} R$$

is defined, and its elements are fractions a/s with $s \notin \mathfrak{p}$. It makes every element outside $\mathfrak{p}$ invertible, leaving the behavior attached to $\mathfrak{p}$ visible; informally, it focuses attention at $\mathfrak{p}$.

Theorem 4.23. For a prime ideal $\mathfrak{p}$, $R_{\mathfrak{p}}$ is local with unique maximal ideal

$$\mathfrak{p}R_{\mathfrak{p}} = \{a/s : a \in \mathfrak{p},\ s \notin \mathfrak{p}\}.$$

Proof. $R \setminus \mathfrak{p}$ is multiplicatively closed by primality. The displayed set is an ideal: numerators in $\mathfrak{p}$ remain there under addition and multiplication. It is proper, since an equality $1 = a/s$ with $a \in \mathfrak{p}$ would say $u(s - a) = 0$ for some $u \notin \mathfrak{p}$, contradicting primality. If $a \notin \mathfrak{p}$, then a/s has inverse s/a. If $a \in \mathfrak{p}$, it cannot be a unit by the same argument applied to $(a/s)(b/t) = 1$. Hence this ideal is exactly the nonunits; every proper ideal is contained in it. □

For $\mathbb{Z}_{(p)}$, the unique maximal ideal is $p\mathbb{Z}_{(p)}$. Concretely,

$$\mathbb{Z}_{(p)} = \{a/b \in \mathbb{Q} : p \nmid b\};$$

every integer not divisible by p is a unit, and its unique maximal ideal is formed by the fractions whose numerator is divisible by p.

S	localization	elements made invertible
$\{1, s, s^2, \ldots\}$	$R[1/s]$	one chosen element s
$R \setminus \mathfrak{p}$	$R_{\mathfrak{p}}$	everything outside $\mathfrak{p}$
$R \setminus \{0\}$ (R a domain)	$\mathrm{Frac}(R)$	every nonzero element

Exercises

Exercise 4.1 (Basic)**.** Determine the units and zero divisors of $\mathbb{Z}/10\mathbb{Z}$.

Exercise 4.2 (Core)**.** Prove that the preimage of an ideal under a ring homomorphism is an ideal. Use correspondence to describe ideals of $\mathbb{Z}/18\mathbb{Z}$.

Exercise 4.3 (Core)**.** Show (x) and $(x-1)$ are comaximal in $F[x]$, and describe $F[x]/(x(x-1))$.

Exercise 4.4 (Core)**.** Determine which of $(2), (3), (4), (6)$ are prime or maximal in $\mathbb{Z}/12\mathbb{Z}$.

Exercise 4.5 (Further)**.** Construct $S^{-1}\mathbb{Z} \cong \mathbb{Z}[1/2]$ for $S = \{1, 2, 2^2, \ldots\}$, and identify its units.

Exercise 4.6 (Further)**.** Prove $\mathbb{Z}_{(p)}/p\mathbb{Z}_{(p)} \cong \mathbb{F}_p$.

Chapter 5

Polynomial Rings and Factorization

Polynomials turn arithmetic in a ring into a tool for studying roots, maps, and later linear operators. Over a field, division of polynomials recovers the Euclidean algorithm; this leads from fields to PIDs and UFDs.

5.1 Polynomial Rings and Evaluation

For a commutative ring R, $R[x]$ consists of finite sums $a_0 + a_1x + \cdots + a_nx^n$, with coefficientwise addition and convolution multiplication. If R is a domain and $f, g \neq 0$, then $\deg(fg) = \deg f + \deg g$, because the product of leading coefficients is nonzero.

Proposition 5.1 (Degree Rule)**.** If R is a domain and $f, g \in R[x]$ are nonzero, then $\deg(fg) = \deg f + \deg g$. The units of $R[x]$ are exactly the constant polynomials whose value is a unit of R.

Proof. Writing $f = a_mx^m + \cdots$, $g = b_nx^n + \cdots$, the coefficient of x^{m+n} in fg is a_mb_n, which is nonzero precisely because a domain has no zero divisors. Thus a product can have degree zero only when both factors have degree zero, proving the assertion about units. $\square$

Proposition 5.2 (Evaluation)**.** For $a \in R$, evaluation $\mathrm{ev}_a : R[x] \to R$, $f \mapsto f(a)$, is a ring homomorphism. If R is a field, its kernel is $(x - a)$.

Proof. The first assertion follows by distributing finite sums. The division algorithm below writes $f = (x - a)q + r$, where r is constant; then $r = f(a)$, proving the kernel statement. $\square$

5.2 Division Algorithms and Roots

Theorem 5.3 (Division Algorithm)**.** If F is a field and $f, g \in F[x]$, $g \neq 0$, then there are unique $q, r \in F[x]$ with

$$f = qg + r, \qquad r = 0 \text{ or } \deg r < \deg g.$$

Proof. Write $g = b_n x^n + \cdots$, with $b_n \neq 0$. If the current dividend is $h = c_m x^m + \cdots$ with $m \geq n$, subtract $(c_m b_n^{-1}) x^{m-n} g$. This is possible because F is a field, and it cancels the x^m-term, leaving a polynomial of smaller degree. Iterating must stop, since nonnegative degrees cannot decrease forever; collecting the subtracted terms gives $f = qg + r$. If two such expressions exist, then $(q - q')g = r' - r$. Unless both sides are zero, the left side has degree at least $\deg g$, while the right side has smaller degree, a contradiction. □

Example 5.4 (Long Division)**.** Dividing $x^3 - 2x + 4$ by $x - 1$, first subtract $x^2(x-1)$, then $x(x-1)$, then $-1(x-1)$. Thus

$$x^3 - 2x + 4 = (x^2 + x - 1)(x - 1) + 3.$$

Corollary 5.5 (Factor Theorem)**.** For $f \in F[x]$, $f(a) = 0$ iff $x - a$ divides f.

Proof. Apply the division algorithm to f by $x - a$ and evaluate. □

Corollary 5.6. A nonzero polynomial of degree n over a field has at most n roots.

Proof. Each root supplies a distinct linear factor; induction on degree proves the assertion. □

5.3 Euclidean Domains and Principal Ideal Domains

Definition 5.7. A domain R is **Euclidean** if it has a function $\delta : R \setminus \{0\} \to \mathbb{N}$ such that division with remainder reduces δ. A **principal ideal domain** (PID) is a domain in which every ideal is principal.

Definition 5.8. For a, b in a domain, a **greatest common divisor** is an element d such that $d \mid a$, $d \mid b$, and every common divisor of a, b divides d. A gcd, when it exists, is unique only up to associates. In $\mathbb{Z}$ one chooses the positive representative, but a general domain need not have a preferred associate.

Theorem 5.9. Every Euclidean domain is a PID. In particular $F[x]$ is a PID for every field F.

Proof. Let $I \neq 0$ be an ideal and choose $d \in I$ with minimal $\delta(d)$. Divide any $a \in I$ by d: $a = qd + r$. Then $r \in I$, so minimality gives $r = 0$, hence $I = (d)$. Polynomial degree is a Euclidean function by the division algorithm. □

Proposition 5.10 (Bézout and gcd)**.** In a PID, $(a, b) = (d)$ if and only if d is a gcd of a, b, up to associates. Consequently every pair has a gcd and

$$d = ra + sb$$

for suitable r, s.

Proof. If $(a, b) = (d)$, then $a, b \in (d)$, so $d \mid a, b$. A common divisor c divides every linear combination of a, b, and hence divides $d \in (a, b)$. Thus d is a gcd. Conversely, write $(a, b) = (g)$, which is possible because R is a PID. The forward direction shows that g is a gcd. If d is another gcd, then $d \mid g$, since d is a common divisor, and $g \mid d$, since g is a common divisor. Thus d and g are associates, so $(d) = (g) = (a, b)$. Consequently $d \in (a, b)$, giving the displayed Bézout representation. This connects gcds, generated ideals, and Bézout identities. □

5.4 Unique Factorization Domains and Gauss's Lemma

The factorization of an integer into primes is a model for a property enjoyed by many domains. The central question here is whether the property survives the passage from a UFD R to $R[x]$. The answer is yes, and Gauss's lemma explains why coefficients do not introduce hidden factorizations.

Definition 5.11. Let R be a domain. Nonzero $a, b \in R$ are **associates** if $a = ub$ for a unit u. A nonzero nonunit p is **irreducible** if $p = ab$ implies that a or b is a unit. It is **prime** if $p \mid ab$ implies $p \mid a$ or $p \mid b$. A **UFD** is a domain where every nonzero nonunit has a factorization into irreducibles, unique up to ordering and replacement by associates.

Every prime is irreducible: if $p = ab$, then $p \mid a$ or $p \mid b$; cancellation makes the other factor a unit. The converse need not hold in an arbitrary domain, but it does hold in a PID.

Proposition 5.12 (Irreducibles are prime in a PID)**.** Every irreducible element of a principal ideal domain is prime.

Proof. Let p be irreducible, suppose $p \mid ab$, and assume $p \nmid a$. The gcd d of p, a cannot be associated to p, so it is a unit. Thus $(p, a) = R$, and Bézout gives $rp + sa = 1$. After multiplication by b, both terms on the left of $rpb + sab = b$ are divisible by p. Hence $p \mid b$. □

Theorem (Irreducible polynomials and field quotients)**.** Let F be a field and let $f \in F[x]$ be nonconstant. The following are equivalent:

f is irreducible, $\quad$ (f) is a maximal ideal of $F[x]$, $\quad$ $F[x]/(f)$ is a field.

Proof. Since $F[x]$ is a PID, every ideal containing (f) has the form (g), where $g \mid f$. If f is irreducible, then g is either a unit or an associate of f. Thus there is no ideal strictly between (f) and $F[x]$, so (f) is maximal. Conversely, if $f = gh$ with both g and h nonunits, then

$$(f) \subsetneq (g) \subsetneq F[x],$$

contradicting maximality. The equivalence between maximal ideals and field quotients is the quotient characterization of maximal ideals from Chapter 4. □

Irreducibility therefore does more than forbid a factorization: it is exactly what makes the quotient obtained by adjoining a formal root into a field.

Theorem 5.13 (Every PID is a UFD)**.** Every principal ideal domain is a unique factorization domain.

Proof Idea. Existence comes from the ascending-chain condition on ideals: an element that could not be factored would force a strictly increasing chain of principal ideals. Uniqueness follows by cancelling prime factors.

Proof. An ascending chain of ideals in a PID stabilizes, since its union is an ideal (d), and d belongs to one member of the chain. Suppose some nonzero nonunit does not factor into irreducibles, and choose a with (a) maximal among ideals generated by such elements. Then $a = bc$, with both b, c nonunits. Therefore $(a) \subsetneq (b)$ and $(a) \subsetneq (c)$, so maximality gives factorizations of b, c, a contradiction. Thus factorizations exist.

If $p_1 \cdots p_m = q_1 \cdots q_n$ are products of irreducibles, p_1 is prime by the proposition, hence divides some q_j. Since q_j is irreducible, they are associates. Reorder, absorb the unit, and cancel. Induction gives uniqueness. □

Let R now be a UFD. A gcd of coefficients is only determined up to a unit, precisely the ambiguity that factorization theory permits.

Gcds exist in every UFD, although they need not have Bézout representations. After choosing representatives of irreducible associate classes, write

$$a = u\prod p_i^{\alpha_i}, \qquad b = v\prod p_i^{\beta_i}.$$

Then, up to associates,

$$\gcd(a,b) \sim \prod p_i^{\min(\alpha_i,\beta_i)}.$$

The exponents show at once that this element divides both a and b, and that every common divisor divides it. In a PID, gcds are Bézout combinations; in a general UFD this is false. For example, $\gcd(x,y) = 1$ in $F[x,y]$, but $(x,y) \neq F[x,y]$: an identity $fx+gy = 1$ would become $0 = 1$ after evaluation at $x = y = 0$. Thus factorization-coprime and comaximal agree in a PID but not in every UFD.

Definition 5.14 (Content)**.** For $0 \neq f = a_0+\cdots+a_nx^n \in R[x]$, its **content** is

$$\operatorname{cont}(f) = \gcd(a_0,\ldots,a_n),$$

where the gcd is understood up to multiplication by a unit. The polynomial is **primitive** exactly when $\operatorname{cont}(f)$ is a unit, equivalently when no nonunit divides all of its coefficients.

The content is therefore simply the gcd of the coefficients, whose existence is supplied by the UFD property just discussed.

Every nonzero f has the form

$$f = \operatorname{cont}(f)f_0$$

up to multiplication by a unit, with f_0 primitive. For example,

$$6x^2 - 9x + 3 = 3(2x^2 - 3x + 1),$$

and the parenthesized factor is primitive in $\mathbb{Z}[x]$.

Lemma 5.15 (Primitive Product Lemma)**.** The product of two primitive polynomials in $R[x]$ is primitive.

Proof. If a nonunit divided every coefficient of fg, some irreducible factor p would do so. In a UFD, irreducibles are prime, so (p) is a prime ideal: $ab \in (p)$ means $p \mid ab$. Hence $R/(p)$ is a domain by Chapter 4. The reductions of f and g modulo p are nonzero, because f,g are primitive, but their product is zero. This contradicts the degree rule in the polynomial ring over a domain. $\square$

Corollary 5.16 (Content formula)**.** For nonzero $f, g \in R[x]$,

$$\mathrm{cont}(fg) \sim \mathrm{cont}(f)\,\mathrm{cont}(g),$$

where $\sim$ denotes association. Equality is naturally only up to associates in a general UFD. For $R = \mathbb{Z}$, normalizing content to be positive gives the literal equality $\mathrm{cont}(fg) = \mathrm{cont}(f)\,\mathrm{cont}(g)$.

Proof. Write $f = \mathrm{cont}(f)f_0$, $g = \mathrm{cont}(g)g_0$, where f_0, g_0 are primitive. Then $fg = \mathrm{cont}(f)\,\mathrm{cont}(g)f_0g_0$, and the lemma says the final factor has unit content. □

Let $K = \mathrm{Frac}(R)$. The next statement is the precise form of clearing denominators.

Lemma 5.17 (Gauss's factorization lemma)**.** If $f \in R[x]$ is primitive and $f = gh$ in $K[x]$, then there are primitive $g_0, h_0 \in R[x]$ and a unit $\varepsilon \in R$ such that

$$f = (\varepsilon g_0)h_0.$$

Thus a positive-degree factorization over K yields one over R.

Proof. Choose nonzero $a, b \in R$ such that $G = ag$ and $H = bh$ lie in $R[x]$. Write $G = \mathrm{cont}(G)g_0$, $H = \mathrm{cont}(H)h_0$ with primitive parts. Since $GH = abf$, the content formula gives

$$\mathrm{cont}(G)\,\mathrm{cont}(H) \sim \mathrm{cont}(GH) = \mathrm{cont}(abf) \sim ab.$$

Thus $\mathrm{cont}(G)\,\mathrm{cont}(H) = \varepsilon ab$ for a unit ε, and

$$abf = GH = \varepsilon ab g_0 h_0.$$

Cancellation in the domain $R[x]$ gives the result. Scaling does not change degree. □

Corollary 5.18 (Irreducibility transfer)**.** For primitive $f \in R[x]$,

$$f \text{ is irreducible in } R[x] \quad \Longleftrightarrow \quad f \text{ is irreducible in } K[x].$$

Proof. A factorization in $R[x]$ is one in $K[x]$. Conversely, a nontrivial factorization in $K[x]$ has two positive-degree factors; the lemma turns it into one in $R[x]$. □

Lemma 5.19. If primitive $g, h \in R[x]$ are associates in $K[x]$, then they are associates in $R[x]$.

Proof. Write $h = (a/b)g$ with $a, b \in R \setminus \{0\}$. Then $bh = ag$. Taking contents gives $b \sim a$, so a/b is a unit of R. $\square$

Theorem 5.20 (Polynomial UFD Theorem)**.** If R is a UFD, then $R[x]$ is a UFD. Hence $R[x_1, \ldots, x_n]$ is a UFD for every $n \geq 1$.

Proof. The field K has $K[x]$ a PID, hence a UFD. For $0 \neq f \in R[x]$, factor its content in R, then factor its primitive part in $K[x]$. The factorization lemma clears the denominators of the latter factors; the irreducibility-transfer corollary makes the resulting primitive factors irreducible in $R[x]$. Irreducible constants of R remain irreducible in $R[x]$, by the degree rule. This proves existence.

For uniqueness, compare two factorizations in the UFD $K[x]$. Its uniqueness pairs the primitive factors up to association, and the preceding lemma upgrades these associations to $R[x]$. Unique factorization of the remaining contents occurs in R. Repeated application of the one-variable result gives the multivariable assertion. $\square$

Corollary 5.21. $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$ are UFDs. A primitive polynomial in $\mathbb{Z}[x]$ is irreducible over $\mathbb{Z}$ exactly when it is irreducible over $\mathbb{Q}$.

Proof. Apply the theorem to the UFD $\mathbb{Z}$, whose fraction field is $\mathbb{Q}$, and use irreducibility transfer. $\square$

5.5 Irreducibility Criteria and Factorization over Fields

Proposition 5.22. A polynomial of degree two or three over a field is reducible if and only if it has a root in that field.

Proof. A root yields a linear factor by the Factor Theorem. Conversely, a proper factorization of degree two or three has a factor of degree one, which has a root. $\square$

Theorem 5.23 (Eisenstein's Criterion)**.** Let $f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$. If a prime p divides every a_i for $i < n$, does not divide a_n, and $p^2 \nmid a_0$, then f is irreducible over $\mathbb{Q}$.

Proof. By Gauss's Lemma it is enough to rule out a factorization in $\mathbb{Z}[x]$; the factors may be chosen primitive. Hence neither has all coefficients divisible by p, so their reductions modulo p are nonzero. Reducing a hypothetical $f = gh$ modulo p gives $\overline{g}\,\overline{h} = \overline{a_n} x^n$. Since the reduced factors are nonzero and $\mathbb{F}_p[x]$ is a domain, each is a nonzero scalar times a power of x. Thus every nonleading coefficient of g, h is divisible by p. Their constant terms are therefore both divisible by p, contradicting $p^2 \nmid a_0$. $\square$

Example 5.24. Eisenstein at $p = 2$ proves $x^n - 2$ irreducible over $\mathbb{Q}$ for every $n \geq 1$.

Exercises

Exercise 5.1 (Basic)**.** Divide $x^3 - 1$ by $x - 1$ and factor the result over $\mathbb{Q}$.

Exercise 5.2 (Core)**.** Use the Euclidean algorithm in $\mathbb{Q}[x]$ to find a gcd of $x^3 - 1$ and $x^2 - 1$, and express it as a Bézout combination.

Exercise 5.3 (Core)**.** Determine whether $x^3 - 2$ and $x^4 + 1$ are irreducible over $\mathbb{Q}$.

Exercise 5.4 (Core)**.** Find the content and primitive part of each of $12x^3 - 18x^2 + 6x$ and $15x^2 + 10x + 5$. Verify the content formula for their product.

Exercise 5.5 (Core)**.** Prove directly from the definition that a primitive polynomial in $\mathbb{Z}[x]$ cannot have all of its coefficients divisible by the same prime. Use reduction modulo 2 to show that $x^3 + x + 1$ is irreducible over $\mathbb{Q}$. **Hint:** use Gauss's Lemma; a nontrivial factorization of this primitive polynomial would reduce to one in $\mathbb{F}_2[x]$, where the cubic has no root.

Exercise 5.6 (Further)**.** Use Bézout to show that coprime polynomials in $F[x]$ have no common root in any extension field.

Exercise 5.7 (Further, Challenging)**.** Let R be a UFD and let $f \in R[x]$ be primitive. Show that if f divides g in $\mathrm{Frac}(R)[x]$ and $g \in R[x]$, then there is a primitive polynomial $h \in R[x]$ and $a \in R$ such that $g = ahf$ up to multiplication by a unit.

Part III

Modules and Linear Algebra

Chapter 6

Modules and Linear Algebra

Vector spaces combine addition with scalar multiplication by numbers from a field. Modules keep the same formal operations while allowing scalars from a ring. This modest-looking change includes abelian groups, ideals, quotient rings, and vector spaces in one language. It also explains why linear algebra is the first important special case of module theory: over a field one can divide by every nonzero scalar, whereas a general module can have torsion and need not possess coordinates.

Throughout R is a ring with identity, and a module means a unital left R-module unless a different convention is stated. For a noncommutative ring this is important: a right module has its scalar multiplication on the other side. From §6.8 onward, F denotes a field and V, W denote finite-dimensional F-vector spaces.

6.1 Modules and Submodules

The vector-space axioms make sense before one assumes that nonzero scalars are invertible. That observation gives the correct setting for relations such as $n\bar{a} = 0$ in $\mathbb{Z}/n\mathbb{Z}$, which cannot occur in a nonzero vector space.

Definition 6.1. An R**-module** is an abelian group $(M, +)$ equipped with a map $R \times M \to M$, $(r, m) \mapsto rm$, such that, for $r, s \in R$ and $m, n \in M$,

$$\begin{aligned} (r+s)m &= rm + sm, & r(m+n) &= rm + rn, \\ (rs)m &= r(sm), & 1m &= m. \end{aligned}$$

A subset $N \subseteq M$ is a **submodule**, written $N \leq M$, if $m - n \in N$ and $rm \in N$ whenever $m, n \in N$ and $r \in R$. An R**-module homomorphism** $f : M \to P$ preserves addition and scalar multiplication. Its **kernel** and **image** are $\ker f = \{m : f(m) = 0\}$ and $\operatorname{im} f = \{f(m) : m \in M\}$. An isomorphism is a bijective module homomorphism.

The principal examples should be kept in view. A vector space is a module over its scalar field; abelian groups are exactly $\mathbb{Z}$-modules; R and R^n are left R-modules; and every left ideal is a submodule of the regular module R. If R is commutative, every ideal has this role. The module $\mathbb{Z}/n\mathbb{Z}$ is not free over $\mathbb{Z}$, because its nonzero elements are killed by a nonzero scalar.

There is a useful one-line test hidden in the definition. A nonempty subset N of a module is a submodule precisely when it is closed under subtraction and under multiplication by every scalar. The subtraction test contains both additive closure and additive inverses, just as it did for subgroups. For instance, $2\mathbb{Z} \leq \mathbb{Z}$ as a $\mathbb{Z}$-module, while the set of odd integers is not a submodule: the difference of two odd integers is even. Such examples are elementary, but they train the reader to distinguish closure conditions from merely familiar-looking subsets.

Proposition 6.2. For a module homomorphism $f : M \to P$, both $\ker f$ and $\operatorname{im} f$ are submodules.

Proof. If $m, n \in \ker f$, then $f(m - n) = 0$, and $f(rm) = rf(m) = 0$. Thus the kernel is a submodule. If $f(m), f(n) \in \operatorname{im} f$, then $f(m)-f(n) = f(m-n)$ and $rf(m) = f(rm)$, so the image is one as well. $\square$

6.2 Quotient Modules and the Isomorphism Theorems

Submodules are precisely the pieces that can be collapsed to zero without destroying scalar multiplication. Quotient modules therefore generalize the quotient groups and quotient rings developed earlier.

Definition 6.3. If $N \leq M$, the **quotient module** M/N is the additive quotient group with scalar multiplication $r(m + N) = rm + N$.

Proof. If $m + N = m' + N$, then $m - m' \in N$, whence $r(m - m') \in N$. Therefore $rm+N = rm'+N$, so the operation is well-defined. Each module axiom follows from the corresponding axiom in M. $\square$

Theorem 6.4 (Module isomorphism theorems)**.** Let $f : M \to P$ be linear.

1. $M/\ker f \cong \operatorname{im} f$.
2. If $A, B \leq M$, then $(A + B)/B \cong A/(A \cap B)$.
3. If $N \leq L \leq M$, then $(M/N)/(L/N) \cong M/L$.
4. Submodules of M/N correspond bijectively to submodules of M containing N, under inverse image and quotient by N.

Proof. For (1), $m + \ker f \mapsto f(m)$ is well-defined, onto the image, and has zero kernel. For (2), restrict $M \to M/B$ to A: its kernel is $A \cap B$, and its image is $(A+B)/B$. For (3), map $(m+N)+(L/N)$ to $m+L$, checking directly that the map is well-defined and has zero kernel. Finally, if $\pi : M \to M/N$ is the quotient map, then $Q \mapsto \pi^{-1}(Q)$ and $K \mapsto K/N$ are inverse operations. □

Theorem 6.5 (Universal property of a quotient)**.** If $f : M \to P$ is linear and $N \subseteq \ker f$, there is a unique linear map $\bar{f} : M/N \to P$ with $f = \bar{f}\pi$, where π is the quotient map.

$$\begin{tikzcd} M \arrow[r, "\pi"] \arrow[dr, "f"'] & M/N \arrow[d, "\bar{f}"] \\ & P \end{tikzcd}$$

Proof. Set $\bar{f}(m+N) = f(m)$. If $m - m' \in N$, then $f(m-m') = 0$, so this does not depend on the representative. It is linear, and surjectivity of π forces uniqueness. □

Thus the group, ring, and module isomorphism theorems are not unrelated coincidences: they express one recurring quotient pattern.

Example 6.6 (A quotient module)**.** The quotient $\mathbb{Z}/6\mathbb{Z}$ is a $\mathbb{Z}$-module, and its submodule generated by $\bar{2}$ is $\{\bar{0}, \bar{2}, \bar{4}\}$. Collapsing this submodule gives a quotient with two elements,

$$(\mathbb{Z}/6\mathbb{Z})/\langle\bar{2}\rangle \cong \mathbb{Z}/2\mathbb{Z}.$$

The example is the module version of reducing a group by a normal subgroup; the point of the module language is that multiplication by every integer also descends automatically.

6.3 Direct Sums and Products

Products collect arbitrary coordinate choices; direct sums collect finite linear combinations of contributions from separate modules. The latter is the algebraic construction needed for bases and for decompositions.

Definition 6.7. For a family $(M_i)_{i\in I}$, the product

$$\prod_{i\in I} M_i$$

consists of all tuples (m_i), with coordinatewise operations. The direct sum

$$\bigoplus_{i\in I} M_i$$

is its submodule consisting of tuples with **finite support**: only finitely many m_i are nonzero. For a finite family, product and direct sum have the same elements and are both denoted

$$M_1 \oplus \cdots \oplus M_n.$$

There are canonical injections and projections

$$\iota_i : M_i \longrightarrow \bigoplus_{i\in I} M_i, \qquad \pi_i : \prod_{i\in I} M_i \longrightarrow M_i.$$

A map

$$X \longrightarrow \prod_{i\in I} M_i$$

is equivalent to a family of maps $X \to M_i$, obtained by composing with the projections. Conversely, a family $M_i \to Y$ determines a unique map

$$\bigoplus_{i\in I} M_i \longrightarrow Y,$$

because every element is a finite sum of injected components. For example, $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z} \cong \mathbb{Z}/6\mathbb{Z}$.

For an infinite family the finite-support requirement matters. The element

$$(1, 1, 1, \ldots) \in \prod_{n\geq 1} \mathbb{Z}$$

does not belong to

$$\bigoplus_{n\geq 1} \mathbb{Z}.$$

A map out of the direct sum can be defined by adding only finitely many contributions, whereas there is no reason an infinite sum should make sense in an arbitrary module. This is why direct sums, rather than products, are the natural home for bases.

6.4 Cyclic Modules, Annihilators, Torsion, and Exact Sequences

The simplest module is generated by one element. Understanding such modules is useful because a generator has exactly one kind of obstruction: some scalars may kill it. The annihilator records those scalar relations and turns cyclic modules into quotients of the ring itself.

Definition 6.8. An R-module is **cyclic** if $M = Rm$ for some $m \in M$. The **annihilator** of m is

$$\operatorname{Ann}_R(m) = \{r \in R : rm = 0\};$$

the annihilator of M consists of scalars that kill every element of M. If R is a domain, an element m is **torsion** if $rm = 0$ for some nonzero $r \in R$. A module is torsion-free if its only torsion element is zero.

For a left module, an annihilator is a left ideal; in the commutative rings used later it is an ideal in the unqualified sense. Over $\mathbb{Z}$, torsion means finite additive order. Thus $\mathbb{Z}^n$ is torsion-free, whereas every element of $\mathbb{Z}/n\mathbb{Z}$ is torsion.

When R is a domain, the set $T(M)$ of torsion elements is a submodule, called the **torsion submodule**. Indeed, if $am = 0$ and $bn = 0$ with $a, b \neq 0$, then $ab(m-n) = 0$; and $a(rm) = 0$ for every $r \in R$. The domain hypothesis ensures $ab \neq 0$. This construction will make the free part of a finitely generated PID module intrinsic in Chapter 7.

Theorem 6.9 (Cyclic-module classification)**.** For $m \in M$, there is an isomorphism

$$Rm \cong R/\operatorname{Ann}_R(m).$$

Consequently every cyclic module over a commutative ring is isomorphic to R/I for an ideal I.

Proof. The map $R \to Rm$, $r \mapsto rm$, is a surjective module homomorphism, and its kernel is exactly $\operatorname{Ann}_R(m)$. The first isomorphism theorem supplies the stated isomorphism. Conversely, R/I is generated by $1 + I$. □

Exact sequences provide a compact language for saying that a module is built from a submodule and a quotient. They will later organize module decompositions and further constructions.

Definition 6.10. A sequence $A \xrightarrow{u} B \xrightarrow{v} C$ is **exact at** B if $\operatorname{im} u = \ker v$. A **short exact sequence**

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{q} C \longrightarrow 0$$

is exact everywhere; equivalently, i is injective, q is surjective, and $\operatorname{im} i = \ker q$.

Definition (Cokernel)**.** For a module homomorphism $f : M \to N$, its **cokernel** is

$$\operatorname{coker}(f) = N / \operatorname{im} f.$$

The canonical map $N \to \operatorname{coker}(f)$ is the quotient map.

Kernels measure what a map kills; cokernels measure what remains in its target after the image is collapsed. Concretely, if $g : N \to P$ satisfies $gf = 0$, then g vanishes on $\operatorname{im} f$, so there is a unique map $\widetilde{g} : \operatorname{coker}(f) \to P$ with

$$\begin{tikzcd} N \arrow[r] \arrow[dr, "g"'] & \operatorname{coker}(f) \arrow[d, "\widetilde{g}"] \\ & P. \end{tikzcd}$$

The quotient sequence $0 \to N \to M \to M/N \to 0$ is the basic example. It is useful to distinguish equality of short exact sequences from the weaker fact that their three terms happen to be isomorphic.

Definition 6.11. An **isomorphism of short exact sequences** is a commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r, "i"] \arrow[d, "\alpha", "\sim"'] & B \arrow[r, "q"] \arrow[d, "\beta", "\sim"'] & C \arrow[r] \arrow[d, "\gamma", "\sim"'] & 0 \\ 0 \arrow[r] & A' \arrow[r, "i'"] & B' \arrow[r, "q'"] & C' \arrow[r] & 0 \end{tikzcd}$$

in which the three vertical maps are module isomorphisms.

Proposition 6.12 (The canonical split sequence)**.** For modules A and C, the sequence

$$0 \longrightarrow A \xrightarrow{a \mapsto (a,0)} A \oplus C \xrightarrow{(a,c) \mapsto c} C \longrightarrow 0$$

is short exact.

Proof. The first map is injective and the second is surjective. Its kernel consists precisely of the pairs $(a, 0)$, which are exactly the image of the first map. Thus exactness says, quite literally, that $A \oplus C$ contains A as its first coordinate and has quotient C. $\square$

6.5 Splitting and Diagram Chases

The preceding observation asks when a short exact sequence contains no extension information beyond its two ends. A map $s : C \to B$ with $qs = 1_C$ is a **section**; a map $r : B \to A$ with $ri = 1_A$ is a **retraction**.

Theorem 6.13 (Splitting Lemma)**.** For a short exact sequence

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{q} C \longrightarrow 0,$$

the following are equivalent: q has a section; i has a retraction; and the given short exact sequence is isomorphic, as a short exact sequence, to the canonical split sequence of Proposition 6.12. Concretely, the end maps are 1_A and 1_C, and the middle isomorphism $\Phi : A \oplus C \to B$ satisfies

$$\Phi(a, 0) = i(a), \qquad q\Phi(a, c) = c.$$

Proof. A section gives the map

$$\Phi : A \oplus C \longrightarrow B, \qquad \Phi(a, c) = i(a) + s(c).$$

It is onto: for $b \in B$, the element $b - sq(b)$ lies in $\ker q$, hence is $i(a)$ for some a. If $\Phi(a, c) = 0$, applying q gives $c = 0$, and then injectivity of i gives $a = 0$. Thus Φ is an isomorphism, and it gives an isomorphism to the canonical sequence in Proposition 6.12. The map $r = \mathrm{pr}_A \Phi^{-1}$ is a retraction of i.

Conversely, suppose that $ri = 1_A$. For $c \in C$, choose $b \in B$ with $q(b) = c$, and put

$$s(c) = b - i(r(b)).$$

If $b' = b + i(a)$ is another lift, then

$$b' - i(r(b')) = b + i(a) - i(r(b) + a) = b - i(r(b)),$$

so the formula is independent of the lift. It is linear, and $q(s(c)) = c$. Hence it is a section. □

Commutative diagrams organize a **diagram chase**: start with an element, use exactness to replace a kernel element by a preimage, and use commutativity to transport the resulting equality vertically. The Five Lemma is the basic instance.

Theorem 6.14 (Five Lemma)**.** In a commutative diagram with exact rows

$$\begin{tikzcd} A_1 \arrow[r] \arrow[d, "f_1"] & A_2 \arrow[r] \arrow[d, "f_2"] & A_3 \arrow[r] \arrow[d, "f_3"] & A_4 \arrow[r] \arrow[d, "f_4"] & A_5 \arrow[d, "f_5"] \\ B_1 \arrow[r] & B_2 \arrow[r] & B_3 \arrow[r] & B_4 \arrow[r] & B_5 \end{tikzcd}$$

if f_1 is surjective, f_2, f_4 are isomorphisms, and f_5 is injective, then f_3 is an isomorphism.

Proof. Write the horizontal maps as $d_i : A_i \to A_{i+1}$ and $e_i : B_i \to B_{i+1}$. Suppose first that $f_3(x) = 0$. Then $f_4 d_3(x) = e_3 f_3(x) = 0$. Since f_4 is injective, $d_3(x) = 0$, so $x = d_2(y)$. As

$$0 = f_3 d_2(y) = e_2 f_2(y),$$

there is $z \in B_1$ with $e_1(z) = f_2(y)$. Choose $w \in A_1$ with $f_1(w) = z$. Then

$$f_2(y - d_1(w)) = 0.$$

Injectivity of f_2 gives $y = d_1(w)$, hence $x = d_2 d_1(w) = 0$. This proves the **injectivity** of f_3.

Surjectivity. Now take $b \in B_3$. Since $e_3(b) \in B_4$, choose $a_4 \in A_4$ with $f_4(a_4) = e_3(b)$. Exactness gives $d_4(a_4) = 0$, because $f_5 d_4(a_4) = e_4 f_4(a_4) = 0$ and f_5 is injective. Choose $a_3 \in A_3$ with $d_3(a_3) = a_4$. Then

$$e_3(b - f_3(a_3)) = 0,$$

so $b - f_3(a_3) = e_2(b_2)$ for some $b_2 \in B_2$. Choose $a_2 \in A_2$ with $f_2(a_2) = b_2$. Replacing a_3 by $a_3 + d_2(a_2)$ gives $f_3(a_3) = b$. This proves the **surjectivity** of f_3.

Isomorphism. A module homomorphism is an isomorphism precisely when it is both injective and surjective. The preceding two parts prove the claim. ☐

Theorem 6.15 (Snake Lemma)**.** In a commutative diagram with exact rows

$$\begin{tikzcd} 0 \arrow[r] & A' \arrow[r, "i'"] \arrow[d, "f_A"] & B' \arrow[r, "q'"] \arrow[d, "f_B"] & C' \arrow[r] \arrow[d, "f_C"] & 0 \\ 0 \arrow[r] & A \arrow[r, "i"] & B \arrow[r, "q"] & C \arrow[r] & 0, \end{tikzcd}$$

there is an exact sequence

$$\begin{aligned} 0 \longrightarrow \ker f_A \longrightarrow \ker f_B \longrightarrow \ker f_C \xrightarrow{\delta} \operatorname{coker} f_A \\ \longrightarrow \operatorname{coker} f_B \longrightarrow \operatorname{coker} f_C \longrightarrow 0. \end{aligned}$$

Proof. For $c' \in \ker f_C$, choose $b' \in B'$ with $q'(b') = c'$. Then

$$q(f_B(b')) = f_C(q'(b')) = 0,$$

so $f_B(b') = i(a)$ for a unique $a \in A$. Define

$$\delta(c') = a + \operatorname{im} f_A.$$

If b' is replaced by $b' + i'(a')$, then a is replaced by $a + f_A(a')$; hence δ is well-defined. Choosing lifts of two elements and adding them also proves that δ is linear.

The maps before and after δ are the maps induced by the vertical arrows. The first one is injective because i' is injective. Exactness at the first two kernels follows immediately by chasing the two squares and using injectivity of i. If $c' = q'(b')$ comes from $\ker f_B$, then $f_B(b') = 0$, so $\delta(c') = 0$. Conversely, $\delta(c') = 0$ means that $a = f_A(a')$; then $b' - i'(a')$ lies in $\ker f_B$ and still maps to c'. This proves exactness at $\ker f_C$.

If $a + \operatorname{im} f_A$ maps to zero in $\operatorname{coker} f_B$, then $i(a) = f_B(b')$ for some b'. Hence $f_C(q'(b')) = 0$, and the construction gives $\delta(q'(b')) = a + \operatorname{im} f_A$. This is exactness at $\operatorname{coker} f_A$. Finally, exactness at $\operatorname{coker} f_B$ and $\operatorname{coker} f_C$ follows by the same two-square chase: a class $b + \operatorname{im} f_B$ maps to zero precisely when $q(b) = f_C(c')$, and a lift of c' corrects b to an element of $i(A)$. The last induced map is surjective because q is surjective. This also proves the asserted endpoint exactness. □

The construction of δ is the model diagram chase: choose a lift, measure the obstruction in the next kernel, and record it modulo the ambiguity forced by the first vertical map. It is natural with respect to maps of such diagrams, a fact used later in homological algebra.

6.6 Free Modules and Module Bases

Coordinates are useful only when expansions are unique. Free modules record that condition over a general ring; vector spaces are the especially well behaved free modules over fields.

Definition 6.16. A subset $S \subseteq M$ **spans** M if every element is a finite linear combination of members of S. It is **linearly independent** if every relation

$$\sum_i r_i s_i = 0,$$

with distinct $s_i \in S$, has all coefficients zero. A **basis** is an independent spanning set, and a module with a basis is **free**. The free module on a set X is the module $R^{(X)}$ of finitely supported functions $X \to R$, with basis $\{e_x : x \in X\}$.

Theorem 6.17 (Universal property of a free module)**.** For every function $u : X \to M$, there is a unique linear map $R^{(X)} \to M$ taking e_x to $u(x)$.

Proof. Every element of $R^{(X)}$ has a unique finite expression

$$\sum_{x \in X} r_x e_x.$$

Send it to

$$\sum_{x \in X} r_x u(x).$$

This is linear, and any linear map with the required values has this formula. □

Not every module is free: $\mathbb{Z}/n\mathbb{Z}$ has no $\mathbb{Z}$-basis for $n > 1$. Indeed, if $\bar{a}$ belonged to a $\mathbb{Z}$-basis, then $n\bar{a} = 0$ would be a nontrivial relation. This is the first warning that the familiar basis theory of vector spaces cannot simply be imported into module theory. In particular, a submodule of a free module need not be free over an arbitrary ring, and the apparent number of generators of R^n requires justification before it is called a rank.

6.7 Projective and Injective Modules

Free modules make lifting possible because a map out of a free module is controlled by its values on a basis. Projective modules are precisely the modules that retain this lifting property, even when they need not themselves have a basis.

Definition 6.18. A module P is **projective** if, for every surjection $\pi : M \twoheadrightarrow N$ and every map $f : P \to N$, there is a map $\widetilde{f} : P \to M$ such that $\pi\widetilde{f} = f$. A module E is **injective** if, for every injection $j : A \hookrightarrow B$ and every map $f : A \to E$, there is a map $\widetilde{f} : B \to E$ such that $\widetilde{f}j = f$.

$$\begin{tikzcd}
 & P \arrow[dl, dashed, "\widetilde{f}"'] \arrow[d, "f"] & \\
M \arrow[r, two heads, "\pi"'] & N \arrow[r] & 0
\end{tikzcd}
\qquad
\begin{tikzcd}
 & E & \\
0 \arrow[r] & A \arrow[u, "f"] \arrow[r, hook, "j"'] & B \arrow[ul, dashed, "\widetilde{f}"']
\end{tikzcd}$$

These matching triangles display the dual geometry of the two definitions: projectivity is a lifting problem, while injectivity is an extension problem.

Definition 6.19. For left R-modules M, N, write

$$\mathrm{Hom}_R(M, N)$$

for the set of R-linear maps $M \to N$. It is an abelian group under pointwise addition:

$$(f+g)(m) = f(m) + g(m).$$

If R is commutative, it is also an R-module by

$$(rf)(m) = rf(m).$$

For a noncommutative ring the displayed scalar formula need not define a left-module map, since $rf(sm)$ need not equal $srf(m)$. Accordingly, the exact sequences below are always sequences of abelian groups, and only under commutativity are they automatically sequences of R-modules.

Proposition 6.20 (Hom detects exactness)**.** Suppose

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{q} C \longrightarrow 0$$

satisfies $qi = 0$, without yet assuming exactness. Define

$$\begin{aligned} i_* &: \mathrm{Hom}_R(M, A) \longrightarrow \mathrm{Hom}_R(M, B), \quad f \longmapsto i \circ f, \\ q_* &: \mathrm{Hom}_R(M, B) \longrightarrow \mathrm{Hom}_R(M, C), \quad g \longmapsto q \circ g, \end{aligned}$$

$$\begin{aligned} q^* &: \mathrm{Hom}_R(C, N) \longrightarrow \mathrm{Hom}_R(B, N), \quad h \longmapsto h \circ q, \\ i^* &: \mathrm{Hom}_R(B, N) \longrightarrow \mathrm{Hom}_R(A, N), \quad k \longmapsto k \circ i. \end{aligned}$$

Then

$$0 \longrightarrow \mathrm{Hom}_R(M, A) \xrightarrow{i_*} \mathrm{Hom}_R(M, B) \xrightarrow{q_*} \mathrm{Hom}_R(M, C)$$

is exact for every R-module M if and only if

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{q} C$$

is exact. Dually,

$$0 \longrightarrow \mathrm{Hom}_R(C, N) \xrightarrow{q^*} \mathrm{Hom}_R(B, N) \xrightarrow{i^*} \mathrm{Hom}_R(A, N)$$

is exact for every R-module N if and only if

$$A \xrightarrow{i} B \xrightarrow{q} C \longrightarrow 0$$

is exact. Thus the original sequence is short exact precisely when both displayed Hom tests are exact for all M and N.

Proof. Assume first that $0 \to A \to B \to C$ is exact. A map $M \to B$ has zero composite with q exactly when its image lies in $\ker q = \operatorname{im} i$, and it then factors uniquely through A; this proves exactness of the first Hom sequence. Conversely, suppose this Hom sequence is exact for every M. With $K = \ker i$, the inclusion $K \to A$ has zero image under i_*, hence is zero; so i is injective. With $K = \ker q$, the inclusion $K \to B$ has zero image under q_*, hence factors through i. Thus $\ker q \subseteq \operatorname{im} i$, and the reverse inclusion follows from $qi = 0$.

Now assume that $A \to B \to C \to 0$ is exact. A map $B \to N$ vanishes on $\operatorname{im} i = \ker q$ exactly when it factors uniquely through C, proving exactness of the second Hom sequence. Conversely, suppose that sequence is exact for every N. Taking

$$N = C/\operatorname{im} q$$

and the quotient map $C \to N$, injectivity of q^* forces that quotient map to be zero; hence q is surjective. Taking

$$N = B/\operatorname{im} i$$

and the quotient map $B \to N$, exactness gives a factorization through q. It vanishes on $\ker q$, so $\ker q \subseteq \operatorname{im} i$; the opposite inclusion again follows from $qi = 0$. ☐

For a short exact sequence, both Hom sequences are exact at their initial terms. This familiar property is often called left exactness of Hom. Exactness through the final Hom term is strictly stronger: with $M = P$ it is the lifting property defining projectivity, and with $N = E$ it is the extension property defining injectivity.

Theorem 6.21 (Projective-module characterizations)**.** For an R-module P, the following are equivalent:

1. P is projective;
2. every short exact sequence ending in P splits;
3. P is isomorphic to a direct summand of a free module;
4. for every short exact sequence, the first sequence in Proposition 6.20, with $M = P$, is exact at its final term.

In particular, every free module is projective.

Proof. If P is projective and

$$0 \longrightarrow A \longrightarrow B \xrightarrow{q} P \longrightarrow 0$$

is exact, apply the lifting property to $1_P : P \to P$. The resulting section of q splits the sequence by the Splitting Lemma.

Suppose every such sequence splits. Choose a generating set X of P. The universal property of $R^{(X)}$ gives a surjection $R^{(X)} \twoheadrightarrow P$. Its kernel produces a short exact sequence ending in P, so it splits. Thus

$$R^{(X)} \cong \ker(R^{(X)} \to P) \oplus P,$$

and P is a direct summand of a free module.

Finally, let P be a direct summand of a free module F, with inclusion $u : P \to F$ and retraction $r : F \to P$. Given $\pi : M \twoheadrightarrow N$ and $f : P \to N$, choose a basis of F. For each basis vector e_x, choose a lift in M of $f(r(e_x))$. The universal property of F gives $g : F \to M$ with $\pi g = fr$. Then $\widetilde{f} = gu$ satisfies $\pi\widetilde{f} = f$. Thus P is projective. Taking $P = F$ proves the final assertion. Finally, Proposition 6.20 says exactly that the final Hom map is surjective precisely when every map $P \to C$ lifts through B. This proves the equivalence of (1) and (4), and distinguishes full exactness here from the left exactness enjoyed by every test module. □

Example. For $n > 1$, the $\mathbb{Z}$-module $\mathbb{Z}/n\mathbb{Z}$ is not projective. If

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0$$

split, then $\mathbb{Z}$ would contain a subgroup isomorphic to $\mathbb{Z}/n\mathbb{Z}$. This is impossible because $\mathbb{Z}$ has no nonzero torsion.

Theorem 6.22 (Injective-module characterization)**.** An R-module E is injective if and only if every short exact sequence

$$0 \longrightarrow E \longrightarrow B \longrightarrow C \longrightarrow 0$$

splits. Equivalently, for every short exact sequence, the second sequence in Proposition 6.20, with $N = E$, is exact at its final term.

Proof. If E is injective, apply its extension property to the identity map of E and the first injection in the sequence. The resulting retraction splits the sequence. Conversely, let $j : A \hookrightarrow B$ and $f : A \to E$ be given. Form

$$S = \{(f(a), -j(a)) : a \in A\} \subseteq E \oplus B, \qquad Q = (E \oplus B)/S.$$

The set S is a submodule: it is the image of the linear map $A \to E \oplus B$, $a \mapsto (f(a), -j(a))$. The map $E \to Q$, $e \mapsto (e, 0) + S$, is injective: if

$$(e, 0) = (f(a), -j(a)) \in S,$$

then $j(a) = 0$, so $a = 0$ and $e = 0$. The map

$$Q \longrightarrow B/j(A), \qquad (e, b) + S \longmapsto b + j(A)$$

is well-defined, since adding $(f(a), -j(a))$ changes b by an element of $j(A)$. Its kernel is the image of E: if $b = j(a)$, then

$$(e, b) + S = (e + f(a), 0) + S,$$

and the reverse inclusion is immediate. It is onto, so the First Isomorphism Theorem identifies the quotient of Q by the image of E with $B/j(A)$. By hypothesis the resulting short exact sequence splits, giving a retraction $Q \to E$. Composing $B \to Q$, $b \mapsto (0, b)+S$, with that retraction gives an extension of f. Hence E is injective. The final Hom characterization follows from Proposition 6.20: surjectivity of its final map is exactly the assertion that maps $A \to E$ extend to B. $\square$

Proposition 6.23 (Enough projectives)**.** Every R-module is a quotient of a free, hence projective, R-module.

Proof. Let X be the underlying set of M. The free module

$$R^{(X)}$$

has basis $\{e_x : x \in X\}$. The rule $e_x \mapsto x$ extends uniquely to a surjective map $R^{(X)} \to M$. The final assertion follows from Theorem 6.21. $\square$

Theorem 6.24 (Baer's criterion)**.** An R-module E is injective if and only if every R-linear map

$$f : J \longrightarrow E$$

from a left ideal J of R extends to a map $R \to E$.

Proof. The forward implication applies injectivity to $J \hookrightarrow R$. Conversely, given $A \subseteq B$ and $f : A \to E$, order extensions

$$(C, g), \qquad A \subseteq C \subseteq B, \qquad g|_A = f,$$

by extension. Unions of chains give extensions, so Zorn's Lemma gives a maximal pair (C, g). If $b \notin C$, then

$$J = \{r \in R : rb \in C\}$$

is a left ideal. The map $r \mapsto g(rb)$ extends to $r \mapsto re$ on R. Hence

$$g'(c + rb) = g(c) + re$$

is well-defined: if $c + rb = c' + r'b$, then

$$(r - r')b = c' - c \in C,$$

so $r - r' \in J$. The chosen extension agrees with $s \mapsto g(sb)$ on J, hence

$$(r - r')e = g((r - r')b) = g(c' - c).$$

Rearranging gives $g(c) + re = g(c') + r'e$. Thus g' extends g to $C + Rb$, contradicting maximality. Therefore $C = B$. □

Remark 6.25. The standard converse proof of Baer's criterion uses Zorn's Lemma; its forward implication does not. Over Noetherian rings, left ideals are finitely generated, so many applications reduce ideal-level arguments to finite-generation arguments. The precise foundational strength of the full theorem is not pursued here. Looking ahead, every module embeds in an injective module; standard proofs use additional constructions, often injective abelian groups such as $\mathbb{Q}/\mathbb{Z}$ and suitable Hom modules.

Proposition 6.26 (Injective abelian groups)**.** An abelian group is injective as a $\mathbb{Z}$-module if and only if it is **divisible**: multiplication by every nonzero integer is surjective.

Proof. If E is injective, extend

$$n\mathbb{Z} \longrightarrow E, \qquad nk \longmapsto ka$$

along $n\mathbb{Z} \hookrightarrow \mathbb{Z}$; the image of 1 is an n-th root of a. Conversely, every ideal of $\mathbb{Z}$ is $n\mathbb{Z}$, and divisibility extends any map from it by sending 1 to an n-th root of the image of n. Apply Baer's criterion. □

Example 6.27. $\mathbb{Q}$, $\mathbb{Q}/\mathbb{Z}$, and

$$\mathbb{Q}^r$$

are divisible, hence injective $\mathbb{Z}$-modules; $\mathbb{Z}$ is not.

Proposition 6.28 (Enough injectives over $\mathbb{Z}$)**.** Every abelian group embeds in a divisible abelian group. Equivalently, every $\mathbb{Z}$-module embeds in an injective $\mathbb{Z}$-module.

Proof. The group $\mathbb{Q}/\mathbb{Z}$ is divisible, hence injective by Proposition 6.26. For every nonzero $a \in A$, there is a homomorphism

$$\chi_a : A \longrightarrow \mathbb{Q}/\mathbb{Z}$$

with $\chi_a(a) \neq 0$. Indeed, first define such a map on the cyclic subgroup generated by a: send a generator of a finite cyclic group of order n to $1/n + \mathbb{Z}$, and send an infinite cyclic generator to $1/2 + \mathbb{Z}$. Injectivity of $\mathbb{Q}/\mathbb{Z}$ extends this map to A.

Now form the evaluation map

$$A \longrightarrow \prod_{\chi \in \mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Q}/\mathbb{Z})} \mathbb{Q}/\mathbb{Z}, \qquad a \longmapsto \big(\chi(a)\big)_\chi .$$

It is injective because the coordinate χ_a detects every nonzero a. The product is divisible coordinatewise: an n-th root is chosen in each coordinate. Proposition 6.26 then makes it injective as a $\mathbb{Z}$-module. ☐

Remark 6.29. The implication injective $\Rightarrow$ divisible is elementary. The converse, and hence the equivalence used above, is established here through Baer's criterion and its Zorn-lemma argument. Thus the presentation relies on the usual choice principles.

6.8 Vector Spaces and Bases

We now specialize to the case in which the scalar ring is a field F. An F-vector space is simply an F-module, so the notions of span, linear independence, and basis from Sections 6.1 and 6.4 apply without change. The additional strength of vector-space theory comes from the fact that every nonzero scalar is invertible.

Definition 6.30. An **F-vector space** is an F-module, and a **subspace** is an F-submodule.

Theorem 6.31 (Basis existence)**.** Every vector space has a basis.

Proof. Let $\mathcal{P}$ be the set of linearly independent subsets of V, ordered by inclusion. It is nonempty because it contains the empty set. If $\mathcal{C}$ is a chain in $\mathcal{P}$, then

$$U = \bigcup_{S \in \mathcal{C}} S$$

is linearly independent: a finite relation among distinct members of U involves only finitely many members of the chain, hence all of them belong to one member of $\mathcal{C}$, where their coefficients vanish. Thus every chain has an upper bound in $\mathcal{P}$. Zorn's lemma gives a maximal linearly independent set B.

If $v \notin \operatorname{span}(B)$, then $B \cup \{v\}$ is linearly independent: in a relation involving v, a nonzero coefficient of v would solve for v as a linear combination of members of B, while a zero coefficient leaves a relation in B. This contradicts maximality. Hence B spans V, and therefore is a basis. □

Remark 6.32. It applies also to infinite-dimensional spaces. Its proof uses Zorn's lemma and hence the usual Choice principles. The finite replacement argument below is elementary and uses no such principle.

Theorem 6.33 (Replacement theorem)**.** Let

$$G = \{w_1, \ldots, w_n\}$$

span V, and let

$$L = \{v_1, \ldots, v_m\}$$

be linearly independent. Then $m \leq n$, and exactly $n - m$ vectors from G can be chosen so that, together with $v_1, \ldots, v_m$, they span V. Equivalently, one may replace m vectors of the original spanning list by $v_1, \ldots, v_m$ and retain a spanning list of size n.

Proof. We argue by induction on m. For $m = 0$, the original list spans. Assume the result for $m-1$. The induction hypothesis gives a spanning list

$$\{v_1, \ldots, v_{m-1}, w_{i_1}, \ldots, w_{i_{n-m+1}}\}.$$

If $n-m+1 = 0$, then $v_1, \ldots, v_{m-1}$ already span V, so v_m lies in their span, contradicting independence. Hence

$$n - m + 1 \geq 1,$$

and in particular $m \leq n$. We may therefore write

$$v_m = a_1 v_1 + \cdots + a_{m-1} v_{m-1} + b_1 w_{i_1} + \cdots + b_{n-m+1} w_{i_{n-m+1}}.$$

Not every b_j is zero, for otherwise v_m belongs to the span of $v_1, \ldots, v_{m-1}$. Choose $b_k \neq 0$. Since F is a field,

$$w_{i_k} = b_k^{-1}\left(v_m - \sum_{\ell=1}^{m-1} a_\ell v_\ell - \sum_{j \neq k} b_j w_{i_j}\right).$$

Replacing w_{i_k} by v_m therefore preserves spanning. □

Corollary 6.34 (Extension and extraction)**.** If V has a finite spanning set, every finite linearly independent subset extends to a basis, and every finite spanning set contains a basis.

Proof. Apply the replacement theorem to an independent list and a fixed finite spanning list. Its retained members, together with the independent list, form a finite spanning list; discard a retained member whenever it lies in the span of the others. Conversely, applying the same deletion process to the original spanning list leaves an independent spanning list. $\square$

Corollary 6.35 (Uniqueness of finite basis size)**.** Any two finite bases of a vector space have the same cardinality.

Proof. Apply the replacement theorem to either basis as an independent set and the other as a spanning set. The two resulting inequalities are opposite. $\square$

Definition 6.36. A vector space is **finite-dimensional** if it has a finite basis. Its **dimension**, denoted by $\dim_F V$, is the common cardinality of its finite bases. When the scalar field is clear, we write

$$\dim V := \dim_F V.$$

Thus a basis expansion exists because a basis spans and is unique because a basis is independent. An **ordered basis** is a basis displayed as an ordered finite list; this extra order becomes essential when coordinates are introduced in Section 6.11.

Proposition 6.37 (Subspaces of finite-dimensional spaces)**.** If V is finite-dimensional and $W \leq V$, then W has a finite basis and $\dim W \leq \dim V$.

Proof. Begin with the empty independent list in W, and adjoin an element of W not in its span whenever one exists. The replacement theorem, applied to the resulting independent list and a fixed basis of V, shows that at most $\dim V$ choices can be made. When the process stops its list spans W, so it is a basis. The same theorem gives the inequality. $\square$

Corollary 6.38 (IBN for fields)**.** If

$$F^m \cong F^n$$

as F-vector spaces, then $m = n$.

Proof. An isomorphism carries the standard basis of F^m to a basis of F^n. Uniqueness of finite basis size gives $m = n$. $\square$

6.9 Invariant Basis Number and Finite Free Rank

The preceding corollary proves IBN for fields. A finite free module over a general ring needs a separate argument. This distinction is essential in Chapter 7, where the free part of a module over a PID must have a uniquely determined rank.

Definition 6.39. A ring R has **invariant basis number** (IBN) if $R^m \cong R^n$ for finite m, n implies $m = n$. When R has IBN, the **rank** of a finite free R-module is the integer n for which it is isomorphic to R^n.

Theorem 6.40 (Commutative rings have IBN)**.** Every nonzero commutative ring with identity has invariant basis number.

Proof. Suppose $R^m \cong R^n$. Since R is nonzero, (0) is proper; by the maximal-ideal theorem, choose a maximal ideal $\mathfrak{m}$ of R. An isomorphism $f : R^m \to R^n$ carries $\mathfrak{m}R^m$ onto $\mathfrak{m}R^n$, since $f(rx) = rf(x)$, and the same assertion for f^{-1} proves equality. It consequently induces an isomorphism

$$R^m/\mathfrak{m}R^m \;\cong\; R^n/\mathfrak{m}R^n.$$

Coordinatewise reduction identifies these quotient modules with $(R/\mathfrak{m})^m$ and $(R/\mathfrak{m})^n$. The quotient is a field because $\mathfrak{m}$ is maximal, so IBN for fields gives $m = n$. ☐

The nonzero hypothesis is essential: for the zero ring,

$$R^m = R^n = 0$$

for all finite m, n, so invariant basis number fails. Arbitrary noncommutative rings need not have IBN, so no unqualified rank is attached to their finite free modules.

6.10 Linear Transformations and Dimension

The module homomorphisms of vector spaces are the familiar linear transformations. Finite dimension turns the abstract kernel–image picture into a numerical conservation law.

Definition 6.41. A **linear transformation** $T : V \to W$ is an F-module homomorphism: $T(av + bw) = aT(v) + bT(w)$. Its kernel and image are the submodules already defined in §6.1, now called respectively its **kernel** and **range**. It is injective, surjective, bijective, or an isomorphism in the same sense as a module map. An endomorphism is a map $V \to V$, and an automorphism is an invertible endomorphism. The endomorphisms form the F-vector space and ring $\mathcal{L}_F(V)$ of linear operators $V \to V$; when the field is clear, we write

$$\mathcal{L}(V) := \mathcal{L}_F(V).$$

For finite-dimensional V, define

$$\operatorname{nullity} T = \dim \ker T, \qquad \operatorname{rank} T = \dim \operatorname{im} T.$$

The kernel and image are subspaces by Proposition 6.1, and the preceding subspace theorem supplies their finite bases. Thus the numerical words in the definition are legitimate before rank–nullity is invoked.

Theorem 6.42 (Rank–nullity)**.** If $T : V \to W$ is linear and V is finite-dimensional, then

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

Proof. Choose a basis $k_1, \ldots, k_a$ of $\ker T$. Extend it to the basis

$$k_1, \ldots, k_a, v_1, \ldots, v_b$$

of V. The vectors $Tv_1, \ldots, Tv_b$ span the image: applying T removes the k_i-part of any coordinate expression. If

$$\sum_i c_i T v_i = 0,$$

then

$$\sum_i c_i v_i \in \ker T,$$

and uniqueness of the displayed basis expansion gives every $c_i = 0$. Hence they form a basis of the image, so $\dim V = a + b$. □

Thus a linear map between equal finite dimensions is injective exactly when it is surjective, and hence exactly when it is an isomorphism: rank–nullity shows that one of kernel and image has the necessary dimension precisely when the other does.

6.11 Matrices and Change of Basis

Once ordered bases are chosen, a linear transformation becomes a table of coordinates. The matrix is a useful representative, but it is not the transformation itself: changing coordinates changes its matrix.

Let $\beta = (v_1, \ldots, v_n)$ be an ordered basis. Every $v \in V$ has one and only one expansion $v = x_1 v_1 + \cdots + x_n v_n$; define its **coordinate column** by

$$[v]_\beta = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

The order matters: exchanging two basis vectors generally exchanges two coordinates.

Definition 6.43. Let R be a ring. An $m \times n$ **matrix over** R is a rectangular array

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \qquad a_{ij} \in R.$$

The set of all such matrices is $M_{m\times n}(R)$, and

$$M_n(R) = M_{n\times n}(R).$$

Matrices are equal precisely when their corresponding entries are equal. For matrices of the same size, define

$$A + B = (a_{ij} + b_{ij}).$$

If R is commutative, scalar multiplication is defined entrywise by

$$rA = (ra_{ij}),$$

making $M_{m\times n}(R)$ an R-module.

Matrices are arrays of scalars; they encode linear maps only after ordered bases have been chosen. If

$$A = (a_{ij}) \in M_{m\times n}(R),$$

its **transpose** is the matrix

$$A^t = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix} \in M_{n\times m}(R), \qquad (A^t)_{ij} = a_{ji}.$$

Thus transpose visibly interchanges rows and columns.

For $A = (a_{ij}) \in M_{m\times n}(F)$ and a column

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

define matrix–vector multiplication by

$$(Ax)_i = \sum_{j=1}^{n} a_{ij} x_j.$$

For instance,

$$\begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ 2 \end{pmatrix} = \begin{pmatrix} 8 \\ 23 \end{pmatrix}.$$

For

$$A = (a_{ij}) \in M_{m\times n}(R), \qquad B = (b_{jk}) \in M_{n\times r}(R),$$

define

$$AB = (c_{ik}) \in M_{m\times r}(R), \qquad c_{ik} = \sum_{j=1}^{n} a_{ij} b_{jk}.$$

In expanded form,

$$AB = \begin{pmatrix} \sum_{k=1}^{n} a_{1k} b_{k1} & \cdots & \sum_{k=1}^{n} a_{1k} b_{kr} \\ \vdots & \ddots & \vdots \\ \sum_{k=1}^{n} a_{mk} b_{k1} & \cdots & \sum_{k=1}^{n} a_{mk} b_{kr} \end{pmatrix}.$$

Thus the (i, j)-entry comes from row i of A and column j of B: multiply corresponding entries and add. This explains why the number of columns of A must equal the number of rows of B, and why the dimension rule is

$$(m \times n)(n \times r) \longrightarrow (m \times r).$$

For example,

$$\begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 11 & 4 \end{pmatrix}.$$

Associativity follows by distributing the finite sums; no commutativity of R is required. The **identity matrix**

$$I_n = (\delta_{ij}) \in M_n(R)$$

has entries 1 on the diagonal and 0 elsewhere. A square matrix A is **invertible** if some square matrix B satisfies $AB = BA = I_n$; then B is unique and is denoted by A^{-1}.

Proposition 6.44 (Transpose identities)**.** For compatible matrices over a commutative ring,

$$(A^t)^t = A, \qquad (A+B)^t = A^t + B^t, \qquad (rA)^t = rA^t, \qquad (AB)^t = B^t A^t.$$

Proof. The first three identities follow by comparing entries. For the product,

$$\left((AB)^t\right)_{ij} = (AB)_{ji} = \sum_k a_{jk} b_{ki} = \sum_k b_{ki} a_{jk} = (B^t A^t)_{ij}.$$

Commutativity is used in the third equality. □

Now let $T : V \to W$, let

$$\beta = (v_1, \ldots, v_n)$$

be an ordered basis of V, and let

$$\gamma = (w_1, \ldots, w_m)$$

be an ordered basis of W. Write

$$\begin{aligned} T(v_1) &= a_{11} w_1 + a_{21} w_2 + \cdots + a_{m1} w_m, \\ T(v_2) &= a_{12} w_1 + a_{22} w_2 + \cdots + a_{m2} w_m, \\ &\vdots \\ T(v_n) &= a_{1n} w_1 + a_{2n} w_2 + \cdots + a_{mn} w_m. \end{aligned}$$

The **matrix of T from β to γ** is

$$[T]_\beta^\gamma = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

Equivalently,

$$\boxed{\text{the } j\text{-th column of } [T]_\beta^\gamma \text{ is } [T(v_j)]_\gamma.}$$

If $T : V \to V$ and the same ordered basis β is used on both sides, write

$$[T]_\beta := [T]_\beta^\beta.$$

This shorthand is used below for determinants, eigenvalues, diagonalization, and canonical forms, and later for inner-product operators. Thus the matrix depends on the chosen ordered bases, though the linear map does not. If

$$v = x_1 v_1 + \cdots + x_n v_n,$$

then linearity and the displayed expansions give

$$T(v) = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} x_j \right) w_i.$$

The inner sum is the i-th entry of $[T]_\beta^\gamma [v]_\beta$, and hence

$$[T(v)]_\gamma = [T]_\beta^\gamma [v]_\beta. \tag{6.1}$$

For $S : W \to X$, with an ordered basis δ of X, this immediately explains the order of matrix multiplication:

$$[S \circ T]_\beta^\delta = [S]_\gamma^\delta [T]_\beta^\gamma.$$

Formula (6.1) also shows that a square matrix representing an endomorphism is invertible exactly when that endomorphism is an automorphism.

Example 6.45 (Constructing a representing matrix)**.** Let

$$T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2, \qquad T(x, y) = (x + y, x - y),$$

and choose the ordered bases

$$\beta = ((1, 1), (1, 0)), \qquad \gamma = ((1, 0), (1, 1)).$$

Then

$$T(1, 1) = (2, 0) = 2(1, 0) + 0(1, 1), \qquad T(1, 0) = (1, 1) = 0(1, 0) + 1(1, 1).$$

The coordinate columns therefore give

$$[T]_\beta^\gamma = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}.$$

For

$$v = 2(1,1) - (1,0) = (1,2),$$

we have

$$[v]_\beta = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \qquad [T(v)]_\gamma = \begin{pmatrix} 4 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

If β' is another ordered basis of V, define the change-of-coordinate matrix, with its direction displayed, by $P = [I_V]_{\beta'}^{\beta}$. Formula (6.1) for the identity says

$$[v]_\beta = P[v]_{\beta'};$$

thus P converts β'-coordinates into β-coordinates. The reverse change-of-coordinate matrix $Q = [I_V]_{\beta}^{\beta'}$ satisfies $PQ = QP = I$, so $P^{-1} = Q$. For an endomorphism T, applying the composition formula to the two identity maps gives

$$[T]_{\beta'} = P^{-1}[T]_\beta P.$$

Matrices related in this way are **similar**; they represent one operator in different ordered bases.

Finally, the matrices E_{ij}, having a single 1 in position (i,j), form a basis of $M_{m\times n}(F)$. Hence this matrix space has dimension mn. For fixed ordered bases, $T \mapsto [T]_\beta^\gamma$ is a linear bijection

$$\mathrm{Hom}_F(V,W) \longrightarrow M_{m\times n}(F)$$

by the column construction above; this identification depends on the chosen bases. In particular, $\dim_F \mathcal{L}_F(V) = (\dim_F V)^2$.

Example 6.46 (A change of basis)**.** Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be given by $T(x,y) = (3x+y, y)$. In the standard basis its matrix is $A = \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$. Put $v_1 = (1,0)$, $v_2 = (1,-2)$, and let $\beta = (v_1, v_2)$. The change-of-coordinate matrix has these vectors as columns,

$$P = \begin{pmatrix} 1 & 1 \\ 0 & -2 \end{pmatrix}, \qquad P^{-1} = \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & -\frac{1}{2} \end{pmatrix}.$$

Therefore

$$[T]_\beta = P^{-1}AP = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

The diagonal entries were not produced by a new operator: the chosen basis consists of eigenvectors. This computation is the concrete meaning of similarity and anticipates diagonalization.

6.12 Determinants

An endomorphism of an n-dimensional space either preserves oriented n-dimensional volume up to a scalar or collapses some direction. The determinant is that scalar. Its most useful definition is therefore structural, not computational. Permutations and their signs, used below, were proved in Chapter 2.

Definition 6.47. Let V be an n-dimensional vector space over F, with ordered basis $\beta = (v_1, \ldots, v_n)$. A map $\omega : V^n \to F$ is **n-multilinear** if it is linear in each argument separately, and **alternating** if it vanishes whenever two arguments agree. The **determinant form associated with** β is an alternating n-multilinear map ω_β satisfying

$$\omega_\beta(v_1, \ldots, v_n) = 1.$$

Multilinearity models how volume changes when one edge is changed; alternation models the collapse caused by two equal directions; normalization fixes the unit oriented parallelepiped determined by the chosen basis.

Lemma 6.48 (Alternating sign-change lemma)**.** Let $\omega : V^n \to F$ be alternating and multilinear. Interchanging any two arguments changes its value by a sign. More precisely, if every argument other than those in two fixed positions is held fixed, then

$$\omega(\ldots, x, \ldots, y, \ldots) = -\omega(\ldots, y, \ldots, x, \ldots).$$

Proof. Alternation gives

$$0 = \omega(\ldots, x + y, \ldots, x + y, \ldots).$$

Expanding the two displayed positions by multilinearity gives four terms. The terms with two copies of x or two copies of y vanish by alternation. Thus

$$0 = \omega(\ldots, x, \ldots, y, \ldots) + \omega(\ldots, y, \ldots, x, \ldots),$$

which is the asserted identity. □

Consequently, each transposition of arguments contributes a factor of -1. Chapter 2 proves that the parity of a permutation is well-defined, so a permutation $\sigma \in S_n$ changes the value of ω by the factor $\operatorname{sgn}(\sigma)$. The lemma supplies the effect on an alternating multilinear map; the earlier permutation theory supplies the rule by which the signs of successive transpositions combine.

Theorem 6.49 (Existence and uniqueness of the determinant form)**.** There is exactly one determinant form ω_β.

Proof. We first prove uniqueness. Write

$$x_j = \sum_{i=1}^{n} a_{ij} v_i \qquad (1 \le j \le n).$$

Repeated multilinearity forces

$$\omega(x_1, \dots, x_n) = \sum_{i_1, \dots, i_n} a_{i_1,1} \cdots a_{i_n,n}\, \omega(v_{i_1}, \dots, v_{i_n}).$$

Alternation kills every term with repeated indices. The surviving tuples are exactly

$$(i_1, \dots, i_n) = (\sigma(1), \dots, \sigma(n)), \qquad \sigma \in S_n.$$

By Lemma 6.48, each adjacent transposition of arguments contributes a factor of -1. The Chapter 2 parity theory then shows that the term indexed by σ has the factor $\operatorname{sgn}(\sigma)$, and normalization gives

$$\omega_\beta(x_1, \dots, x_n) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} \cdots a_{\sigma(n),n}. \tag{6.2}$$

Thus there can be at most one such map.

Conversely, define the right side of (6.2) to be $\omega_\beta(x_1, \dots, x_n)$. Each coefficient in one column occurs linearly, so the expression is multilinear in the corresponding argument. If $x_r = x_s$, pair a summand indexed by σ with the summand obtained by exchanging $\sigma(r)$ and $\sigma(s)$. The products agree because columns r and s agree, while the signs are opposite; every summand therefore cancels. Finally, for $(x_1, \dots, x_n) = (v_1, \dots, v_n)$, only the identity permutation contributes, and the value is 1. Hence the candidate is a determinant form. $\square$

Formula (6.2), the **Leibniz formula**, is consequently not a separate definition: it is the forced coordinate expression for the unique normalized alternating multilinear form.

Definition 6.50. For $A \in M_n(F)$, with columns $A_1, \dots, A_n$, define

$$\det(A) = \omega_e(A_1, \dots, A_n),$$

where e is the standard ordered basis of F^n. Thus

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} \cdots a_{\sigma(n),n}.$$

This is a column convention. The row convention follows below from transpose invariance.

Definition 6.51. For $T \in \mathcal{L}_F(V)$, define $\det(T)$ by

$$\omega_\beta(Tx_1, \dots, Tx_n) = \det(T)\,\omega_\beta(x_1, \dots, x_n) \qquad (x_1, \dots, x_n \in V).$$

The left side is alternating and multilinear in the x_i. The expansion used in the proof of Theorem 6.49 shows that every such form is its value on β times ω_β, so the displayed scalar exists and is unique; evaluating at β shows that it equals $\omega_\beta(Tv_1, \dots, Tv_n)$. The identity itself holds for every tuple, so evaluating in another ordered basis proves that $\det(T)$ is independent of β. Expanding the columns Tv_j in β gives

$$\det(T) = \det([T]_\beta). \tag{6.3}$$

This separates the intrinsic operator from its coordinate matrix.

Theorem 6.52 (Basic determinant identities)**.** For endomorphisms S, T of a finite-dimensional vector space over a field,

$$\det(I) = 1, \qquad \det(S \circ T) = \det(S)\det(T).$$

For square matrices over F,

$$\det(AB) = \det(A)\det(B), \qquad \det(A^t) = \det(A).$$

Proof. The defining identity for determinants gives

$$\omega_\beta(STx_1, \dots, STx_n) = \det(S)\det(T)\omega_\beta(x_1, \dots, x_n),$$

which proves multiplicativity and $\det(I) = 1$. Matrix multiplicativity follows from (6.3) and the composition formula proved in §6.11. Replacing σ by σ^{-1} in (6.2) gives $\det(A^t) = \det(A)$. ☐

Thus adding a multiple of one column to another leaves the determinant unchanged, scaling a column scales the determinant by the same scalar, and interchanging two columns changes its sign; these are immediate consequences of multilinearity and alternation. Transpose invariance supplies the corresponding row operations.

Definition 6.53. For $A = (a_{ij}) \in M_n(F)$, let A_{ij} be the matrix obtained by deleting row i and column j. Its **cofactor** is

$$C_{ij} = (-1)^{i+j} \det(A_{ij}).$$

Theorem 6.54 (Laplace expansion and adjugate)**.** For every row i and column j,

$$\det A = \sum_{k=1}^{n} a_{ik} C_{ik}, \qquad \det A = \sum_{k=1}^{n} a_{kj} C_{kj}.$$

If $\operatorname{adj}(A) = (C_{ji})$, then

$$A \operatorname{adj}(A) = \operatorname{adj}(A) A = (\det A) I_n. \tag{6.4}$$

Proof. Group the terms of the Leibniz formula according to the column selected in row i. Moving that selected entry to the first position requires $i+j$ parity changes, giving the first expansion; transpose invariance gives the column version. The (i, j)-entry of $A \operatorname{adj}(A)$ is the row-i cofactor expansion if $i = j$. If $i \neq j$, it is the same expansion after replacing row j by row i, which is zero by alternation. The column calculation proves the identity on the other side. □

Corollary 6.55 (Invertibility criterion over a field)**.** For $A \in M_n(F)$, equivalently for an endomorphism of V,

$$A \text{ is invertible} \quad \Longleftrightarrow \quad \det(A) \neq 0.$$

Proof. If A is invertible, determinant multiplicativity gives

$$1 = \det(A) \det(A^{-1}).$$

Conversely, if $\det(A) \neq 0$, then $\det(A)^{-1} \operatorname{adj}(A)$ is an inverse by (6.4). □

Example 6.56. For a 2×2 matrix,

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Thus $\det \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix} = 0$, while

$$\det \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = 1, \qquad \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Laplace expansion and the adjugate are computational consequences of the same structural determinant form, not competing definitions.

6.12.1 Determinants over commutative rings

The Leibniz expression uses only addition, multiplication, commutativity, and the elements 1 and -1. It therefore has a useful scope beyond fields.

Theorem 6.57 (Determinants over commutative rings)**.** Let R be a commutative ring with identity. For $A \in M_n(R)$, define

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} \cdots a_{\sigma(n),n}.$$

On R^n, this is the unique normalized alternating n-multilinear form with respect to the standard ordered basis. The identities

$$\det(I) = 1, \qquad \det(AB) = \det(A)\det(B), \qquad \det(A^t) = \det(A)$$

and the Laplace and adjugate identities (6.4) remain valid.

Proof. The uniqueness, cancellation, and multilinearity arguments in Theorem 6.49 use no division, only commutativity and the sign of a permutation. The structural proof of multiplicativity and the cofactor proof use the same operations, and hence carry over word for word. □

If P is an invertible change-of-basis matrix of a finite free R-module F, then

$$\det(P)\det(P^{-1}) = 1.$$

Consequently, for an endomorphism $T : F \to F$, the value

$$\det(T) = \det([T]_\beta)$$

is independent of the ordered basis β: two matrices are related by $P^{-1}[T]_\beta P$, and determinant multiplicativity cancels the two unit determinants. Thus the preceding field construction extends exactly to finite free modules, not to arbitrary finitely generated modules.

Corollary 6.58 (Invertibility over a commutative ring)**.** For $A \in M_n(R)$,

$$A \text{ is invertible} \quad \Longleftrightarrow \quad \det(A) \in R^\times.$$

Proof. An inverse of A makes $\det(A)\det(A^{-1}) = 1$. Conversely, if $\det(A)$ is a unit, $(\det A)^{-1} \operatorname{adj}(A)$ is an inverse by (6.4). □

Over a field, nonzero and unit are equivalent. Over $\mathbb{Z}$, they are not: the 1×1 matrix (2) has nonzero determinant but is not invertible. In particular, the field criterion involving linear dependence must not be transferred unqualified to general commutative rings. For $A \in M_n(R)$, the polynomial $\chi_A(t) = \det(tI - A) \in R[t]$ is nevertheless defined, and the adjugate identity over $R[t]$ gives $\chi_A(A) = 0$. Chapter 7 uses this argument in its operator-theoretic field setting.

6.13 Eigenvalues, Eigenvectors, and Invariant Subspaces

An eigenvector is a direction whose direction is not changed by an operator: the operator merely scales it. Such directions are the first indication that a complicated transformation might have a simple coordinate system.

Definition 6.59. For $T : V \to V$, a nonzero v is an **eigenvector** with **eigenvalue** $\lambda \in F$ if $Tv = \lambda v$. The **eigenspace** is $E_\lambda = \ker(T - \lambda I)$. A subspace $W \leq V$ is T**-invariant** if $T(W) \subseteq W$.

Because eigenspaces are kernels, they are subspaces before any dimension is attached to them. Now let β be an ordered basis and $A = [T]_\beta$. Define the **characteristic polynomial** by

$$\chi_T(x) = \det(xI - A) \in F[x].$$

This definition is independent of β. Indeed, if $A' = P^{-1}AP$, then, in the polynomial ring $F[x]$,

$$xI - A' = P^{-1}(xI - A)P.$$

Determinant multiplicativity, already proved above, gives $\det(xI - A') = \det(P^{-1})\det(xI - A)\det(P) = \det(xI - A)$.

Proposition 6.60. For $\lambda \in F$, λ is an eigenvalue of T if and only if $\chi_T(\lambda) = 0$.

Proof. The equation $Tv = \lambda v$ for a nonzero v says that $T - \lambda I$ is not injective. In finite dimension it is therefore not invertible. Its matrix is $A - \lambda I$, so the determinant criterion above says precisely that $\det(A - \lambda I) = 0$, equivalently $\det(\lambda I - A) = 0$. This is $\chi_T(\lambda) = 0$. The reverse chain of equivalences proves the converse. $\square$

The eigenspace is a subspace because it is a kernel. It can have dimension larger than one: for the identity operator, the unique eigenspace is all of V. By contrast, a rotation of the real plane through a nonzero angle less than π has no real eigenvector at all. The field therefore matters: the same rotation has complex eigenvalues, a point returned to in Chapter 7.

6.14 Diagonalization

The best possible matrix of an operator is diagonal: each coordinate evolves independently. The question is therefore not whether a matrix happens to look diagonal, but whether the underlying operator has a basis made of eigenvectors.

Theorem 6.61 (Diagonalization criterion)**.** For an endomorphism $T : V \to V$, the following are equivalent:

1. T is represented by a diagonal matrix in some basis;
2. V has a basis of eigenvectors of T;
3. $$V = \bigoplus_{\lambda} E_{\lambda},$$
   where the sum ranges over the eigenvalues that occur.

Proof. The columns of a diagonal matrix say exactly that its basis vectors are eigenvectors, proving (1) iff (2). Eigenvectors belonging to distinct eigenvalues are independent. For if a shortest nontrivial relation

$$\sum_{i=1}^{r} c_i v_i = 0$$

involved distinct eigenvalues λ_i, then applying $T - \lambda_r I$ gives a shorter relation

$$\sum_{i<r} c_i(\lambda_i - \lambda_r) v_i = 0;$$

its coefficients are nonzero whenever the original ones are, a contradiction. Thus the sum of the eigenspaces is direct. A basis of eigenvectors spans precisely this direct sum, proving (2) iff (3). □

In particular, if $\dim V = n$ and T has n distinct eigenvalues, then it is diagonalizable. Diagonalization is powerful but not universal. The next chapter explains the additional structure that remains when no eigenbasis exists.

Example 6.62 (Three first tests for diagonalization)**.** On $\mathbb{R}^2$, the operator with matrix $\begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$ has distinct eigenvalues 3 and 1, so the preceding worked calculation diagonalizes it. The identity operator has the repeated eigenvalue 1, yet is diagonalizable because every nonzero vector is an eigenvector. In contrast,

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

has characteristic polynomial $(x-1)^2$, but $\ker(J-I)$ is one-dimensional. It has only one independent eigenvector and therefore is not diagonalizable. Thus a split characteristic polynomial does not by itself supply an eigenbasis; the missing information is precisely what the module structure in Chapter 7 records.

6.15 Dual Spaces and Dual Maps

Coordinates record a vector after a basis has been chosen. There is another useful way to probe a vector: apply every scalar-valued linear measurement to it. This viewpoint is intrinsic, and it is the starting point for bilinear forms and alternating forms.

Definition 6.63. A **linear functional** on an F-vector space V is a linear map $\varphi : V \to F$. The **algebraic dual space** of V is

$$V^\vee = \mathrm{Hom}_F(V, F).$$

The notation $V^\vee$ is deliberately different from the notation T^* used for the inner-product adjoint in Chapter 8. The construction below is algebraic and requires no inner product.

Proposition 6.64 (Dual basis)**.** Let $\beta = (v_1, \dots, v_n)$ be an ordered basis of a finite-dimensional vector space V. There are unique linear functionals

$$v_i^\vee : V \longrightarrow F$$

such that

$$v_i^\vee(v_j) = \delta_{ij}.$$

Moreover,

$$\beta^\vee = (v_1^\vee, \dots, v_n^\vee)$$

is a basis of $V^\vee$. In particular,

$$\dim_F V^\vee = \dim_F V.$$

Proof. For

$$v = \sum_{j=1}^{n} a_j v_j,$$

define $v_i^\vee(v) = a_i$. The uniqueness of basis coordinates proves that this is well-defined and linear, and gives the displayed values. If

$$\sum_{i=1}^{n} c_i v_i^\vee = 0,$$

evaluation at v_j gives $c_j = 0$, so the functionals are independent. For $\varphi \in V^\vee$, linearity gives

$$\varphi = \sum_{i=1}^{n} \varphi(v_i) v_i^\vee.$$

Thus they span and form a basis. □

Consequently V and $V^\vee$ are isomorphic when V is finite-dimensional, but this proof used the chosen basis. In general there is no preferred isomorphism

$$V \cong V^\vee.$$

An inner product can supply one, but that is additional structure.

Definition 6.65. For a linear map

$$T : V \longrightarrow W,$$

its **dual map**, also called its algebraic transpose, is

$$T^\vee : W^\vee \longrightarrow V^\vee, \qquad T^\vee(\varphi) = \varphi \circ T.$$

The direction reverses because a measurement on W can be pulled back along T to a measurement on V.

Proposition 6.66 (Functorial identities and transpose matrix)**.** For linear maps $T : V \to W$ and $S : W \to U$,

$$(\mathrm{id}_V)^\vee = \mathrm{id}_{V^\vee}, \qquad (S \circ T)^\vee = T^\vee \circ S^\vee.$$

If β and γ are ordered bases of V and W, then

$$[T^\vee]_{\gamma^\vee}^{\beta^\vee} = \left([T]_\beta^\gamma\right)^t.$$

Proof. For $\varphi \in V^\vee$,

$$(\mathrm{id}_V)^\vee(\varphi) = \varphi \circ \mathrm{id}_V = \varphi.$$

Likewise, for $\psi \in U^\vee$,

$$(S \circ T)^\vee(\psi) = \psi \circ S \circ T = T^\vee(S^\vee(\psi)).$$

Write

$$T(v_j) = \sum_{i=1}^{m} a_{ij} w_i, \qquad [T]_\beta^\gamma = (a_{ij}).$$

For the i-th dual basis vector of $W^\vee$,

$$T^\vee(w_i^\vee)(v_j) = w_i^\vee(T(v_j)) = a_{ij}.$$

Hence

$$T^\vee(w_i^\vee) = \sum_{j=1}^{n} a_{ij} v_j^\vee.$$

Its matrix therefore has (j,i)-entry a_{ij}, which is the transpose of the matrix of T. $\square$

Definition 6.67. The **evaluation map** is

$$\iota_V : V \longrightarrow V^{\vee\vee}, \qquad \iota_V(v)(\varphi) = \varphi(v).$$

Proposition 6.68 (Double dual)**.** The evaluation map ι_V is linear and injective. If V is finite-dimensional, it is an isomorphism.

Proof. For $a, b \in F$, $v, w \in V$, and $\varphi \in V^\vee$,

$$\iota_V(av + bw)(\varphi) = \varphi(av + bw) = a\iota_V(v)(\varphi) + b\iota_V(w)(\varphi),$$

so ι_V is linear. If $v \neq 0$, extend v to a basis and take the corresponding first coordinate functional. It does not vanish on v, so $\iota_V(v) \neq 0$. Thus ι_V is injective. In finite dimension, Proposition 6.64 gives

$$\dim V^{\vee\vee} = \dim V,$$

so an injective map between these finite-dimensional spaces is surjective. $\square$

Thus finite-dimensional vector spaces have a canonical isomorphism

$$V \cong V^{\vee\vee},$$

even though an isomorphism with the first dual is usually noncanonical.

Definition 6.69. For a subspace $W \leq V$, its **annihilator** is

$$W^\circ = \{\varphi \in V^\vee : \varphi(w) = 0 \text{ for every } w \in W\}.$$

Proposition 6.70. If V is finite-dimensional, then

$$\dim W^\circ = \dim V - \dim W.$$

Proof. Choose a basis $w_1, \ldots, w_r$ of W and extend it to a basis

$$(w_1, \ldots, w_r, u_1, \ldots, u_s)$$

of V. Its dual basis shows that W° has basis

$$(u_1^\vee, \ldots, u_s^\vee).$$

Since $r + s = \dim V$, the result follows. □

6.16 Bilinear and Quadratic Forms

Duality turns a vector into a linear measurement. A bilinear form instead assigns a scalar to an ordered pair of vectors, linearly in each argument. This is the algebraic framework behind dot products, but it does not include positivity or conjugation.

Definition 6.71. A **bilinear form** on an F-vector space V is a map

$$B : V \times V \longrightarrow F$$

that is linear in each variable. It is **symmetric** if

$$B(x, y) = B(y, x),$$

skew-symmetric if

$$B(x, y) = -B(y, x),$$

and **alternating** if

$$B(v, v) = 0 \qquad (v \in V).$$

Every alternating bilinear form is skew-symmetric, by expanding

$$0 = B(x + y, x + y).$$

Conversely, a skew-symmetric form is alternating only when the characteristic is not 2. This distinction is essential in characteristic 2.

Proposition 6.72 (Matrix and congruence)**.** For an ordered basis $\beta = (v_1, \dots, v_n)$, put

$$[B]_\beta = (B(v_i, v_j)).$$

Then

$$B(x, y) = [x]_\beta^t [B]_\beta [y]_\beta.$$

If

$$P = [I_V]_{\beta'}^{\beta},$$

so that

$$[x]_\beta = P[x]_{\beta'},$$

then

$$[B]_{\beta'} = P^t [B]_\beta P.$$

Proof. Write

$$x = \sum_i a_i v_i, \qquad y = \sum_j b_j v_j.$$

By bilinearity,

$$B(x, y) = \sum_{i,j} a_i B(v_i, v_j) b_j,$$

which is the stated matrix product. The columns of P are the β-coordinates of the vectors of β'. Substitution of

$$[x]_\beta = P[x]_{\beta'}, \qquad [y]_\beta = P[y]_{\beta'}$$

into the coordinate formula gives

$$B(x, y) = [x]_{\beta'}^t P^t [B]_\beta P [y]_{\beta'},$$

and uniqueness of a matrix representing the form in β' proves the claim. $\square$

This is **congruence**, not similarity. For an operator, one input is converted into output coordinates and change of basis gives

$$[T]_{\beta'} = P^{-1} [T]_\beta P.$$

A bilinear form has two vector inputs, so both coordinates are substituted and the formula has P^t on the left instead.

Definition 6.73. The **left radical** and **right radical** of B are, respectively,

$$\mathrm{rad}_L(B) = \{v \in V : B(v, w) = 0 \text{ for every } w \in V\},$$

$$\mathrm{rad}_R(B) = \{w \in V : B(v, w) = 0 \text{ for every } v \in V\}.$$

For a general bilinear form these are conceptually distinct. The associated linear maps are

$$\Phi_B : V \longrightarrow V^\vee, \qquad \Phi_B(v) = B(v, -).$$

and

$$\Psi_B : V \longrightarrow V^\vee, \qquad \Psi_B(w) = B(-, w).$$

Proposition 6.74 (Nondegeneracy criterion)**.** For a finite-dimensional vector space, the following are equivalent:

1. $\mathrm{rad}_L(B) = \{0\}$;
2. $\mathrm{rad}_R(B) = \{0\}$;
3. $[B]_\beta$ is invertible for one, equivalently every, basis β.

When these conditions hold, B is called **nondegenerate**.

Proof. By their definitions,

$$\ker \Phi_B = \mathrm{rad}_L(B), \qquad \ker \Psi_B = \mathrm{rad}_R(B).$$

In the basis β and its dual, the matrices of these maps are

$$[\Phi_B]_\beta^{\beta^\vee} = [B]_\beta^t, \qquad [\Psi_B]_\beta^{\beta^\vee} = [B]_\beta.$$

Indeed, the j-th coordinate of $\Phi_B(v_i)$ is $B(v_i, v_j)$, whereas that of $\Psi_B(v_i)$ is $B(v_j, v_i)$. Thus either map is injective exactly when $[B]_\beta$ is invertible. Since

$$\dim V = \dim V^\vee,$$

injectivity is equivalent to bijectivity for both maps. Finally, the congruence formula in Proposition 6.72 shows that matrix invertibility is independent of the chosen basis. □

For a symmetric bilinear form, $B(v, w) = B(w, v)$, so the two radicals coincide. In that case we write their common value as $\mathrm{rad}(B)$.

Definition 6.75. Assume $\operatorname{char} F \neq 2$. A quadratic form associated with a symmetric bilinear form B is

$$q(v) = B(v, v).$$

The form can be recovered from its quadratic function by polarization:

$$B(x, y) = \frac{1}{2}(q(x + y) - q(x) - q(y)).$$

Characteristic 2 requires a separate theory: a quadratic form must not there be identified blindly with a symmetric bilinear form.

Theorem 6.76 (Diagonalization of symmetric bilinear forms)**.** Suppose that $\operatorname{char} F \neq 2$. Every symmetric bilinear form on a finite-dimensional F-vector space has a basis in which its matrix is diagonal. Equivalently, its quadratic form has coordinates

$$q(x_1, \ldots, x_n) = a_1 x_1^2 + \cdots + a_n x_n^2.$$

Proof. Induct on $\dim V$. If $B = 0$, there is nothing to prove. Otherwise some v satisfies $B(v, v) \neq 0$: if every such value were zero, then polarization would give $B = 0$. Let

$$W = \{w \in V : B(v, w) = 0\}.$$

For $x \in V$,

$$x = \frac{B(x, v)}{B(v, v)} v + \left(x - \frac{B(x, v)}{B(v, v)} v \right),$$

and the parenthesized vector lies in W. Thus

$$V = Fv \oplus W.$$

The restriction of B to W is symmetric, so induction supplies a diagonalizing basis of W; adjoining v gives the desired basis. ☐

Over $\mathbb{R}$, inner products are precisely positive-definite symmetric bilinear forms. Over $\mathbb{C}$, the inner products of Chapter 8 are Hermitian: they are linear in the first variable and conjugate-linear in the second. Thus Chapter 8 is not merely the complex special case of this bilinear discussion.

Exercises

Exercise 6.1 (Basic)**.** Verify directly that abelian groups are precisely $\mathbb{Z}$-modules, and determine the submodules of $\mathbb{Z}/12\mathbb{Z}$.

Exercise 6.2 (Core)**.** Prove the universal property of the direct sum for an arbitrary family of modules.

Exercise 6.3 (Core)**.** Let $T : F^4 \to F^3$ have rank 2. Find its nullity and decide whether it can be injective or surjective.

Exercise 6.4 (Core)**.** Compute the matrix of differentiation on polynomials of degree at most 3, using the basis $1, x, x^2, x^3$.

Exercise 6.5 (Core)**.** Show that similar matrices have the same characteristic polynomial and determinant.

Exercise 6.6 (Further)**.** Decide which of the matrices $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ are diagonalizable.

Exercise 6.7 (Further, Challenging)**.** Prove that an operator is diagonalizable exactly when the sum of the dimensions of its eigenspaces is $\dim V$.

Exercise 6.8 (Core)**.** For the standard basis of F^3, compute its dual basis and the matrix of the dual of a specified 3×3 matrix. Then verify the congruence formula for the bilinear form whose matrix is

$$\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$$

after the basis change $(e_1, e_2) \mapsto (e_1 + e_2, e_2)$.

Chapter 7

Modules over PIDs and Canonical Forms

Chapter 6 showed that diagonalization is possible exactly when an operator has enough eigenvectors. This chapter asks what remains when it does not. The answer comes from viewing a linear operator as multiplication by x, and from the remarkable fact that finitely generated modules over a principal ideal domain have a complete decomposition theory. The same theorem therefore classifies finitely generated abelian groups and produces rational and Jordan canonical forms.

Throughout the first six sections R is a principal ideal domain (PID). A PID need not be a Euclidean domain. We use only principal ideals, gcds and Bézout identities; no division-with-remainder argument is used over an arbitrary PID. The polynomial ring $F[x]$, used later, is Euclidean, so its ordinary polynomial division algorithm is available there.

7.1 Submodules of Free Modules over a PID

Over a field, every subspace has a basis. The analogous statement fails over general rings, but a PID retains just enough divisibility control for finite free modules to behave similarly.

Theorem 7.1 (Submodules of finite free modules)**.** Let R be a PID. Every submodule of R^n is free of rank at most n.

Proof. We argue by induction on n. The assertion is clear for $n = 0$. Let $N \leq R^n$ and project to the first coordinate. Its image is an ideal, say (d). If $d = 0$, then $N \subseteq 0 \oplus R^{n-1}$, and induction applies. Otherwise choose $x \in N$ whose first coordinate is d. The kernel K of the restricted projection lies in $0 \oplus R^{n-1}$ and is free by induction.

For $y \in N$, the first coordinate is rd for some $r \in R$, so $y - rx \in K$. Hence $N = Rx + K$. If $ax + k = 0$, with $k \in K$, then the first coordinate gives $ad = 0$. Since a PID is a domain and $d \neq 0$, $a = 0$, and then $k = 0$. Thus this sum is direct, and a basis of K together with x is a basis of N. $\square$

We will need the following direct termination principle for ideals.

Lemma 7.2 (Ascending chains of ideals in a PID)**.** If $I_1 \subseteq I_2 \subseteq \cdots$ is an ascending chain of ideals in a PID, then $I_n = I_N$ for every sufficiently large n.

Proof. The union
$$I = \bigcup_n I_n$$
is an ideal: two elements lie together in a large enough member of the chain, where their difference and all scalar multiples lie. Write $I = (a)$. Since $a \in I$, it lies in some I_N. Then $(a) \subseteq I_N \subseteq I = (a)$, so $I_N = I$, and all later ideals are equal to it. $\square$

7.2 Structure Theorem: Existence in Invariant-Factor Form

A finitely generated module is built from generators subject to relations. Over a PID, the relations can be separated into independent one-variable relations. The next diagonal-reduction lemma is the technical heart of that separation; it is deliberately stated before Smith normal form, which will be introduced later as a matrix interpretation of the result.

Lemma 7.3 (PID diagonal reduction)**.** Let A be a finite matrix over a PID R. By invertible row and column operations, A can be transformed into a diagonal matrix
$$\mathrm{diag}(d_1, \ldots, d_t, 0, \ldots, 0)$$
with nonzero entries satisfying $d_1 \mid d_2 \mid \cdots \mid d_t$.

Proof. We first construct one pivot. Move any nonzero entry a to the upper-left corner. If a does not divide an entry b, use row and column swaps to bring b into the first row or first column. Let g generate (a, b) and write $a = ga'$, $b = gb'$. There are $u, v \in R$ with $ua + vb = g$, hence $ua' + vb' = 1$. The matrix
$$\begin{pmatrix} u & v \\ -b' & a' \end{pmatrix}$$

has determinant one and sends the column $(a, b)^t$ to $(g, 0)^t$. Applying this invertible two-row or two-column operation replaces the pivot by an associate of g. If $a \nmid b$, then $(a) \subsetneq (a, b) = (g)$. Repeating therefore creates a strictly ascending chain of pivot ideals, and the preceding lemma says that it stops.

At that point the pivot divides every entry. Subtract suitable multiples of the first row and first column to clear its row and column. Induction on the smaller remaining matrix gives a Smith diagonal $\operatorname{diag}(d_2, \dots, d_t, 0, \dots, 0)$. Every entry in that remaining submatrix is divisible by d_1, and this remains true under its row and column operations. Thus $d_1 \mid d_2$, while induction gives $d_2 \mid \cdots \mid d_t$. This produces the displayed divisibility chain. Every operation used is invertible, so the claim follows. □

Theorem 7.4 (Structure theorem: invariant-factor existence)**.** Let M be a finitely generated module over a PID R. There are an integer $r \geq 0$ and nonzero nonunits $d_1, \dots, d_k$ satisfying $d_1 \mid d_2 \mid \cdots \mid d_k$ such that

$$M \cong R^r \oplus R/(d_1) \oplus \cdots \oplus R/(d_k).$$

Proof. Choose generators $m_1, \dots, m_n$ of M. The map $R^n \to M$ taking the i-th standard vector to m_i is onto, with relation module N. By §7.1, N is free, so choose a finite basis and write the inclusion $N \hookrightarrow R^n$ as a matrix. PID diagonal reduction chooses bases in source and target for which the image is generated by $d_1 e_1, \dots, d_k e_k$, with the displayed divisibility condition. Quotienting R^n by this image gives $R/(d_i)$ in the first k coordinates and one free copy of R in each remaining coordinate. Since $M \cong R^n/N$, this is the desired decomposition. □

7.3 Elementary-Divisor Existence via the UFD Property

Invariant factors package the torsion part into a divisibility chain. Prime factorization lets us instead see its independent prime-power pieces. This is an existence statement; uniqueness will be proved only after cancellation.

Proposition 7.5 (Elementary-divisor existence)**.** The torsion part of a finitely generated PID module is a direct sum of modules $R/(p^e)$, where p is a prime element and $e \geq 1$.

Proof. Every PID is a UFD, by Chapter 5. Factor an invariant factor as

$$d = u \prod_i p_i^{e_i},$$

with distinct p_i. The ideals $(p_i^{e_i})$ are pairwise comaximal: a common divisor of two distinct prime powers is a unit, so a Bézout identity gives their sum as R. The Chinese Remainder Theorem therefore gives

$$R/(d) \cong \bigoplus_i R/(p_i^{e_i}).$$

Apply this to every invariant factor. $\square$

7.4 Cancellation and Uniqueness

Before cancelling free summands, we must know that their ranks are meaningful. This is supplied by the invariant basis number theorem in Chapter 6: a PID is a nonzero commutative ring, so its finite free rank is well-defined.

For a prime p, write

$$M[p^\infty] = \{m \in M : p^e m = 0 \text{ for some } e \geq 1\}.$$

This is a submodule intrinsic to M, called its p-primary part.

Lemma 7.6 (Primary layers)**.** For a finitely generated p-primary torsion module P, put

$$a_j(P) = \dim_{R/(p)}(p^{j-1}P/p^jP) \qquad (j \geq 1).$$

The displayed quotient is naturally an $R/(p)$-vector space, $a_j(P)$ is invariant under isomorphism and additive under direct sums, and, if

$$P \cong \bigoplus_i R/(p^{e_i}),$$

then

$$a_j(P) = \#\{i : e_i \geq j\}.$$

Proof. Define $(r + (p))(z + p^jP) = rz + p^jP$ for $z \in p^{j-1}P$. This is well-defined: changing z changes rz by an element of p^jP, and if $r - r' = ps$, then $(r - r')z \in p^jP$. Thus the quotient is an $R/(p)$-module, hence a vector space because $R/(p)$ is a field. Isomorphisms preserve the two powers of p, and direct sums commute with them, proving invariance and additivity. For one summand $R/(p^e)$, the layer is zero for $e < j$, and otherwise is generated by the class of p^{j-1} with annihilator (p), hence is one copy of $R/(p)$. Elementary-divisor existence and additivity give the formula. $\square$

Lemma 7.7 (Special cyclic prime-power cancellation)**.** Let M, N be finitely generated p-primary torsion modules and let $e \geq 1$. If

$$M \oplus R/(p^e) \cong N \oplus R/(p^e),$$

then $M \cong N$.

Proof. Put $C = R/(p^e)$. The preceding lemma gives $a_j(C) = 1$ for $j \leq e$ and $a_j(C) = 0$ for $j > e$. Additivity and the assumed isomorphism therefore give $a_j(M) = a_j(N)$ for every j. By elementary-divisor existence, write M and N as sums of modules $R/(p^q)$. The formula above implies

$$a_j(P) - a_{j+1}(P) = \#\{\text{summands } R/(p^j) \text{ in } P\}.$$

Thus the two decompositions have the same multiplicity at every exponent, and hence give an isomorphism $M \cong N$. This uses existence of elementary divisors, but no uniqueness statement. $\square$

Theorem 7.8 (Structure theorem: uniqueness)**.** In the invariant-factor decomposition, the free rank and the ideals $(d_1), \ldots, (d_k)$ are uniquely determined. Equivalently, elementary divisors are unique up to associates and order.

Proof. The torsion submodule $T(M)$ is intrinsic: over a domain, sums and scalar multiples of torsion elements are torsion. Thus $M/T(M) \cong R^r$, and IBN proves that the free rank r is unique. The intrinsic submodule $M[p^\infty]$ separates the remaining torsion into its prime-primary components, so fix a prime p and a nonzero component P. We induct on the number of cyclic summands in one elementary-divisor decomposition, the zero-summand case being clear.

Let e be its largest exponent. The annihilator of

$$\bigoplus_i R/(p^{e_i})$$

is (p^e): an element kills every summand exactly when it is divisible by each p^{e_i}, equivalently by p^e. Hence e is intrinsic. If s is the number of summands $R/(p^e)$, then $p^{e-1}P$ is naturally an $R/(p)$-vector space: define scalar multiplication just as in the primary-layer lemma. Shorter summands vanish there, while each largest summand contributes one dimension, so $s = \dim_{R/(p)} p^{e-1}P$ is intrinsic. Two decompositions therefore have the forms

$$P \cong M' \oplus (R/(p^e))^s \cong N' \oplus (R/(p^e))^s.$$

Apply special cyclic prime-power cancellation s times to obtain $M' \cong N'$. The induction hypothesis then determines the remaining summands. This proves uniqueness of all elementary divisors.

Finally, line up the prime-power lists by increasing exponent, padding on the left by units when necessary, and multiply entries in corresponding positions. This recovers the unique divisibility-chain invariant factors. $\square$

7.5 Finitely Generated Abelian Groups

The familiar classification of finitely generated abelian groups is now a corollary, rather than a separate theorem: abelian groups are precisely $\mathbb{Z}$-modules.

Corollary 7.9 (Finitely generated abelian groups)**.** Every finitely generated abelian group is uniquely isomorphic to

$$\mathbb{Z}^r \oplus \mathbb{Z}/(d_1) \oplus \cdots \oplus \mathbb{Z}/(d_k), \qquad 1 < d_1 \mid \cdots \mid d_k,$$

or, equivalently, to a direct sum of a free abelian group and cyclic prime-power groups.

Indeed, abelian groups are precisely $\mathbb{Z}$-modules and $\mathbb{Z}$ is a PID, so this is the PID module structure theorem applied to $R = \mathbb{Z}$. Factoring the invariant factors into prime powers gives the elementary-divisor form.

For example, $\mathbb{Z}/12\mathbb{Z} \oplus \mathbb{Z}/18\mathbb{Z}$ has elementary divisors $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/3\mathbb{Z}$, $\mathbb{Z}/2\mathbb{Z}$, and $\mathbb{Z}/9\mathbb{Z}$; regrouping gives the invariant factors $\mathbb{Z}/6\mathbb{Z} \oplus \mathbb{Z}/36\mathbb{Z}$.

7.6 Smith Normal Form

The structure theorem classified modules before mentioning a named matrix normal form. Smith normal form now makes the same information visible in one presentation matrix, and provides a separate uniqueness check.

Definition 7.10. A matrix over a PID is in **Smith normal form** if it is

$$\operatorname{diag}(d_1, \ldots, d_t, 0, \ldots, 0), \qquad d_1 \mid \cdots \mid d_t, \quad d_i \neq 0.$$

Theorem 7.11 (Smith normal form)**.** For every matrix A over a PID, there are invertible matrices U, V such that UAV is in Smith normal form.

Proof. The PID diagonal-reduction lemma proves existence using only gcd and Bézout row and column operations and the directly proved ascending-chain lemma. Such operations multiply on the left or right by an invertible elementary matrix, so their accumulated products are the required U and V. No Euclidean division is used. $\square$

Worked example. Over $\mathbb{Z}$, elementary operations give

$$\begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix} \xrightarrow{R_2 \mapsto R_2 - 3R_1} \begin{pmatrix} 2 & 4 \\ 0 & -4 \end{pmatrix} \xrightarrow{C_2 \mapsto C_2 - 2C_1} \begin{pmatrix} 2 & 0 \\ 0 & -4 \end{pmatrix}.$$

Multiplying the second row by -1 yields

$$\operatorname{diag}(2, 4).$$

Consequently its cokernel is

$$\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}.$$

For a matrix, let $D_j(A)$ be the ideal generated by its $j \times j$ minors, with $D_0(A) = R$. Elementary row and column operations preserve each $D_j(A)$, because each new minor is an R-linear combination of old minors, and the inverse operation gives the reverse inclusion. In Smith form,

$$D_j(A) = (d_1 \cdots d_j).$$

Thus the determinantal ideals recover each (d_j) successively. This is an alternative proof of uniqueness, logically subsequent to the annihilator-and-induction proof above.

7.7 Linear Operators as $F[x]$-Modules

We return to the concrete invariant subspaces of Chapter 6. An operator can be encoded by declaring that the polynomial variable x acts as that operator; polynomial relations then become module relations.

Let $T : V \to V$ be linear over a field F. Define

$$p(x) \cdot v = p(T)v, \qquad p(T) = a_0 I + a_1 T + \cdots + a_n T^n.$$

Because $(pq)(T) = p(T)q(T)$, the module axioms hold.

Proposition 7.12. A subspace $W \subseteq V$ is T-invariant if and only if it is an $F[x]$-submodule for this action.

Proof. If W is a submodule, then $Tw = x \cdot w \in W$. Conversely, if $T(W) \subseteq W$, every polynomial in T preserves W, so W is an $F[x]$-submodule. $\square$

7.8 Minimal Polynomial and Cayley–Hamilton

The characteristic polynomial introduced in Chapter 6 detects eigenvalues. The minimal polynomial records the smallest polynomial relation satisfied by the whole operator and will control the sizes of canonical-form blocks.

Lemma 7.13. If V is finite-dimensional, then $\operatorname{Ann}_{F[x]}(V) = \{p \in F[x] : p(T) = 0\}$ is nonzero.

Proof. The vector space $\mathcal{L}_F(V)$ has dimension $(\dim V)^2$. The $(\dim V)^2 + 1$ endomorphisms $I, T, \ldots, T^{(\dim V)^2}$ are therefore linearly dependent. Their nontrivial linear relation is $p(T) = 0$ for a nonzero polynomial p. □

Since $F[x]$ is a PID, this annihilator has a unique monic generator, called the **minimal polynomial** μ_T. Every polynomial annihilating T is a multiple of μ_T.

Theorem 7.14 (Cayley–Hamilton)**.** Every linear operator satisfies its characteristic polynomial:

$$\chi_T(T) = 0.$$

Proof. Choose a basis and write $A = [T]$. In the commutative polynomial ring $F[A][x]$, whose coefficients are matrices commuting with A, the adjugate identity reads

$$(xI - A)\operatorname{adj}(xI - A) = \chi_T(x)I.$$

Evaluation $x \mapsto A$ is a ring homomorphism from this commutative subring to $F[A]$. It sends the left side to 0, so it sends the right side to $\chi_T(A) = 0$. Conjugating back to T proves the result. This justifies the substitution: the coefficient matrices lie in the commuting algebra $F[A]$, not in an arbitrary matrix-coefficient polynomial ring. □

Polynomial division in $F[x]$ now gives $\chi_T = q\mu_T + r$, with $\deg r < \deg \mu_T$. Evaluating at T and using Cayley–Hamilton gives $r(T) = 0$; minimality forces $r = 0$. Thus $\mu_T \mid \chi_T$.

7.9 Rational Canonical Form

The invariant-factor theorem over the PID $F[x]$ classifies an arbitrary operator without requiring its characteristic polynomial to split. If $d = x^n +$

$a_{n-1}x^{n-1} + \cdots + a_0$ is monic, multiplication by x on $F[x]/(d)$, in the basis $1, x, \ldots, x^{n-1}$, has the companion matrix C_d

$$\begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix}.$$

Indeed, multiplication shifts each basis vector one place, except that $x^n \equiv -a_{n-1}x^{n-1} - \cdots - a_0$ modulo d.

Theorem (Characteristic polynomial of a companion matrix)**.** For a monic polynomial

$$d(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0,$$

the companion matrix C_d has characteristic polynomial

$$\chi_{C_d}(x) = d(x).$$

Proof. The matrix C_d represents multiplication by x on

$$F[x]/(d).$$

The polynomial d annihilates this operator. No nonzero polynomial of smaller degree does: if p did, then $p + (d) = 0$ in $F[x]/(d)$, so $d \mid p$, which is impossible when $\deg p < \deg d$. Thus the minimal polynomial of C_d is d. Cayley–Hamilton gives

$$d \mid \chi_{C_d}.$$

Both polynomials are monic of degree n, so they are equal. $\square$

Applying the invariant-factor decomposition to V as an $F[x]$-module gives a block diagonal matrix of these companion matrices. This is the **rational canonical form**; its ordered invariant factors give existence and uniqueness up to block order. The minimal polynomial of the direct sum is the least common multiple of the block minimal polynomials, so

$$\mu_T = d_r.$$

The characteristic polynomial of a block diagonal matrix is the product of the block characteristic polynomials; the companion-matrix theorem therefore gives

$$\chi_T = d_1 \cdots d_r.$$

Example 7.15 (From invariant factors to rational canonical form)**.** Suppose an $F[x]$-module has invariant factors

$$d_1 = x - 1, \qquad d_2 = (x-1)(x^2+1).$$

Then it is the direct sum $F[x]/(d_1) \oplus F[x]/(d_2)$. The first summand contributes the 1×1 companion matrix (1), and the second contributes the companion matrix of $x^3 - x^2 + x - 1$. Their block diagonal sum is the rational canonical form. At once we read

$$\mu_T = d_2 = (x-1)(x^2+1), \qquad \chi_T = d_1 d_2 = (x-1)^2(x^2+1).$$

The example illustrates the division of labor: invariant factors describe the cyclic summands, while matrices make that description visible in a chosen basis.

Example 7.16 (One matrix through both canonical forms)**.** Over $\mathbb{C}$, begin with the actual matrix

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Let T be its operator on $\mathbb{C}^3$, and give the space its $\mathbb{C}[x]$-module structure from §7.7. On the first two standard basis vectors, $(T-I)e_2 = e_1$ and $(T-I)e_1 = 0$, so this plane is the cyclic module $\mathbb{C}[x]/((x-1)^2)$. The last basis vector gives $\mathbb{C}[x]/(x+1)$. Thus the elementary divisors are $(x-1)^2$ and $x+1$. They are coprime, so CRT combines them into the single invariant factor

$$d = (x-1)^2(x+1) = x^3 - x^2 - x + 1.$$

Consequently $\chi_T = d$ and $\mu_T = d$, the rational canonical form is the companion matrix of d,

$$\begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix},$$

and the Jordan form is $J_2(1) \oplus (-1)$. The original matrix happens already to be in Jordan form, but the calculation shows how its module invariants produce both canonical descriptions rather than merely recognize one by inspection.

7.10 Jordan Canonical Form

When the relevant polynomials split, elementary divisors refine the rational blocks into the familiar Jordan blocks. This is the structural completion of the diagonalization discussion in Chapter 6.

If the characteristic polynomial splits over F, elementary divisors have the form $(x-\lambda)^e$. On $F[x]/((x-\lambda)^e)$, use the basis $(x-\lambda)^{e-1}, \dots, (x-\lambda), 1$. Multiplication by x has a Jordan block with eigenvalue λ and ones immediately above the diagonal. Hence T is similar over F to a block diagonal matrix of Jordan blocks, uniquely determined up to their order.

In particular, T is diagonalizable exactly when every Jordan block has size 1, equivalently when μ_T splits over F into distinct linear factors. The splitting hypothesis is essential: real rotation of the plane has no real eigenbasis, even though it has a rational canonical form.

Example 7.17 (From elementary divisors to Jordan form)**.** Over $\mathbb{C}$, suppose the elementary divisors are

$$(x-2)^3, \qquad (x-2), \qquad (x+1)^2.$$

There are therefore two Jordan blocks for eigenvalue 2, of sizes 3 and 1, and one block for eigenvalue -1, of size 2. The Jordan form is

$$J_3(2) \oplus J_1(2) \oplus J_2(-1).$$

Its characteristic polynomial is $(x-2)^4(x+1)^2$, whereas its minimal polynomial is $(x-2)^3(x+1)^2$: the largest block for each eigenvalue determines the corresponding exponent in the minimal polynomial.

Remark 7.18 (How the polynomial invariants fit together). If $d_1 \mid \cdots \mid d_r$ are invariant factors, then $\chi_T = d_1 \cdots d_r$ and $\mu_T = d_r$. Factoring the d_i's and using CRT yields elementary divisors. When these factors are powers of linear polynomials, their exponents are exactly the Jordan-block sizes. Thus the characteristic polynomial records total algebraic multiplicities, the minimal polynomial records the largest required exponents, invariant factors produce rational canonical form over every field, and elementary divisors produce Jordan form when splitting is available.

Exercises

Exercise 7.1 (Basic)**.** Classify

$$\mathbb{Z}/8\mathbb{Z} \oplus \mathbb{Z}/12\mathbb{Z}$$

in invariant-factor and elementary-divisor form.

Exercise 7.2 (Core)**.** Find the Smith normal form of $\begin{pmatrix} 2 & 4 & 6 \\ 4 & 10 & 14 \end{pmatrix}$ over $\mathbb{Z}$, regarded as a homomorphism $\mathbb{Z}^3 \to \mathbb{Z}^2$, and determine its cokernel.

Exercise 7.3 (Core)**.** For $T(x, y) = (-y, x)$ on $\mathbb{R}^2$, find the minimal polynomial and explain why no real Jordan form exists.

Exercise 7.4 (Core)**.** Prove directly that the companion matrix of a monic polynomial f has characteristic polynomial f.

Exercise 7.5 (Further)**.** Show that the invariant factors of a diagonalizable operator are square-free.

Exercise 7.6 (Further, Challenging)**.** Determine the rational and Jordan canonical forms of an operator whose elementary divisors are $(x - 1)^3, (x - 1), (x + 2)^2$.

Chapter 8

Inner Product Spaces and Spectral Theory

Chapters 6 and 7 develop linear maps algebraically: kernels, images, eigenvalues, and canonical forms do not require a notion of length. An inner product adds geometry. It lets us ask stronger questions: when is a direct sum orthogonal, when is a matrix a genuine change of orthonormal coordinates, and when can an eigenbasis be chosen orthonormally? The answers lead to the finite-dimensional spectral theorem and singular value decomposition.

Throughout this chapter V and W are finite-dimensional vector spaces over $F = \mathbb{R}$ or $F = \mathbb{C}$. In the complex case our convention is that the inner product is linear in the first variable and conjugate-linear in the second. This convention agrees with the one used for adjoints below.

8.1 Inner Products and Norms

Length and angle in $\mathbb{R}^n$ are encoded by the dot product. The following axioms retain its useful properties while incorporating the complex conjugation that makes a complex squared length real and nonnegative.

Definition 8.1. An **inner product** on V is a map

$$\langle\ ,\ \rangle : V \times V \longrightarrow F$$

such that, for all $u, v, w \in V$ and $a, b \in F$,

$$\langle au + bv, w\rangle = a\langle u, w\rangle + b\langle v, w\rangle,$$

$$\langle v, u\rangle = \overline{\langle u, v\rangle},$$

and

$$\langle v, v\rangle \geq 0, \qquad \langle v, v\rangle = 0 \Longleftrightarrow v = 0.$$

The associated **norm** is

$$\|v\| = \sqrt{\langle v, v\rangle}.$$

Conjugate symmetry turns first-variable linearity into

$$\langle u, av + bw\rangle = \overline{a}\langle u, v\rangle + \overline{b}\langle u, w\rangle.$$

In particular,

$$\|av\| = |a|\|v\|.$$

Thus the complex convention is not cosmetic: it is what makes a scalar act on length by its ordinary absolute value.

Example 8.2 (Standard inner products)**.** On F^n, the standard inner product is

$$\langle u, v\rangle = \sum_{i=1}^{n} u_i \overline{v_i}.$$

On $M_{m,n}(F)$, the formula

$$\langle A, B\rangle = \operatorname{tr}(AB^*) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}\overline{b_{ij}}, \qquad B^* = \overline{B}^{\,t},$$

is an inner product. On the vector space $P_n(F)$ of polynomials of degree at most n,

$$\langle p, q\rangle = \int_0^1 p(t)\overline{q(t)}\, dt$$

is also an inner product. Positivity in the last example follows because a nonzero polynomial is nonzero on some interval, so its squared absolute value has strictly positive integral.

Example 8.3 (Why conjugation is necessary)**.** The formula

$$\sum_{i=1}^{n} u_i v_i$$

is not an inner product on $\mathbb{C}^n$: for $u = (1, i)$, it gives

$$1 + i^2 = 0$$

although $u \neq 0$. The standard formula instead gives

$$\langle u, u\rangle = |1|^2 + |i|^2 = 2.$$

Conjugation is exactly what turns the complex dot product into a genuine squared length.

Theorem 8.4 (Cauchy–Schwarz inequality)**.** For all $u, v \in V$,

$$|\langle u, v\rangle| \le \|u\|\|v\|.$$

Equality holds, for nonzero u and v, if and only if they are linearly dependent.

Proof. If $v = 0$, the inequality is immediate. Otherwise put

$$c = \frac{\langle u, v\rangle}{\|v\|^2}.$$

Positivity gives

$$0 \le \|u - cv\|^2.$$

Using the linearity convention in the two variables, the right-hand side expands to

$$\|u\|^2 - \frac{|\langle u, v\rangle|^2}{\|v\|^2}.$$

Multiplication by $\|v\|^2$ proves the inequality. Equality holds exactly when $u - cv = 0$, which is exactly linear dependence when both vectors are nonzero. □

Proposition 8.5 (Elementary norm identities)**.** For $u, v \in V$,

$$\|u + v\| \le \|u\| + \|v\|,$$

$$u \perp v \quad \Longrightarrow \quad \|u + v\|^2 = \|u\|^2 + \|v\|^2,$$

and

$$\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2.$$

The last identity is the **parallelogram identity**.

Proof. Expanding gives

$$\|u + v\|^2 = \|u\|^2 + 2\,\mathrm{Re}\langle u, v\rangle + \|v\|^2.$$

Cauchy–Schwarz bounds this by

$$(\|u\| + \|v\|)^2,$$

which proves the triangle inequality after taking nonnegative square roots. When $u \perp v$, the middle term vanishes, giving Pythagoras. Adding the corresponding expansions for $u + v$ and $u - v$ cancels the two cross terms and proves the parallelogram identity. □

Proposition 8.6 (Polarization)**.** The norm induced by an inner product determines the inner product. More precisely, over $\mathbb{R}$,

$$\langle u, v\rangle = \frac{1}{4}(\|u+v\|^2 - \|u-v\|^2),$$

whereas over $\mathbb{C}$, with our first-variable-linear convention,

$$\langle u, v\rangle = \frac{1}{4}(\|u+v\|^2 - \|u-v\|^2) + \frac{i}{4}(\|u+iv\|^2 - \|u-iv\|^2).$$

Proof. The first difference is

$$\|u+v\|^2 - \|u-v\|^2 = 4\,\mathrm{Re}\langle u, v\rangle.$$

Over $\mathbb{C}$, direct expansion of the second difference gives

$$\|u+iv\|^2 - \|u-iv\|^2 = 4\,\mathrm{Im}\langle u, v\rangle.$$

The displayed formulas follow by combining real and imaginary parts. The sign of the second term reflects the convention that the first, rather than the second, variable is linear. □

8.2 Orthonormal Sets and Gram–Schmidt

Orthogonality makes coordinates behave like independent directions: all cross terms disappear. Gram–Schmidt shows that this convenience is available in every finite-dimensional inner-product space.

Definition 8.7. Vectors $u, v \in V$ are **orthogonal**, written $u \perp v$, if

$$\langle u, v\rangle = 0.$$

A set is **orthogonal** if distinct members are orthogonal and **orthonormal** if, in addition, every member has norm one. For a subset $S \subseteq V$, its **orthogonal complement** is

$$S^\perp = \{v \in V : \langle v, s\rangle = 0 \text{ for every } s \in S\}.$$

Proposition 8.8 (Orthogonal coordinates)**.**
Every orthogonal set of nonzero vectors is independent. If

$$(e_1, \ldots, e_n)$$

is an orthonormal basis of V, then every $v \in V$ has the expansion

$$v = \sum_{j=1}^{n} \langle v, e_j \rangle e_j,$$

and

$$\|v\|^2 = \sum_{j=1}^{n} |\langle v, e_j \rangle|^2.$$

Proof. If

$$\sum_{j=1}^{n} a_j e_j = 0$$

for an orthogonal set of nonzero vectors, take the inner product with e_k. This gives

$$a_k \|e_k\|^2 = 0,$$

so every $a_k = 0$. For an orthonormal basis, write

$$v = \sum_{j=1}^{n} a_j e_j.$$

Taking the inner product with e_k gives $a_k = \langle v, e_k \rangle$. Pythagoras applied to this orthogonal sum gives the norm formula. $\square$

Corollary 8.9 (Bessel and Parseval)**.** If $e_1, \ldots, e_r$ is an orthonormal set in V, then

$$\sum_{j=1}^{r} |\langle v, e_j \rangle|^2 \leq \|v\|^2 \qquad (v \in V).$$

If the set is an orthonormal basis, equality holds; this equality is **Parseval's identity** in the finite-dimensional setting.

Proof. Put

$$w = \sum_{j=1}^{r} \langle v, e_j \rangle e_j.$$

Then $v - w$ is orthogonal to every e_j, hence to w. Pythagoras gives

$$\|v\|^2 = \|w\|^2 + \|v - w\|^2 = \sum_{j=1}^{r} |\langle v, e_j \rangle|^2 + \|v - w\|^2.$$

This proves Bessel's inequality. If the e_j's form a basis, then $w = v$, so equality holds. $\square$

Theorem 8.10 (Gram–Schmidt orthogonalization)**.** Let

$$(v_1, \dots, v_n)$$

be a linearly independent list. Define first

$$u_1 = v_1, \qquad e_1 = \frac{u_1}{\|u_1\|},$$

then

$$u_2 = v_2 - \langle v_2, e_1 \rangle e_1, \qquad e_2 = \frac{u_2}{\|u_2\|},$$

then

$$u_3 = v_3 - \langle v_3, e_1 \rangle e_1 - \langle v_3, e_2 \rangle e_2, \qquad e_3 = \frac{u_3}{\|u_3\|}.$$

In general,

$$u_k = v_k - \sum_{j=1}^{k-1} \langle v_k, e_j \rangle e_j, \qquad e_k = \frac{u_k}{\|u_k\|}.$$

Every u_k is nonzero, the resulting list is orthonormal, and

$$\operatorname{span}(e_1, \dots, e_k) = \operatorname{span}(v_1, \dots, v_k) \qquad (1 \le k \le n).$$

Proof. Assume that $e_1, \dots, e_{k-1}$ have been constructed. For $i < k$,

$$\langle u_k, e_i \rangle = \langle v_k, e_i \rangle - \sum_{j=1}^{k-1} \langle v_k, e_j \rangle \langle e_j, e_i \rangle = 0.$$

Thus u_k is orthogonal to the preceding e_i's. If $u_k = 0$, the definition would put v_k in the span of $e_1, \dots, e_{k-1}$, which, by the induction hypothesis, is the span of $v_1, \dots, v_{k-1}$. This contradicts independence. Hence e_k is defined and has norm one.

The formula defining u_k shows that u_k lies in the span of the first k vectors v_i. Conversely, it can be rearranged to express v_k in the span of $u_k, e_1, \dots, e_{k-1}$. The two successive spans are therefore equal. Induction proves every assertion. □

Corollary 8.11 (Orthonormal bases)**.**
Each finite-dimensional inner-product space admits an orthonormal basis. Every orthonormal set can be enlarged to one.

Proof. Apply Gram–Schmidt to any basis. For an orthonormal set, first extend it to an ordinary basis using the finite-dimensional basis theorem of Chapter 6, then apply Gram–Schmidt to the additional vectors. The first members are unchanged because they are already orthonormal. □

Example 8.12 (Gram–Schmidt in $\mathbb{R}^2$)**.** Starting with

$$v_1 = (1,1), \qquad v_2 = (1,0),$$

we obtain

$$e_1 = \frac{1}{\sqrt{2}}(1,1),$$

and

$$u_2 = (1,0) - \langle (1,0), e_1 \rangle\, e_1 = \left(\frac{1}{2}, -\frac{1}{2}\right).$$

Thus

$$e_2 = \frac{1}{\sqrt{2}}(1,-1).$$

The original and orthonormal lists span the same plane, but the latter makes lengths, coordinates, and projections immediate.

8.3 Orthogonal Complements and Projections

An arbitrary direct-sum complement of a subspace is seldom canonical. The orthogonal complement is canonical because it is determined by the inner product alone; it separates a vector into a component in the subspace and a perpendicular error.

Proposition 8.13 (Orthogonal complements)**.** If $W \le V$, then $W^\perp$ is a subspace. In finite dimensions,

$$V = W \oplus W^\perp, \qquad \dim W + \dim W^\perp = \dim V, \qquad (W^\perp)^\perp = W.$$

Proof. Closure of $W^\perp$ under linear combinations follows from linearity in the first variable. Choose an orthonormal basis

$$(e_1, \dots, e_r)$$

of W. For $v \in V$, define

$$P_W v = \sum_{i=1}^{r} \langle v, e_i \rangle e_i.$$

Then $P_W v \in W$, while, for each i,

$$\langle v - P_W v, e_i \rangle = 0.$$

Thus $v - P_W v \in W^\perp$, proving $V = W + W^\perp$. A vector in the intersection is orthogonal to itself and is therefore zero, so the sum is direct. Taking dimensions gives the formula. Since $W \subseteq (W^\perp)^\perp$ and the two spaces have the same dimension, they are equal. $\square$

Definition 8.14. The linear operator

$$P_W : V \longrightarrow V, \qquad P_W v = \sum_{i=1}^{r} \langle v, e_i \rangle e_i,$$

where $e_1, \dots, e_r$ is any orthonormal basis of W, is the **orthogonal projection onto** W; its image is W.

The preceding proof shows that the formula depends only on the unique decomposition

$$v = P_W v + (v - P_W v), \qquad P_W v \in W, \quad v - P_W v \in W^{\perp},$$

and therefore not on the chosen orthonormal basis.

Theorem 8.15 (Best approximation)**.** For $v \in V$ and $w \in W$,

$$\|v - w\|^2 = \|v - P_W v\|^2 + \|P_W v - w\|^2.$$

Consequently $P_W v$ is the unique point of W nearest to v.

Proof. The vector $v - P_W v$ lies in $W^{\perp}$, whereas $P_W v - w$ lies in W. They are orthogonal and their sum is $v - w$, so Pythagoras gives the formula. The second summand is nonnegative and vanishes exactly when $w = P_W v$, proving uniqueness. □

Example 8.16 (Projection onto a line)**.** Let

$$W = \operatorname{span}(1, 1, 1) \subseteq \mathbb{R}^3.$$

The unit vector $e = (1, 1, 1)/\sqrt{3}$ gives

$$P_W(1, 2, 3) = \langle (1, 2, 3), e \rangle e = 2(1, 1, 1).$$

The error

$$(-1, 0, 1)$$

has dot product zero with $(1, 1, 1)$, visibly confirming the orthogonal decomposition.

8.4 Linear Functionals and Adjoints

The algebraic dual $V^\vee$, developed in Section 6.15, consists of all linear measurements on V. In finite-dimensional inner-product spaces, the Riesz theorem says that every such measurement is obtained by taking an inner product with one uniquely determined vector. It is this theorem that makes the adjoint an intrinsic operator rather than merely a matrix operation.

Theorem 8.17 (Finite-dimensional Riesz representation)**.** For every linear functional

$$\varphi : V \longrightarrow F,$$

there is a unique $u \in V$ such that

$$\varphi(v) = \langle v, u\rangle \qquad (v \in V).$$

Proof. Choose an orthonormal basis $e_1, \ldots, e_n$. Put

$$u = \sum_{j=1}^{n} \overline{\varphi(e_j)} e_j.$$

If

$$v = \sum_{j=1}^{n} x_j e_j,$$

then

$$\langle v, u\rangle = \sum_{j=1}^{n} x_j \varphi(e_j) = \varphi(v).$$

If both u and u' represent φ, then

$$\langle v, u - u'\rangle = 0$$

for every v. Taking $v = u - u'$ proves $u = u'$. □

Theorem 8.18 (Adjoint)**.** For every linear map

$$T : V \longrightarrow W,$$

there is a unique linear map

$$T^* : W \longrightarrow V$$

satisfying

$$\langle Tv, w\rangle = \langle v, T^* w\rangle \qquad (v \in V,\ w \in W).$$

Proof. For fixed $w \in W$, the function

$$v \longmapsto \langle Tv, w\rangle$$

is a linear functional on V. Riesz representation supplies a unique vector, denoted T^*w, satisfying the displayed equality. For $a, b \in F$ and $w_1, w_2 \in W$, conjugate-linearity in the second variable gives

$$\langle Tv, aw_1 + bw_2\rangle = \overline{a}\langle Tv, w_1\rangle + \overline{b}\langle Tv, w_2\rangle = \langle v, aT^*w_1 + bT^*w_2\rangle.$$

Uniqueness in Riesz representation proves the linearity of T^*. The same uniqueness proves uniqueness of the adjoint. □

Proposition 8.19 (Adjoint identities and matrices)**.** For compatible linear maps S, T and $a \in F$,

$$(S + T)^* = S^* + T^*, \qquad (aT)^* = \overline{a}\,T^*, \qquad (ST)^* = T^*S^*, \qquad (T^*)^* = T.$$

If T is invertible, then

$$(T^{-1})^* = (T^*)^{-1}.$$

Relative to orthonormal bases, the matrix of T^* is

$$[T^*] = [T]^* = \overline{[T]}^{\,t}.$$

Proof. Each identity follows by putting the proposed right-hand side into the defining equation for an adjoint and invoking uniqueness. For example,

$$\langle STv, w\rangle = \langle Tv, S^*w\rangle = \langle v, T^*S^*w\rangle.$$

The inverse formula follows from

$$I = (T^{-1}T)^* = T^*(T^{-1})^*.$$

For orthonormal coordinate columns x, y and a matrix $A = [T]$,

$$\langle Ax, y\rangle = x^tA^t\overline{y} = \langle x, \overline{A}^{\,t}y\rangle.$$

Thus the matrix of the adjoint is the conjugate transpose. Over $\mathbb{R}$, this is the ordinary transpose. □

Theorem 8.20 (Kernels, images, and adjoints)**.** For $T : V \to W$,

$$\ker T^* = (\operatorname{im} T)^\perp, \qquad \operatorname{im} T^* = (\ker T)^\perp.$$

Consequently,

$$\operatorname{rank} T = \operatorname{rank} T^*,$$

and the equation $Tv = w$ is solvable if and only if

$$w \perp \ker T^*.$$

Proof. For $w \in W$,

$$\begin{aligned} w \in \ker T^* &\iff \langle v, T^* w \rangle = 0 \quad \text{for every } v \in V \\ &\iff \langle Tv, w \rangle = 0 \quad \text{for every } v \in V. \end{aligned}$$

This is exactly $w \in (\operatorname{im} T)^{\perp}$, proving the first identity. Applying it to T^*, using $(T^*)^* = T$ and taking a second orthogonal complement in finite dimensions gives

$$\operatorname{im} T^* = (\ker T)^{\perp}.$$

The rank equality follows either by dimensions of these complements or by rank–nullity. Finally,

$$w \in \operatorname{im} T \iff w \in (\ker T^*)^{\perp},$$

which is the asserted solvability criterion. □

Proposition 8.21 (Characterization of orthogonal projections)**.** For a linear operator $P : V \to V$, the following are equivalent:

1. $P = P_W$ for some subspace W;
2. $P^2 = P$ and $P^* = P$.

In this situation,

$$W = \operatorname{im} P, \qquad \ker P = W^{\perp}.$$

Proof. The formula for P_W shows directly that $P_W^2 = P_W$. If $v, w \in V$, expansion in an orthonormal basis of W gives

$$\langle P_W v, w \rangle = \langle v, P_W w \rangle,$$

so $P_W^* = P_W$.

Conversely, suppose $P^2 = P = P^*$. The idempotence gives

$$V = \operatorname{im} P \oplus \ker P.$$

If $x = Pu \in \operatorname{im} P$ and $y \in \ker P$, then

$$\langle x, y \rangle = \langle Pu, y \rangle = \langle u, Py \rangle = 0.$$

Thus $\ker P \subseteq (\operatorname{im} P)^{\perp}$. The direct-sum dimension formula makes the two subspaces equal, so P is the orthogonal projection onto its image. □

An ordinary algebraic projection only requires $P^2 = P$. It need not be self-adjoint, and its kernel need not be perpendicular to its image. The self-adjoint condition is exactly what distinguishes orthogonal projection.

8.5 Self-Adjoint, Normal, Unitary, and Orthogonal Operators

Adjoints identify the maps compatible with inner-product geometry. These operator classes are important not merely for their names: normality is the precise complex hypothesis for orthonormal diagonalization.

Definition 8.22. For an endomorphism $T : V \to V$, we say that T is **self-adjoint** if

$$T = T^*,$$

skew-adjoint if

$$T^* = -T,$$

and **normal** if

$$TT^* = T^*T.$$

It is **positive** if it is self-adjoint and

$$\langle Tv, v\rangle \geq 0 \qquad (v \in V),$$

and **positive-definite** if equality occurs only for $v = 0$. A complex operator is **unitary**, and a real operator is **orthogonal**, if

$$T^*T = I.$$

Self-adjoint matrices are real symmetric or complex Hermitian. The terms unitary and orthogonal distinguish the base field because the latter is the real specialization of the former.

Proposition 8.23 (Isometries and unitary operators)**.** For an endomorphism T of a finite-dimensional inner-product space, the following are equivalent:

1. T is unitary or orthogonal, as appropriate to the field;
2. $T^*T = I$;
3. $$\langle Tu, Tv\rangle = \langle u, v\rangle \qquad (u, v \in V);$$
4. $$\|Tv\| = \|v\| \qquad (v \in V).$$

In this case T is invertible and

$$T^{-1} = T^*.$$

Relative to an orthonormal basis, its matrix A satisfies

$$A^*A = AA^* = I.$$

Thus the columns and rows of A are orthonormal bases.

Proof. The equivalence of 1 and 2 is the definition. Moreover,

$$\langle Tu, Tv\rangle = \langle u, T^*Tv\rangle,$$

so 2 implies 3, and 3 implies 4. Conversely, the polarization identity shows that 4 determines the same inner products, so 4 implies 3. If 3 holds, then

$$\langle u, (T^*T - I)v\rangle = 0$$

for every u, v; taking $u = (T^*T - I)v$ proves 2.

Condition 4 makes T injective. Finite dimensionality then makes it surjective, so it is invertible; from $T^*T = I$ we obtain $T^* = T^{-1}$, and hence $TT^* = I$. The matrix identities follow in orthonormal bases. Their entries state exactly that both the columns and rows are orthonormal. □

Proposition 8.24 (Basic consequences of the adjoint)**.** Every operator T^*T is self-adjoint and positive. If T is normal, then

$$\|Tv\| = \|T^*v\| \qquad (v \in V),$$

and

$$\ker T = \ker T^*.$$

Self-adjoint and unitary or orthogonal operators are normal.

Proof. The adjoint identities give

$$(T^*T)^* = T^*T,$$

and

$$\langle T^*Tv, v\rangle = \langle Tv, Tv\rangle \geq 0.$$

If T is normal, then

$$\|Tv\|^2 = \langle T^*Tv, v\rangle = \langle TT^*v, v\rangle = \|T^*v\|^2.$$

This equality proves the kernel identity. Self-adjointness makes the two products identical, while for a unitary or orthogonal operator they both equal I, proving the final assertion. □

Proposition 8.25 (Eigenvalue geometry)**.** Let T be an endomorphism of a finite-dimensional inner-product space.

1. If T is self-adjoint, every eigenvalue of T is real.
2. If T is unitary or orthogonal, every eigenvalue λ has
$$|\lambda| = 1.$$
3. If T is normal on a complex inner-product space and
$$Tv = \lambda v,$$
then
$$T^* v = \overline{\lambda} v.$$
Consequently, eigenvectors of a normal operator belonging to distinct eigenvalues are orthogonal.

Proof. For a self-adjoint T and a nonzero eigenvector v,

$$\lambda\langle v, v\rangle = \langle Tv, v\rangle = \langle v, Tv\rangle = \overline{\lambda}\langle v, v\rangle,$$

so $\lambda = \overline{\lambda}$. If T is unitary or orthogonal,

$$\|v\| = \|Tv\| = |\lambda|\|v\|,$$

proving the second claim. For normal T, the preceding proposition gives

$$\|(T - \lambda I)v\| = \|(T^* - \overline{\lambda} I)v\|.$$

The left side is zero, hence the third assertion follows. If also

$$Tw = \mu w,$$

then

$$\lambda\langle v, w\rangle = \langle Tv, w\rangle = \langle v, T^* w\rangle = \langle v, \overline{\mu} w\rangle = \mu\langle v, w\rangle.$$

Thus $\langle v, w\rangle = 0$ when $\lambda \neq \mu$. □

Remark 8.26. Normality alone does not imply orthogonal diagonalizability over $\mathbb{R}$. Rotation through 90° on $\mathbb{R}^2$ is orthogonal and hence normal, but it has no real eigenvector. The real spectral theorem therefore concerns self-adjoint operators, while the full normal spectral theorem is naturally complex.

8.6 Spectral Theorems

The spectral theorem identifies exactly when ordinary diagonalization can be made geometric. Its finite-dimensional proof requires no Hilbert-space theory, but the complex result uses the Fundamental Theorem of Algebra to produce an eigenvalue.

Remark 8.27 (The Fundamental Theorem of Algebra). The Fundamental Theorem of Algebra says that every nonconstant polynomial in $\mathbb{C}[x]$ has a root in $\mathbb{C}$. We use it only to ensure that the characteristic polynomial of a complex endomorphism has a root. A Galois-theoretic proof appears later in Chapter 12; no complex analysis is needed elsewhere in this chapter.

Theorem 8.28 (Complex spectral theorem)**.** For an endomorphism T of a finite-dimensional complex inner-product space, the following are equivalent:

1. T is normal;
2. V has an orthonormal basis of eigenvectors of T;
3. the matrix of T in some orthonormal basis is diagonal.

Equivalently, T is unitarily diagonalizable.

Proof. The equivalence of 2 and 3 is the definition of a matrix in a chosen basis. A diagonal matrix commutes with its conjugate transpose, so 2 implies 1. Conversely, suppose T is normal and argue by induction on $\dim V$. The case $\dim V = 0$ is clear. The Fundamental Theorem of Algebra supplies an eigenvalue λ, and let

$$E = \ker(T - \lambda I).$$

If $z \in E^{\perp}$ and $e \in E$, then

$$\langle Tz, e\rangle = \langle z, T^*e\rangle = \langle z, \overline{\lambda}e\rangle = 0.$$

Thus $E^{\perp}$ is T-invariant. The same calculation with T^* shows that it is T^*-invariant, so the restriction of T to $E^{\perp}$ remains normal. It has smaller dimension unless $E = V$, and the induction hypothesis supplies an orthonormal eigenbasis of $E^{\perp}$. An orthonormal basis of E, obtained by Gram–Schmidt, together with that basis gives an orthonormal eigenbasis of V. □

Corollary 8.29 (Real spectral theorem)**.** For an endomorphism T of a finite-dimensional real inner-product space, the following are equivalent:

1. T is self-adjoint;

2. V has an orthonormal basis of real eigenvectors of T;

3. T is orthogonally diagonalizable.

Proof. The last two conditions are equivalent by coordinates, and either one plainly implies self-adjointness. Conversely, complexify V and extend the real inner product to the usual Hermitian inner product on $V_{\mathbb{C}}$. The complexified operator is self-adjoint, hence normal; the complex spectral theorem applies. Its eigenvalues are real by the preceding eigenvalue proposition. If

$$T_{\mathbb{C}}(x+iy) = \lambda(x+iy),$$

then

$$Tx = \lambda x, \qquad Ty = \lambda y.$$

Thus the complex eigenspaces are complexifications of real eigenspaces. They span $V_{\mathbb{C}}$, so the real eigenspaces span V. Distinct eigenspaces are orthogonal, and Gram–Schmidt within each one produces a real orthonormal eigenbasis. □

Example 8.30 (Normal diagonalization)**.** The matrix

$$\begin{pmatrix} 2 & 0 \\ 0 & i \end{pmatrix}$$

is normal and already unitarily diagonal. By contrast,

$$\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$$

is diagonalizable only when $a = 0$. For $a \neq 0$, it does not commute with its conjugate transpose and is not normal. The spectral theorem therefore excludes precisely the Jordan-block obstruction described in Chapter 7.

8.7 Singular Value Decomposition

Eigenvalue diagonalization applies only to endomorphisms. Singular value decomposition applies to maps between spaces of different dimensions by diagonalizing the positive operator T^*T.

Theorem 8.31 (Singular value decomposition)**.** Let

$$T : V \longrightarrow W$$

be a linear map of finite-dimensional real or complex inner-product spaces. There are orthonormal bases of V and W in which the matrix of T is diagonal rectangular with nonnegative entries

$$\sigma_1, \ldots, \sigma_r.$$

For matrices, this says

$$A = U\Sigma V^*,$$

where U and V are unitary in the complex case and orthogonal in the real case. The nonzero σ_i's are the square roots of the nonzero eigenvalues of T^*T.

Proof. The operator T^*T is self-adjoint and positive. By the appropriate spectral theorem, choose an orthonormal eigenbasis of V. Its eigenvalues are nonnegative because

$$\langle T^*Tv, v\rangle = \|Tv\|^2.$$

For each positive eigenvalue σ_i^2, set

$$u_i = \frac{Tv_i}{\sigma_i}.$$

If i, j correspond to positive eigenvalues, then

$$\langle u_i, u_j\rangle = \frac{\langle T^*Tv_i, v_j\rangle}{\sigma_i\sigma_j} = \delta_{ij}.$$

Moreover,

$$\ker(T^*T) = \ker T,$$

because the displayed norm identity shows that $T^*Tv = 0$ exactly when $Tv = 0$. The zero-eigenvalue basis vectors therefore form a basis of $\ker T$. Extend the orthonormal u_i's to an orthonormal basis of W. In the resulting bases, $Tv_i = \sigma_i u_i$ for positive singular values and $Tv_i = 0$ on the kernel, yielding the stated rectangular diagonal matrix. □

Example 8.32 (An SVD calculation)**.** For

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix},$$

we have

$$A^*A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Its orthonormal eigenvectors are

$$v_1 = \frac{1}{\sqrt{2}}(1,1), \qquad v_2 = \frac{1}{\sqrt{2}}(1,-1),$$

with eigenvalues 2 and 0. Hence the singular values are $\sqrt{2}$ and 0. Taking

$$u_1 = \frac{Av_1}{\sqrt{2}} = (1,0)$$

and completing it by $u_2 = (0,1)$ gives

$$A = U \operatorname{diag}(\sqrt{2},0)V^*.$$

Unlike eigenvalue diagonalization, this description works even though the original matrix is not normal.

Exercises

Exercise 8.1 (Basic)**.** Verify that the standard formula on $\mathbb{C}^n$ is an inner product under the convention of this chapter.

Exercise 8.2 (Core)**.** Use the complex polarization identity to recover the standard inner product of two specified vectors in $\mathbb{C}^2$ from their norms.

Exercise 8.3 (Core)**.** Apply Gram–Schmidt to

$$(1,1,0),\ (1,0,1),\ (0,1,1)$$

in $\mathbb{R}^3$.

Exercise 8.4 (Core)**.** Find the orthogonal projection of $(1,2,3)$ onto the span of $(1,1,1)$, and verify its best-approximation property for an arbitrary point on that line.

Exercise 8.5 (Core)**.** Compute the adjoint of

$$\begin{pmatrix} 1 & i \\ 0 & 2 \end{pmatrix}$$

on $\mathbb{C}^2$, and determine its kernel.

Exercise 8.6 (Further)**.** Prove that a positive-definite operator is invertible.

Exercise 8.7 (Further, Challenging)**.** Find an SVD of

$$\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

and identify its nonzero singular subspaces.

Chapter 9

Tensor Products of Modules

Direct sums place modules side by side. Tensor products solve a different problem: they turn expressions that are additive in two variables into ordinary homomorphisms from one abelian group. The price is that sidedness now matters. We begin over an arbitrary ring and only later specialize to the more familiar commutative case.

9.1 Balanced Maps and Binary Tensor Products

Definition 9.1. Let M_R be a right R-module, ${}_RN$ a left R-module, and A an abelian group. A map $b : M \times N \to A$ is **R-balanced** if

$$\begin{aligned} b(m+m', n) &= b(m,n) + b(m',n), \\ b(m, n+n') &= b(m,n) + b(m,n'), \\ b(mr, n) &= b(m, rn). \end{aligned}$$

The last equation is the balancing condition. It says that the scalar inserted between the two variables may be moved across their interface. This is precisely what makes a product such as $m \otimes n$ meaningful when the two module actions lie on opposite sides.

Theorem 9.2 (Tensor-product universal property)**.** There is an abelian group $M \otimes_R N$, with an R-balanced map

$$\tau : M \times N \longrightarrow M \otimes_R N, \qquad (m,n) \longmapsto m \otimes n,$$

such that every R-balanced map $b : M \times N \to A$ has a unique homomorphism $\bar{b} : M \otimes_R N \to A$ satisfying $\bar{b}(m \otimes n) = b(m,n)$.

$$\begin{tikzcd} M \times N \arrow[r, "\tau"] \arrow[dr, "b"'] & M \otimes_R N \arrow[d, "\bar{b}"] \\ & A \end{tikzcd}$$

Proof. Let G be the free abelian group on symbols $[m, n]$, and quotient by the subgroup generated by

$$[m + m', n] - [m, n] - [m', n], \qquad [m, n + n'] - [m, n] - [m, n'],$$
$$[mr, n] - [m, rn].$$

Call the quotient $M \otimes_R N$, and write $m \otimes n$ for the class of $[m, n]$. The displayed relations make τ balanced. Given balanced b, the map from G sending $[m, n]$ to $b(m, n)$ kills every listed relation, so it descends to $\bar{b}$. Since pure tensors generate the quotient, this homomorphism is unique. □

If two groups satisfy this universal property, applying each universal property to the other universal balanced map gives mutually inverse homomorphisms. Thus $M \otimes_R N$ is unique up to a **canonical isomorphism**. It is not a Cartesian product, and an element need not be a single pure tensor; it is a finite sum of pure tensors.

9.2 Bimodules and Tensor Actions

Definition 9.3. For rings R, S, an (R, S)**-bimodule** $_RM_S$ is an abelian group carrying a left R-action and a right S-action such that

$$(rm)s = r(ms) \qquad (r \in R,\ m \in M,\ s \in S).$$

Every ring is an (R, R)-bimodule over itself. A left R-module is an $(R, \mathbb{Z})$-bimodule and a right R-module is an $(\mathbb{Z}, R)$-bimodule. Bimodules retain an action on each outside of a tensor product.

Proposition 9.4 (Induced outside actions)**.** Let $_SM_R$, $_RN_T$ be bimodules. Then $M \otimes_R N$ is an (S, T)-bimodule under

$$s(m \otimes n) = (sm) \otimes n, \qquad (m \otimes n)t = m \otimes (nt).$$

Proof. For fixed s, the map $(m, n) \mapsto (sm) \otimes n$ is balanced; indeed,

$$s(mr) \otimes n = ((sm)r) \otimes n = (sm) \otimes (rn).$$

The universal property therefore gives a well-defined endomorphism. The argument for t is analogous. The module axioms follow on pure tensors, and the actions commute because

$$(s(m \otimes n))t = (sm) \otimes (nt) = s((m \otimes n)t).$$

$\square$

9.3 Finite Associativity

Associativity of tensor products is a canonical isomorphism, not literal equality. Only finitely many tensor factors are considered.

Theorem 9.5 (Associativity)**.** For bimodules ${}_SM_R$, ${}_RN_T$, and ${}_TP$, there is a canonical isomorphism of left S-modules

$$(M \otimes_R N) \otimes_T P \;\cong\; M \otimes_R (N \otimes_T P),$$

given by

$$(m \otimes n) \otimes p \longmapsto m \otimes (n \otimes p).$$

Proof. The proposed formula respects the R-balancing relation because

$$(mr \otimes n) \otimes p \longmapsto mr \otimes (n \otimes p) = m \otimes (rn \otimes p).$$

It respects the outer T-balancing relation because

$$(m \otimes nt) \otimes p \longmapsto m \otimes (nt \otimes p) = m \otimes (n \otimes tp).$$

Additivity is immediate, so the two universal properties produce a well-defined homomorphism. The reversed formula

$$m \otimes (n \otimes p) \longmapsto (m \otimes n) \otimes p$$

satisfies the same checks. Both composites fix pure generators, hence are the identity. $\square$

9.4 Tensor Products and Arbitrary Direct Sums

This is the intended infinite-index construction: tensor product commutes with arbitrary **direct sums**, not arbitrary direct products. The finite-support condition makes the following formula meaningful.

Theorem 9.6 (Arbitrary direct sums)**.** For right R-modules $(M_i)_{i\in I}$ and a left R-module N,

$$\left(\bigoplus_{i\in I} M_i\right) \otimes_R N \;\cong\; \bigoplus_{i\in I}(M_i \otimes_R N).$$

For a right R-module M and left R-modules $(N_i)_{i\in I}$,

$$M \otimes_R \left(\bigoplus_{i\in I} N_i\right) \;\cong\; \bigoplus_{i\in I}(M \otimes_R N_i).$$

Proof. For the first map, send $(m_i) \otimes n$ to the finite sum

$$\sum_{i\in I} m_i \otimes n.$$

It is balanced, so the tensor universal property makes it a homomorphism. For each canonical injection

$$\iota_i : M_i \longrightarrow \bigoplus_{i\in I} M_i,$$

the balanced map $(m, n) \mapsto \iota_i(m) \otimes n$ induces

$$M_i \otimes_R N \longrightarrow \left(\bigoplus_{i\in I} M_i\right) \otimes_R N.$$

The direct-sum universal property combines these maps into the reverse homomorphism. The two maps are inverse on pure tensors and on the injected summands, hence everywhere. The second isomorphism follows by the identical argument with the variables reversed. $\square$

9.5 The Commutative-Ring Specialization

Suppose now that R is commutative. Every left R-module becomes canonically an (R, R)-bimodule by defining $mr = rm$. Thus $M \otimes_R N$ is again an R-module, and

$$r(m \otimes n) = (rm) \otimes n = m \otimes (rn).$$

Proposition 9.7 (Symmetry)**.** For R-modules M, N, there is a canonical isomorphism

$$M \otimes_R N \cong N \otimes_R M, \qquad m \otimes n \longmapsto n \otimes m.$$

Proof. The map $(m,n) \mapsto n \otimes m$ is balanced because

$$n \otimes (mr) = (rn) \otimes m.$$

The universal property gives the displayed map; swapping the factors again gives its inverse. Tensor products are symmetric only up to canonical isomorphism, not by literal equality. □

9.6 Finite Multilinear Tensor Products

For a finite list $M_1, \dots, M_n$ of R-modules and an R-module A, a map

$$b : M_1 \times \cdots \times M_n \longrightarrow A$$

is **R-multilinear** if it is additive and respects scalar multiplication in each variable. Define

$$M_1 \otimes_R \cdots \otimes_R M_n$$

to be the quotient of the free R-module on tuples $(m_1, \dots, m_n)$ by the additivity and scalar-linearity relations in each slot. Write the class as $m_1 \otimes \cdots \otimes m_n$.

Theorem 9.8 (Finite multilinear universal property)**.** For every finite n, R-module homomorphisms

$$M_1 \otimes_R \cdots \otimes_R M_n \longrightarrow A$$

are in bijection with R-multilinear maps

$$M_1 \times \cdots \times M_n \longrightarrow A.$$

For $n = 3$, the direct tensor product is canonically isomorphic to both

$$(M_1 \otimes_R M_2) \otimes_R M_3 \quad \text{and} \quad M_1 \otimes_R (M_2 \otimes_R M_3).$$

Proof. The quotient construction makes the universal tuple map multilinear. A multilinear map from the product kills its defining relations and therefore factors uniquely through the quotient, proving the bijection. For $n = 3$, the map

$$m_1 \otimes m_2 \otimes m_3 \longmapsto (m_1 \otimes m_2) \otimes m_3$$

is well-defined by multilinearity and has the evident inverse on pure tensors. Theorem 9.5 gives the other parenthesization. Induction on n proves the finite case. □

This justifies suppressing parentheses in finite tensor products. Repeated associativity and symmetry also let every permutation of finitely many factors induce a canonical isomorphism. No countable or arbitrary infinite tensor product is being defined here.

9.7 Tensoring Homomorphisms and Concrete Calculations

If $f : M_R \to M'_R$ and $g : {}_RN \to {}_RN'$ are module homomorphisms, then

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$$

defines a homomorphism $f \otimes g : M \otimes_R N \to M' \otimes_R N'$. Indeed, the displayed expression is balanced, so the universal property proves well-definedness. On pure tensors,

$$(f' \otimes g') \circ (f \otimes g) = (f'f) \otimes (g'g), \qquad 1_M \otimes 1_N = 1_{M \otimes_R N}.$$

Return to a commutative ring R. The following calculations anchor the abstract construction.

Proposition 9.9. There are canonical isomorphisms

$$R \otimes_R M \cong M, \qquad R^m \otimes_R R^n \cong R^{mn}.$$

If I is an ideal, then

$$(R/I) \otimes_R M \cong M/IM.$$

Proof. The balanced map $(r, m) \mapsto rm$ induces $R \otimes_R M \to M$, with inverse $m \mapsto 1 \otimes m$. For finite free modules, $e_i \otimes f_j$ map to the (i, j)-th standard vector; expanding both factors proves that this map and its evident inverse are mutually inverse. Finally, $(r + I, m) \mapsto rm + IM$ is balanced and well-defined because changing r by an element of I changes the value by an element of IM. Its inverse is $m + IM \mapsto (1 + I) \otimes m$; balancing kills every generator im. □

Example 9.10. Taking $R = \mathbb{Z}$ and $I = (m)$ gives

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/\gcd(m, n)\mathbb{Z}.$$

Indeed, the preceding proposition identifies this with the quotient of $\mathbb{Z}/n\mathbb{Z}$ by the subgroup generated by m.

9.8 The Tensor–Hom Correspondence

The universal property of a tensor product can be tested against arbitrary abelian groups. This produces the basic tensor–Hom correspondence without assuming commutativity of the ring.

Theorem 9.11 (Tensor–Hom)**.** Let M_R be a right R-module and let ${}_RN$ be a left R-module. Let A be an abelian group. Give

$$\mathrm{Hom}_{\mathbb{Z}}(M, A)$$

the left R-module structure

$$(r \cdot f)(m) = f(mr).$$

Then there is an isomorphism of abelian groups

$$\mathrm{Hom}_{\mathbb{Z}}(M \otimes_R N, A) \cong \mathrm{Hom}_R(N, \mathrm{Hom}_{\mathbb{Z}}(M, A)) .$$

Proof. Given $u : M \otimes_R N \to A$, set

$$\widehat{u}(n)(m) = u(m \otimes n).$$

This is additive in n. Balancing gives

$$\widehat{u}(rn)(m) = u(m \otimes rn) = u(mr \otimes n) = (r \cdot \widehat{u}(n))(m),$$

so $\widehat{u}$ is R-linear. Conversely, for an R-linear map $v : N \to \mathrm{Hom}_{\mathbb{Z}}(M, A)$, the rule

$$(m, n) \longmapsto v(n)(m)$$

is additive in each variable and balanced, since

$$v(n)(mr) = (r \cdot v(n))(m) = v(rn)(m).$$

It therefore induces $M \otimes_R N \to A$. Evaluating each composite on a pure tensor shows that the two constructions are inverse. □

Proposition 9.12 (Compatibility of Tensor–Hom)**.** The correspondence in Theorem 9.11 has the following explicit compatibilities. If $f : M' \to M$ is a map of right modules and $u : M \otimes_R N \to A$, then $u \circ (f \otimes 1_N)$ corresponds to

$$n \longmapsto \widehat{u}(n) \circ f.$$

If $h : A \to A'$ is a homomorphism of abelian groups, then $h \circ u$ corresponds to

$$n \longmapsto h \circ \widehat{u}(n).$$

Finally, if $\ell : N' \to N$ is a map of left modules, then $u \circ (1_M \otimes \ell)$ corresponds to $\widehat{u} \circ \ell$.

Proof. For example, the first map sends n and m' to

$$u(f(m') \otimes n) = \widehat{u}(n)(f(m')).$$

The second and third identities follow in exactly the same way by evaluating on $m \otimes n$ and $m \otimes n'$, respectively. The displayed formulas also show that the resulting maps have the required module linearity. $\square$

Proposition 9.13 (Commutative-ring Tensor–Hom)**.** If R is commutative and M, N, P are R-modules, then

$$\mathrm{Hom}_R(M \otimes_R N, P) \cong \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P)) \cong \mathrm{Hom}_R(N, \mathrm{Hom}_R(M, P))\,.$$

Proof. An R-linear map $M \otimes_R N \to P$ is the same, by the tensor-product universal property, as an R-balanced bilinear map $M \times N \to P$. Sending such a map b to

$$m \longmapsto (n \longmapsto b(m, n))$$

gives the first isomorphism, with inverse obtained by evaluation. Switching the two variables gives the second. Commutativity is what identifies the two sidedness conventions and makes each Hom group an R-module. $\square$

Remark 9.14. The preceding compatibility identities are often summarized by saying that the Tensor–Hom correspondence is natural in its variables. The explicit sidedness in Theorem 9.11 remains essential when R is not commutative.

9.9 Right Exactness

Tensor product preserves cokernels, which is stronger than merely preserving surjections. Indeed, an exact sequence

$$A \longrightarrow B \longrightarrow C \longrightarrow 0$$

says that $C \cong B/\operatorname{im}(A)$, so tensoring preserves this quotient/cokernel even though it may destroy injectivity of $A \to B$.

Theorem 9.15 (Right exactness)**.** Let

$$A_R \xrightarrow{u} B_R \xrightarrow{v} C_R \longrightarrow 0$$

be exact, and let ${}_RN$ be a left R-module. Then

$$A \otimes_R N \xrightarrow{u\otimes 1} B \otimes_R N \xrightarrow{v\otimes 1} C \otimes_R N \longrightarrow 0$$

is exact.

Proof. We prove that $v \otimes 1$ is a cokernel of $u \otimes 1$. Let X be an abelian group and let

$$F : B \otimes_R N \longrightarrow X$$

satisfy $F(u \otimes 1) = 0$. Theorem 9.11 identifies F with

$$\widehat{F} : N \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(B, X).$$

By Proposition 9.12, the vanishing condition says

$$\widehat{F}(n) \circ u = 0$$

for every $n \in N$. Since C is the cokernel of u, each $\widehat{F}(n)$ factors uniquely as

$$\widehat{F}(n) = G_n \circ v$$

for a homomorphism $G_n : C \to X$. The assignment is additive because

$$(G_{n+n'} - G_n - G_{n'}) \circ v = 0,$$

so uniqueness of factorization through the cokernel gives

$$G_{n+n'} = G_n + G_{n'}.$$

For the left R-action on $\mathrm{Hom}_{\mathbb{Z}}(C, X)$,

$$(r \cdot G_n) \circ v = r \cdot (G_n \circ v) = r \cdot \widehat{F}(n) = \widehat{F}(rn).$$

Uniqueness similarly gives $G_{rn} = r \cdot G_n$. Therefore

$$n \longmapsto G_n$$

is an R-linear map $N \to \mathrm{Hom}_{\mathbb{Z}}(C, X)$. Applying Theorem 9.11 backward gives a unique

$$G : C \otimes_R N \longrightarrow X$$

with $G(v(b) \otimes n) = F(b \otimes n)$. Hence

$$F = G \circ (v \otimes 1),$$

and uniqueness follows because pure tensors generate. Thus $v \otimes 1$ is the cokernel of $u \otimes 1$, which is exactly the asserted exactness. □

Corollary 9.16 (Tensoring a quotient)**.** For a map $u : A_R \to B_R$ and a left R-module ${}_R N$, there is a canonical isomorphism

$$(B/\operatorname{im} u) \otimes_R N \cong (B \otimes_R N)/\operatorname{im}(u \otimes 1).$$

Proof. Apply Theorem 9.15 to the quotient map

$$B \longrightarrow B/\operatorname{im} u.$$

The two modules displayed are cokernels of the same map $u \otimes 1$, and the unique map induced by the quotient is the required canonical isomorphism. $\square$

No injectivity statement is implied. Tensoring

$$2\mathbb{Z} \hookrightarrow \mathbb{Z}$$

with $\mathbb{Z}/2\mathbb{Z}$ gives a zero map from a nonzero module. Later homological algebra measures this failure; the next definition isolates when the failure does not occur.

9.10 Flat Modules

Definition 9.17. A left R-module N is **flat** if tensoring with N takes every short exact sequence of right R-modules to a short exact sequence.

Equivalently, tensoring with N preserves injections, since Theorem 9.15 already gives right exactness.

Proposition 9.18. Every free module is flat, and every projective module is flat. Thus

$$\text{free} \Longrightarrow \text{projective} \Longrightarrow \text{flat}.$$

Proof. If F is free on X, then, for every right R-module A,

$$A \otimes_R F \cong \bigoplus_{x \in X} A.$$

Consequently tensoring a short exact sequence of right modules with F is a direct sum of short exact sequences and remains exact. Thus free modules are flat.

By Theorem 6.21, a projective module P is a direct summand of a free module $F = P \oplus P'$. Tensoring distributes over this finite direct sum, so a tensor map involving F is the direct sum of the corresponding tensor maps involving P and P'. If the first is injective, each summand map is injective; if the middle sequence is exact, exactness holds componentwise. Since F is flat, P is flat. $\square$

The converses need not hold in general.

9.11 Exterior Powers and Exterior Algebra

Finite tensor products represent multilinear maps. Exterior powers impose one further relation: a multilinear expression should vanish whenever two inputs agree. This packages alternating forms, oriented volume, and the determinant into one construction.

Throughout this section R is a commutative ring and M is an R-module. A multilinear map

$$f : M^k \longrightarrow N$$

is **alternating** if

$$f(m_1, \ldots, m_k) = 0$$

whenever two arguments coincide. This is the same alternating condition used for determinants in Section 6.12; it is not to be confused with skew-symmetry in characteristic 2.

Theorem 9.19 (Exterior-power universal property)**.** For every integer $k \geq 1$, there is an R-module

$$\bigwedge_R^k M$$

and an alternating multilinear map

$$\omega_k : M^k \longrightarrow \bigwedge_R^k M, \qquad (m_1, \ldots, m_k) \longmapsto m_1 \wedge \cdots \wedge m_k,$$

such that every alternating multilinear map

$$f : M^k \longrightarrow N$$

has a unique linear factorization

$$\widetilde{f} : \bigwedge_R^k M \longrightarrow N.$$

$$\begin{tikzcd}
M^k \arrow[r, "\omega_k"] \arrow[dr, "f"'] & \bigwedge_R^k M \arrow[d, "\widetilde{f}"] \\
& N
\end{tikzcd}$$

Proof. By Theorem 9.8, form the finite tensor power

$$M^{\otimes k} = M \otimes_R \cdots \otimes_R M.$$

Let J be the submodule generated by all pure tensors having two equal entries, and define

$$\bigwedge_R^k M = M^{\otimes k}/J.$$

The quotient of the universal multilinear tensor map is ω_k, and it is alternating by construction. If f is alternating, its associated linear map

$$M^{\otimes k} \longrightarrow N$$

vanishes on every generator of J, so it descends uniquely through the quotient. This gives the required factorization. Any two modules with this property are canonically isomorphic by applying their universal properties to one another's universal alternating maps. □

Proposition 9.20 (Alternating wedge identities)**.** If two entries agree, then

$$m_1 \wedge \cdots \wedge m_k = 0.$$

Interchanging two entries changes sign; consequently

$$m_{\sigma(1)} \wedge \cdots \wedge m_{\sigma(k)} = \operatorname{sgn}(\sigma)\, m_1 \wedge \cdots \wedge m_k$$

for every $\sigma \in S_k$.

Proof. The first assertion is part of the construction. Holding all other entries fixed, alternation and multilinearity give

$$0 = \omega_k(\ldots, x + y, \ldots, x + y, \ldots).$$

The repeated-x and repeated-y terms vanish, leaving the sum of the two wedges with x and y interchanged. Thus a transposition contributes -1. The sign theory of Chapter 2 then gives the formula for an arbitrary permutation. □

Example. If $M = R^3$ has basis e_1, e_2, e_3, then

$$\bigwedge_R^2 M$$

has basis

$$e_1 \wedge e_2, \qquad e_1 \wedge e_3, \qquad e_2 \wedge e_3.$$

Terms $e_i \wedge e_i$ vanish, and reversing two basis vectors changes the sign, so these are precisely the increasing-index wedges.

Theorem 9.21 (Basis of an exterior power)**.** If M is finite free with ordered basis

$$(e_1, \dots, e_n),$$

then the wedges

$$e_{i_1} \wedge \cdots \wedge e_{i_k}, \qquad 1 \le i_1 < \cdots < i_k \le n,$$

form a basis of

$$\bigwedge_R^k M.$$

Hence

$$\operatorname{rank}_R(\bigwedge_R^k M) = \binom{n}{k}.$$

Proof. Multilinearity and Proposition 9.20 show that these wedges span: expand every argument in the basis, discard repeated-index terms, and reorder the rest. To prove independence, fix an increasing k-tuple

$$I = (i_1, \dots, i_k).$$

For

$$m_j = \sum_r a_{rj} e_r,$$

define

$$f_I(m_1, \dots, m_k) = \det(a_{i_r j})_{1 \le r, j \le k}.$$

The determinant theory of Section 6.12 shows that f_I is alternating and multilinear, so it induces a linear map from $\bigwedge_R^k M$ to R. On the displayed wedges it has value 1 for the tuple I and 0 for every other increasing tuple. Applying these maps to a linear relation proves that every coefficient is zero. □

We set

$$\bigwedge_R^0 M = R, \qquad \bigwedge_R^1 M = M.$$

The basis theorem gives

$$\bigwedge_R^k M = 0 \qquad (k > n)$$

when M is finite free of rank n, and it shows that

$$\bigwedge_R^n M$$

is free of rank one.

Proposition 9.22 (Induced exterior maps)**.** An R-linear map

$$T : M \longrightarrow N$$

induces a unique linear map

$$\bigwedge^k T : \bigwedge_R^k M \longrightarrow \bigwedge_R^k N$$

satisfying

$$\bigwedge^k T(m_1 \wedge \cdots \wedge m_k) = Tm_1 \wedge \cdots \wedge Tm_k.$$

Moreover,

$$\bigwedge^k(\mathrm{id}_M) = \mathrm{id}_{\bigwedge_R^k M}, \qquad \bigwedge^k(S \circ T) = (\bigwedge^k S) \circ (\bigwedge^k T).$$

Proof. The map

$$(m_1, \ldots, m_k) \longmapsto Tm_1 \wedge \cdots \wedge Tm_k$$

is alternating and multilinear. By Theorem 9.19, it therefore induces a unique map. The two displayed identities agree on every pure wedge, and pure wedges generate the relevant exterior powers. $\square$

Theorem 9.23 (Top exterior action)**.** Let V be n-dimensional over F, and let $T : V \to V$. Then

$$\bigwedge^n T$$

acts on the one-dimensional space

$$\bigwedge_F^n V$$

by multiplication by $\det(T)$. Equivalently,

$$Tv_1 \wedge \cdots \wedge Tv_n = \det(T)\,(v_1 \wedge \cdots \wedge v_n).$$

Proof. Fix an ordered basis $(v_1, \ldots, v_n)$. The linear functional on $\bigwedge_F^n V$ that takes $v_1 \wedge \cdots \wedge v_n$ to 1 corresponds, by the exterior universal property, to a normalized alternating n-linear form on V. By the uniqueness theorem for determinants in Section 6.12, its value on

$$(Tv_1, \ldots, Tv_n)$$

is $\det(T)$. Since the top exterior power is one-dimensional, this is exactly the scalar by which $\bigwedge^n T$ acts. $\square$

This gives a later, equivalent characterization of the determinant already developed from normalized alternating multilinear forms in Section 6.12. Since

$$\dim_F \bigwedge_F^n V = 1,$$

the determinant of T is the unique scalar $\lambda \in F$ such that

$$\bigwedge^n T = \lambda \operatorname{id}_{\bigwedge_F^n V}.$$

Equivalently, for every ordered basis $(v_1, \ldots, v_n)$ of V,

$$Tv_1 \wedge \cdots \wedge Tv_n = \det(T)\big(v_1 \wedge \cdots \wedge v_n\big).$$

Thus this top-exterior-power viewpoint describes the same invariant as the Leibniz and alternating-multilinear descriptions, rather than defining a new determinant.

For $A \in M_n(F)$, regarded as an endomorphism of F^n, the matrix of

$$\bigwedge^n A$$

with respect to the one-element basis

$$e_1 \wedge \cdots \wedge e_n$$

of $\bigwedge_F^n F^n$ is the 1×1 matrix

$$(\det A).$$

Moreover, the composition law proved in Proposition 9.22 makes determinant multiplicativity conceptually immediate:

$$\bigwedge^n (S \circ T) = (\bigwedge^n S) \circ (\bigwedge^n T)$$

implies

$$\det(S \circ T) = \det(S)\det(T).$$

This is a retrospective explanation of the multiplicativity already proved in Chapter 6, not a replacement for that earlier proof.

9.12 The Exterior Algebra and Alternating Forms

The individual exterior powers fit together into a graded algebra. This uses an arbitrary direct sum of already constructed modules; it does **not** introduce an infinite tensor product.

Definition 9.24. The **exterior algebra** of M is

$$\bigwedge M = \bigoplus_{k\geq 0} \bigwedge_R^k M.$$

For pure wedges, define the product by concatenation:

$$\begin{aligned}(m_1 \wedge \cdots \wedge m_p) \wedge (n_1 \wedge \cdots \wedge n_q)\\ = m_1 \wedge \cdots \wedge m_p \wedge n_1 \wedge \cdots \wedge n_q.\end{aligned}$$

The right-hand side is alternating separately in the m's and in the n's, so the two exterior universal properties make this a well-defined bilinear map

$$\bigwedge_R^p M \times \bigwedge_R^q M \longrightarrow \bigwedge_R^{p+q} M.$$

Pure wedges generate each factor, so associativity follows from associativity of concatenation on pure wedges. The degree-k summand is the k-th graded piece.

Proposition 9.25 (Graded commutativity)**.** For homogeneous elements

$$\alpha \in \bigwedge_R^p M, \qquad \beta \in \bigwedge_R^q M,$$

one has

$$\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha.$$

Proof. For pure wedges, moving the block of p entries past the block of q entries requires pq transpositions. Proposition 9.20 gives the sign. Bilinearity extends the identity to arbitrary homogeneous elements. The argument uses alternation itself, so it remains valid in characteristic 2, where the displayed sign may equal 1. □

Theorem 9.26 (Alternating forms as exterior powers of the dual)**.** Let V be a finite-dimensional vector space over F. There is a natural isomorphism from

$$\bigwedge_F^k V^\vee$$

onto the space of alternating k-linear maps

$$V^k \longrightarrow F.$$

On a decomposable element it is given by

$$\varphi_1 \wedge \cdots \wedge \varphi_k \longmapsto \big((v_1, \ldots, v_k) \mapsto \det(\varphi_i(v_j))_{1 \le i,j \le k}\big).$$

Proof. The displayed determinant is alternating and multilinear in the vectors, so the exterior universal property makes the assignment well-defined and linear. If $(v_1, \ldots, v_n)$ is a basis of V, then the wedges of its dual basis indexed by increasing k-tuples map to the corresponding coordinate alternating forms. These forms are linearly independent and span by evaluating an alternating form on basis tuples. Thus the map is an isomorphism. $\square$

This identifies the algebra developed here with the pointwise algebra behind differential forms. On a smooth manifold, a differential k-form assigns to each point p an alternating k-linear form on the tangent space

$$T_pM,$$

that is, an element of

$$\bigwedge^k (T_pM)^\vee.$$

The wedge product sends a p-form and a q-form at p to an element of

$$\bigwedge^{p+q} (T_pM)^\vee.$$

No manifold theory is needed here; the point is that exterior algebra is the linear-algebraic model that makes those geometric objects possible.

Exercises

Exercise 9.1 (Basic)**.** Verify that multiplication $R \times R \to R$ is balanced when the first copy is the regular right module and the second the regular left module.

Exercise 9.2 (Core)**.** Give an (R, R)-bimodule structure on R^n, and verify the induced actions of Proposition 9.4.

Exercise 9.3 (Core)**.** Construct explicitly the associativity isomorphism for $\mathbb{Z}/2\mathbb{Z}$, $\mathbb{Z}/3\mathbb{Z}$, and $\mathbb{Z}/4\mathbb{Z}$.

Exercise 9.4 (Core)**.** Prove the second arbitrary-direct-sum isomorphism in Theorem 9.6 directly.

Exercise 9.5 (Core)**.** Compute $(R/(a)) \otimes_R (R/(b))$ for a commutative ring R.

Exercise 9.6 (Further)**.** Verify the universal property of the three-factor tensor product directly from the quotient construction.

Exercise 9.7 (Further, Challenging)**.** Show that tensoring the inclusion $2\mathbb{Z} \hookrightarrow \mathbb{Z}$ with $\mathbb{Z}/2\mathbb{Z}$ is not injective.

Exercise 9.8 (Core)**.** If M is free with basis e_1, e_2, e_3, give a basis for

$$\bigwedge_R^2 M.$$

For a diagonal endomorphism of F^3, compute its induced map on this exterior power and recover its determinant from the top exterior power.

Part IV

Fields and Galois Theory

Chapter 10

Field Extensions

Many equations require a larger field before they can be solved: $\mathbb{Q}(\sqrt{2})$ contains a root of x^2-2, and $\mathbb{C}$ contains a root of x^2+1 over $\mathbb{R}$. Extensions also let us place several related quantities in a single field and, later, enlarge the scalars on which a vector space is considered. This chapter makes adjoining elements precise and establishes the results on which separability and Galois theory later depend.

10.1 Extensions and Degree

The vector-space language developed in Chapter 6 is therefore immediately available: once $F \subseteq E$, scalar multiplication by elements of F makes E into an F-vector space, and degree is simply its dimension. For instance, the common field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is where the two square roots can be added and multiplied while still retaining $\mathbb{Q}$ as the base field.

Definition 10.1. An **extension field** E/F, written $F \subseteq E$, is a field E containing F as a subfield. Then E is an F-vector space. Its **degree** is

$$[E : F] = \dim_F E.$$

The extension is **finite** when this dimension is finite. If

$$F \subseteq K \subseteq E,$$

then K is an **intermediate field** of E/F.

This definition deliberately reuses familiar linear algebra. The field operations in E are richer than vector-space operations, but the smaller field F may still be used as a field of scalars. Thus the degree measures how many independent F-directions must be added in passing from F to E. For example,

$$[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2, \qquad [\mathbb{C} : \mathbb{R}] = 2,$$

with bases $1, \sqrt{2}$ and $1, i$, respectively. For elements in a common ambient field, $F(S)$ denotes the smallest subfield containing $F \cup S$. The inclusion $\mathbb{Q} \subseteq \mathbb{R}$, by contrast, has infinite degree: no finite list of real numbers spans all of $\mathbb{R}$ over $\mathbb{Q}$. The degree is therefore a genuine measure of the amount of new scalar information in the extension, not merely a count of symbols. The infinite extension $F(t)/F$, formed by adjoining an indeterminate, will appear in §10.2; its powers $1, t, t^2, \ldots$ cannot fit into any finite basis.

The Tower Law is the numerical bookkeeping principle for successive adjunctions. It says that a basis for the top field is obtained by first choosing coordinates over the middle field and then expanding each of those coordinates over the base field.

Theorem 10.2 (Tower Law)**.** If

$$F \subseteq E \subseteq L$$

and $[E : F]$, $[L : E]$ are finite, then

$$[L : F] = [L : E][E : F].$$

Proof. Let $e_1, \ldots, e_m$ be an F-basis of E and $u_1, \ldots, u_n$ an E-basis of L. The products $e_i u_j$ span: expand first in the u_j, then expand each coefficient in the e_i. If

$$\sum_{i,j} c_{ij} e_i u_j = 0,$$

then grouping by u_j and using their independence gives

$$\sum_{i} c_{ij} e_i = 0$$

for each j. Independence of the e_i makes all c_{ij} zero. Thus the products are a basis of size mn. □

Corollary 10.3. If $F \subseteq E \subseteq L$ and $[L : F]$ is finite, then $[E : F]$ divides $[L : F]$. In particular, a prime-degree extension has no proper intermediate field.

Proof. Apply the Tower Law and factor the prime integer $[L : F]$. □

10.2 Algebraic and Transcendental Elements

Definition 10.4. For an extension E/F and $\alpha \in E$, the element α is **algebraic over** F if $f(\alpha) = 0$ for some nonzero $f \in F[x]$; otherwise it is **transcendental**. The extension is **algebraic** when each of its elements is algebraic.

The evaluation homomorphism

$$\mathrm{ev}_\alpha : F[x] \longrightarrow E, \qquad f \longmapsto f(\alpha)$$

collects all relations on α. If α is algebraic, its nonzero kernel has a unique monic generator, the **minimal polynomial** $m_{\alpha,F}$.

Thus the minimal polynomial is not an arbitrary equation satisfied by α: it is the irreducible relation of least degree, and it records exactly the algebraic information lost when the indeterminate x is replaced by α. The next proposition explains why every other polynomial relation is a consequence of this one.

Proposition 10.5 (Minimal-polynomial properties)**.** The minimal polynomial $m_{\alpha,F}$ is irreducible and divides every polynomial in $F[x]$ which vanishes at α.

Proof. If $m = gh$, then $g(\alpha)h(\alpha) = 0$, so one factor belongs to the evaluation kernel. Minimal positive degree forces the other factor to be a unit. For a polynomial f vanishing at α, divide by m:

$$f = qm + r, \qquad \deg r < \deg m.$$

Then $r(\alpha) = 0$, so $r = 0$ by minimality. $\square$

Definition 10.6. $F[\alpha]$ is the subring consisting of polynomial expressions in α, while $F(\alpha)$ is the smallest subfield containing F and α.

Theorem 10.7 (Simple algebraic extensions)**.** If α is algebraic and $d = \deg m_{\alpha,F}$, then

$$F[\alpha] \cong F[x]/(m_{\alpha,F}) = F(\alpha).$$

The elements

$$1, \alpha, \dots, \alpha^{d-1}$$

are an F-basis of $F(\alpha)$, so

$$[F(\alpha) : F] = d.$$

Proof. Evaluation and the First Isomorphism Theorem give the quotient. Since the minimal polynomial is irreducible, that quotient is a field, and therefore $F[\alpha] = F(\alpha)$. Division by the minimal polynomial gives spanning by the displayed powers. A relation among them is a smaller nonzero polynomial in the evaluation kernel, which is impossible. □

Example 10.8 (Two square roots)**.** The simple-extension theorem now lets us complete the calculation suggested in §10.1. The polynomial $x^2 - 3$ has no root in $\mathbb{Q}(\sqrt{2})$. Indeed, every element of this field has the form $a + b\sqrt{2}$ with $a, b \in \mathbb{Q}$. If

$$(a + b\sqrt{2})^2 = 3,$$

comparison of the rational and $\sqrt{2}$ parts gives

$$a^2 + 2b^2 = 3, \qquad 2ab = 0.$$

If $a = 0$, then $b^2 = 3/2$, impossible in $\mathbb{Q}$; if $b = 0$, then $a^2 = 3$, also impossible in $\mathbb{Q}$. Thus $x^2 - 3$ is irreducible over $\mathbb{Q}(\sqrt{2})$, so it is the minimal polynomial of $\sqrt{3}$ over that field. The theorem gives

$$[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})] = 2.$$

The Tower Law therefore yields

$$[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 2 \cdot 2 = 4.$$

This is the first synthesis of irreducibility, minimal polynomials, simple extension degree, and the Tower Law. A second square root increases the degree precisely when it was absent from the first extension.

Proposition 10.9 (The transcendental case)**.** If t is transcendental over F, then

$$F[t] \cong F[x], \qquad F(t) \cong F(x).$$

In particular $F(t)/F$ is infinite.

Proof. The evaluation kernel is zero. Passing from the polynomial ring to its field of fractions gives the second isomorphism, and the powers of t are linearly independent over F. □

The algebraic and transcendental cases have opposite shapes. An algebraic element has a power that can be reduced to a combination of finitely many earlier powers, whereas the powers of a transcendental element never satisfy such a relation. Passing from $F[t]$ to $F(t)$ adds fractions, just as passing from $\mathbb{Z}$ to $\mathbb{Q}$ adds quotients; it does not create a polynomial relation for t.

	algebraic α	transcendental t
polynomial relations	generated by $m_{\alpha,F}$	none
associated field	$F(\alpha) \cong F[x]/(m_{\alpha,F})$	$F(t) \cong F(x)$
degree over F	finite	infinite

Example 10.10. Eisenstein's criterion makes x^3-2 the minimal polynomial of $\sqrt[3]{2}$ over $\mathbb{Q}$. Thus

$$\mathbb{Q}(\sqrt[3]{2}) = \mathbb{Q} \oplus \mathbb{Q}\sqrt[3]{2} \oplus \mathbb{Q}\sqrt[3]{4}.$$

In contrast, $F(x)$ contains rational functions not in $F[x]$.

Definition 10.11. An extension is **finitely generated** if it has the form

$$F(\alpha_1, \ldots, \alpha_r).$$

Proposition 10.12. Finite extensions are algebraic, finitely generated algebraic extensions are finite, and algebraicity is transitive through towers.

Proof. First suppose $[E : F] = n$, and let $a \in E$. The $n+1$ elements

$$1, a, \ldots, a^n$$

are linearly dependent over F. Their dependence relation is a nonzero polynomial over F which kills a, proving that E/F is algebraic.

Next let

$$E = F(\alpha_1, \ldots, \alpha_r)$$

with every α_i algebraic over F. Each α_i remains algebraic over the intermediate field already constructed, because its polynomial over F is also a polynomial over that field. Thus every successive extension is finite by Theorem 10.7; repeated use of the Tower Law makes E/F finite.

Finally suppose $F \subseteq E \subseteq L$, E/F is algebraic, and $\alpha \in L$ is algebraic over E. Write a monic relation

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_0 = 0, \qquad a_i \in E,$$

and put

$$K = F(a_0, \ldots, a_{n-1}).$$

The elements a_i are algebraic over F, so K/F is finite by the second part. The displayed relation shows that α is algebraic over K; hence $K(\alpha)/K$ is finite. The Tower Law gives

$$[K(\alpha) : F] < \infty.$$

Its first part now says that α is algebraic over F. This proves transitivity. □

10.3 Root Adjunction and Splitting Fields

When an irreducible polynomial has no root in the current field, this is not an obstruction to doing algebra with a root. The quotient construction below creates the smallest possible field in which a formal symbol is required to satisfy that polynomial.

Theorem 10.13 (Root adjunction)**.** If $f \in F[x]$ is irreducible, then

$$F[x]/(f)$$

is a field containing F, and $x + (f)$ is a root of f. Its minimal polynomial is f, and its degree over F is $\deg f$.

Proof. The quotient is the formal way to impose the single relation $f(x) = 0$. Because f is irreducible, the irreducible-quotient theorem of Chapter 5 makes the principal ideal (f) maximal; hence the quotient is a field rather than merely a ring. Constants embed as their residue classes, and the class

$$\alpha = x + (f)$$

is the formal adjoined root since $f(\alpha) = f(x) + (f) = 0$. The evaluation kernel is (f); the remaining claims follow from Theorem 10.7. □

Example 10.14 (A root made concrete)**.** Taking $f = x^2 - 2 \in \mathbb{Q}[x]$ gives

$$\mathbb{Q}[x]/(x^2 - 2).$$

If $\alpha = x + (x^2 - 2)$, then $\alpha^2 = 2$, and every residue class has a unique form

$$a + b\alpha, \qquad a, b \in \mathbb{Q}.$$

For instance,

$$(a + b\alpha)(c + d\alpha) = (ac + 2bd) + (ad + bc)\alpha.$$

Sending α to the usual real number $\sqrt{2}$ identifies this quotient with $\mathbb{Q}(\sqrt{2})$. The quotient construction is useful because it works before a convenient ambient field, such as $\mathbb{R}$, has been chosen.

Lemma 10.15 (Extension of embeddings)**.** Let $\sigma : K \to \Omega$ be an F-embedding. Let α be algebraic over K, with minimal polynomial m. Extensions

$$\widetilde{\sigma} : K(\alpha) \longrightarrow \Omega$$

of σ correspond bijectively to roots in Ω of the coefficientwise image $\sigma(m)$.

Proof. Write

$$m(x) = a_0 + a_1 x + \cdots + a_n x^n, \qquad \sigma(m)(x) = \sigma(a_0) + \sigma(a_1)x + \cdots + \sigma(a_n)x^n.$$

For a root β of this transported polynomial, the map

$$\sum a_i x^i \longmapsto \sum \sigma(a_i)\beta^i$$

has m in its kernel and descends to a homomorphism from

$$K[x]/(m) \cong K(\alpha)$$

to Ω. This is well-defined precisely because the defining relation $m(\alpha) = 0$ is respected by the chosen root β. The source is a field, so it is injective. It is uniquely determined by its values on K and α. Conversely an extension sends $m(\alpha) = 0$ to

$$\sigma(m)(\widetilde{\sigma}(\alpha)) = 0.$$

□

Definition 10.16. A **splitting field** of a nonconstant polynomial $f \in F[x]$ is an extension generated over F by the roots of f, in which f splits into linear factors.

Theorem 10.17 (Splitting fields)**.** Every nonconstant polynomial in $F[x]$ has a splitting field, unique up to an F-isomorphism.

Proof. For existence, induct on $n = \deg f$. There is nothing to prove for $n = 1$. Otherwise choose an irreducible factor p of f, and use Theorem 10.13 to obtain a field K/F containing a root α of p. In $K[x]$, division gives

$$f(x) = (x - \alpha)g(x), \qquad \deg g = n - 1.$$

By induction, g has a splitting field L/K. Then f splits in L, and L is generated over F by α together with the roots of g, which are precisely all roots of f.

For uniqueness, let $L = F(\alpha_1, \ldots, \alpha_r)$ and L' be splitting fields of f. Start with $1_F : F \to L'$. Suppose that an embedding has been defined on $F(\alpha_1, \ldots, \alpha_{j-1})$. The minimal polynomial of α_j over this field divides f, after the coefficients have been transported. Since f splits in L', that transported minimal polynomial has a root in L'. Lemma 10.15 extends the embedding over α_j.

After the last step we have an F-embedding $\sigma : L \to L'$. If

$$f(x) = c\prod_i (x - \alpha_i)$$

in $L[x]$, then applying σ, which fixes the coefficients of f, gives

$$f(x) = c\prod_i (x - \sigma(\alpha_i))$$

in $L'[x]$. Thus the images are the complete roots of f, with multiplicities. Since L' is generated by those roots, it lies in $\sigma(L)$. Therefore σ is onto and is an isomorphism. ☐

Example 10.18. The splitting field of $x^2 - 2$ is $\mathbb{Q}(\sqrt{2})$. The three roots of $x^3 - 2$ are

$$\sqrt[3]{2}, \qquad \zeta_3\sqrt[3]{2}, \qquad \zeta_3^2\sqrt[3]{2},$$

so its splitting field is

$$\mathbb{Q}(\sqrt[3]{2}, \zeta_3).$$

The first adjunction supplies one real root, but it does not supply the two conjugate nonreal roots. The primitive cube root ζ_3 is precisely what rotates $\sqrt[3]{2}$ into those roots, so adjoining it produces the whole splitting field. This is the first indication that the symmetries of all roots together, rather than a single chosen root, are what Galois theory will study. The polynomial $(x-1)(x+2)$ already splits over $\mathbb{Q}$, so its splitting field is $\mathbb{Q}$. In characteristic p, the polynomial $x^p - x$ splits in $\mathbb{F}_p$, because each $a \in \mathbb{F}_p$ satisfies $a^p = a$, and it has degree p.

10.4 Algebraically Closed Fields and Embeddings

Definition 10.19. A field Ω is **algebraically closed** if every nonconstant polynomial in $\Omega[x]$ has a root in Ω.

Proposition 10.20. A field is algebraically closed if and only if every nonconstant polynomial splits into linear factors.

Proof. Repeatedly factor out a root and induct on degree. The converse is immediate. ☐

Definition 10.21. An F**-embedding** $\sigma : E \to \Omega$ is a field homomorphism between extensions of F that fixes F pointwise. Its images of an algebraic element are called its **conjugates**.

Proposition 10.22 (Embeddings and conjugates)**.** If Ω is algebraically closed and α is algebraic over F, then

$$\sigma \longmapsto \sigma(\alpha)$$

is a bijection from F-embeddings

$$F(\alpha) \longrightarrow \Omega$$

to roots of $m_{\alpha,F}$ in Ω.

Proof. An embedding sends the equation $m_{\alpha,F}(\alpha) = 0$ to the required root relation. Thus every image is a root of the minimal polynomial; these are exactly the possible algebraic realizations of α compatible with the base field. Conversely, a root gives the unique embedding by Lemma 10.15. The two constructions are inverse because an embedding of $F(\alpha)$ is determined by its value on α. □

Example 10.23. The conjugates of $\sqrt{2}$ over $\mathbb{Q}$ are

$$\sqrt{2}, \qquad -\sqrt{2}.$$

There are correspondingly two $\mathbb{Q}$-embeddings of $\mathbb{Q}(\sqrt{2})$ into $\mathbb{C}$. The conjugates of $\sqrt[3]{2}$ are its three complex cube roots; Chapter 11 will explain why the number of distinct conjugates can be smaller than the degree in positive characteristic.

More explicitly, the three $\mathbb{Q}$-embeddings of $\mathbb{Q}(\sqrt[3]{2})$ into $\mathbb{C}$ are determined by the three possible images

$$\sqrt[3]{2}, \qquad \zeta_3\sqrt[3]{2}, \qquad \zeta_3^2\sqrt[3]{2}.$$

Only the first has image equal to the original real cubic field. The other two land in different conjugate subfields of its splitting field. This is why embeddings are the right preliminary notion: automorphisms are the special embeddings whose image remains in the same field.

The next extension theorem is the standard place where Zorn's Lemma enters field theory. Start with every partial extension of the prescribed embedding, order these choices by further extension, take the union along a chain, and observe that a maximal partial embedding can be enlarged unless its domain is already all of E. The proof records the details because this pattern will return when algebraic closures and normal extensions are compared.

Theorem 10.24 (Extension of embeddings)**.** If E/K is algebraic, Ω is algebraically closed, and $\sigma : K \to \Omega$ is an F-embedding, then σ extends to an F-embedding $E \to \Omega$.

Proof. Let $\mathcal{P}$ consist of pairs (M, τ) with

$$K \subseteq M \subseteq E$$

and $\tau : M \to \Omega$ an F-embedding extending σ. Define

$$(M, \tau) \leq (N, \rho)$$

when $M \subseteq N$ and $\rho|_M = \tau$. This is a set: the possible domains are subsets of the fixed set E, and possible maps are subsets of $E \times \Omega$.

Let $\mathcal{C}$ be a chain. Its domains have union

$$M_{\mathcal{C}} = \bigcup_{(M,\tau)\in\mathcal{C}} M.$$

For two elements of this union, one chain member contains both, so it contains their sum, product, and inverses. Hence the union is a field. The maps agree on overlaps because the pairs are comparable; their union is therefore a well-defined F-embedding $\tau_{\mathcal{C}} : M_{\mathcal{C}} \to \Omega$. It is an upper bound. Zorn's Lemma supplies a maximal pair (M, τ).

If $M \neq E$, choose $\alpha \in E \setminus M$. Since E/M is algebraic, α has a minimal polynomial over M. The coefficientwise image of that polynomial has a root in the algebraically closed field Ω, and Lemma 10.15 extends τ to $M(\alpha)$, contradicting maximality. Thus $M = E$. $\square$

10.5 Algebraic Closures

Definition 10.25. An **algebraic closure** $\overline{F}/F$ is an algebraic extension $\overline{F}$ that is algebraically closed.

An algebraic closure is a sufficiently large, but still algebraic, universe in which every polynomial over the starting field can be completely split. The construction below first enlarges a field so that every current polynomial has a root, then repeats, and finally retains the elements algebraic over the original field. That final restriction is what keeps the closure algebraic over F.

Theorem 10.26 (Existence of algebraic closures)**.** Every field has an algebraic closure.

Proof. We use a construction that works within ordinary sets throughout. For any field K, let S_K be the set of nonconstant polynomials in $K[x]$. Introduce one indeterminate X_f for each $f \in S_K$, and form

$$R_K = K[X_f : f \in S_K].$$

Let I_K be the ideal generated by the elements

$$f(X_f), \qquad f \in S_K.$$

This ideal is proper. Indeed, any alleged relation $1 \in I_K$ involves only finitely many variables $X_{f_1}, \dots, X_{f_r}$. Adjoin roots one at a time by Theorem 10.13; in the resulting field every $f_i(X_{f_i})$ vanishes, while 1 does not. Thus no such relation can exist.

By the maximal-ideal theorem, choose a maximal ideal $\mathfrak{m}_K \supseteq I_K$, and set

$$K^+ = R_K/\mathfrak{m}_K.$$

The quotient is a field containing a copy of K: the ideal $\mathfrak{m}_K \cap K$ is an ideal of the field K, and it cannot equal K, since then $1 \in \mathfrak{m}_K$. Hence

$$\mathfrak{m}_K \cap K = (0),$$

so the composite $K \to R_K \to K^+$ is injective. Every nonconstant polynomial $f \in K[x]$ has the root $X_f + \mathfrak{m}_K$ in K^+.

Starting with $F_0 = F$, recursively construct

$$F_0 \subseteq F_1 \subseteq F_2 \subseteq \cdots, \qquad F_{n+1} = F_n^+,$$

and set

$$\Omega = \bigcup_{n \geq 0} F_n.$$

This union is a field. If $g \in \Omega[x]$ is nonconstant, its finitely many coefficients lie in some F_n, and then g has a root in $F_{n+1} \subseteq \Omega$. Hence Ω is algebraically closed.

Finally let $\overline{F}$ be the subfield of Ω consisting of elements algebraic over F. This set is a field: if α, β are algebraic over F, then $F(\alpha, \beta)/F$ is finite. Hence

$$\alpha + \beta, \qquad \alpha - \beta, \qquad \alpha\beta, \qquad \alpha^{-1} \quad (\alpha \neq 0)$$

belong to a finite extension of F, and are therefore algebraic over F. If $h \in \overline{F}[x]$ is nonconstant, its coefficients lie in a finite algebraic extension K/F inside $\overline{F}$. A root $b \in \Omega$ of h is algebraic over K, hence algebraic over F; so $b \in \overline{F}$. Therefore $\overline{F}$ is algebraically closed and is an algebraic closure of F. $\square$

Theorem 10.27 (Uniqueness of algebraic closures)**.** Any two algebraic closures of F are F-isomorphic.

Proof. For closures $\overline{F}, \overline{F}'$, Theorem 10.24 extends 1_F to an embedding

$$\sigma : \overline{F} \longrightarrow \overline{F}'.$$

The image is algebraically closed: if a polynomial has coefficients in $\sigma(\overline{F})$, transport its coefficients back through σ, take a root in $\overline{F}$, and transport that root forward. Now take $\beta \in \overline{F}'$. It is algebraic over $\sigma(\overline{F})$. Its irreducible minimal polynomial has a root in the algebraically closed field $\sigma(\overline{F})$, so it is linear. Hence $\beta \in \sigma(\overline{F})$. Therefore the image is all of $\overline{F}'$. □

Remark 10.28. An algebraic closure is thus unique only up to an isomorphism fixing F. There is usually no distinguished isomorphism between two constructions, so one fixes a chosen algebraic closure when it is useful to speak of all conjugates in a single ambient field.

Exercises

Exercise 10.1 (Basic)**.** Prove directly that

$$[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4.$$

Exercise 10.2 (Basic)**.** Find the minimal polynomial of $1 + \sqrt{2}$ over $\mathbb{Q}$.

Exercise 10.3 (Basic)**.** Show that $\mathbb{Q}(\sqrt{2}) = \mathbb{Q}(1 + \sqrt{2})$.

Exercise 10.4 (Basic)**.** Prove that $\mathbb{Q}(\sqrt{2}) \cap \mathbb{Q}(\sqrt{3}) = \mathbb{Q}$.

Exercise 10.5 (Basic)**.** Show that $F[t]$ is not a field when t is transcendental over F, whereas $F(t)$ is a field.

Exercise 10.6 (Core)**.** Show that if α has odd degree over F, then $F(\alpha^2) = F(\alpha)$.

Exercise 10.7 (Core)**.** Let α have minimal polynomial $x^4 + 2x^2 + 2$ over F. Write α^7 as an F-linear combination of $1, \alpha, \alpha^2, \alpha^3$.

Exercise 10.8 (Core)**.** Prove that every element of a finite extension E/F has minimal polynomial of degree at most $[E : F]$.

Exercise 10.9 (Core)**.** If $F \subseteq E \subseteq L$, with $[L : F] = 12$, list the possible degrees of E/F.

Exercise 10.10 (Core)**.** Construct the splitting field of $x^4 - 2$ over $\mathbb{Q}$ and determine its degree.

Exercise 10.11 (Core)**.** Prove that an intermediate field of a degree-six extension cannot have degree four over the base.

Exercise 10.12 (Further)**.** Show that the number of F-embeddings of $F(\alpha)$ into an algebraic closure is at most $\deg m_{\alpha,F}$.

Exercise 10.13 (Further)**.** Let $f \in F[x]$ be irreducible and let α, β be roots of f in an algebraically closed field. Use the root-extension lemma to construct an F-isomorphism

$$F(\alpha) \longrightarrow F(\beta).$$

Exercise 10.14 (Further)**.** Prove that a field which is algebraic over an algebraically closed field is that field itself.

Exercise 10.15 (Further)**.** Show that $\mathbb{C}$ is an algebraic closure of $\mathbb{R}$, using the Fundamental Theorem of Algebra as an external input. A Galois-theoretic proof of that theorem appears later in Chapter 12.

Exercise 10.16 (Looking Ahead)**.** Show that every finite extension of a finite field is a splitting field of a polynomial over that finite field.

Exercise 10.17 (Looking Ahead)**.** Let $f \in F[x]$ be irreducible. Show that every F-embedding of $F(\alpha)$ into an algebraic closure sends α to a root of f, and compare this with the automorphisms of a splitting field of f.

Chapter 11

Separability, Normality, and Finite Fields

Embeddings count genuinely different conjugate roots. Separability measures that distinction; normality says that no conjugate root escapes the field. Only after both ideas are clear will we define a Galois extension.

11.1 Separable Polynomials and Extensions

Definition 11.1. A nonconstant polynomial $f \in F[x]$ is **separable** if its roots in a splitting field are distinct. An algebraic element is **separable** over F when its minimal polynomial is separable; an algebraic extension is separable when each of its elements is separable.

The definition applies to arbitrary polynomials. For an irreducible polynomial it asks whether the possible conjugates of one root occur only once. This is the distinction needed for counting embeddings.

Proposition 11.2 (Derivative criterion)**.** For nonconstant $f \in F[x]$,

$$f \text{ has a repeated root in } \overline{F} \quad \Longleftrightarrow \quad \gcd(f, f') \neq 1.$$

For irreducible f, this is equivalent to $f' = 0$.

Proof. Write $f = (x-a)^r g$ in a splitting field, with $g(a) \neq 0$. The product rule gives

$$f' = (x-a)^{r-1}(rg + (x-a)g').$$

Thus a is repeated exactly when $r \geq 2$, equivalently when $f(a) = f'(a) = 0$. A common divisor of f and f' is therefore equivalent, after splitting, to a common linear factor, hence to a repeated root. For irreducible f, the gcd is either 1 or an associate of f; the latter can occur only if $f' = 0$. ∎

Remark 11.3. In characteristic $p > 0$, $f' = 0$ exactly when every exponent appearing in f is divisible by p, so

$$f(x) = g(x^p).$$

This is the source of inseparability. For example,

$$x^p - t \in \mathbb{F}_p(t)[x]$$

has derivative zero and is irreducible because a rational function whose p-th power is t would have every zero and pole order divisible by p, whereas t has a simple zero. In an algebraic closure it is

$$x^p - t = (x - t^{1/p})^p;$$

it has one distinct root, with multiplicity p.

Example 11.4 (A separable polynomial with repeated-looking coefficients)**.** Over $\mathbb{F}_2$, the polynomial

$$x^3 + x + 1$$

has derivative x^2+1. It has no root in $\mathbb{F}_2$, hence is irreducible, and it cannot share a root with its derivative. It is therefore separable. The contrast with $x^2 - t$ over $\mathbb{F}_2(t)$, whose derivative is zero, shows that characteristic alone is not the issue: inseparability comes from the special p-th-power shape.

11.2 Embeddings, Perfect Fields, and Primitive Elements

The next three ideas answer complementary questions about a finite extension. Separability asks whether the degree is fully visible through distinct conjugates; normality, introduced later, asks whether those conjugates remain inside the field; and the primitive element theorem says that a finite separable extension can often be described by one well-chosen generator. The comparison is useful to keep in view:

property	what it controls
separable	distinct embeddings and conjugate roots
normal	whether all conjugates stay in the extension
Galois	both conditions, hence internal symmetries

Definition 11.5. A field F is **perfect** if every algebraic extension of F is separable, equivalently if every irreducible polynomial in $F[x]$ is separable.

Proposition 11.6. Every characteristic-zero field is perfect. A field F of characteristic $p > 0$ is perfect if and only if Frobenius

$$F \longrightarrow F, \qquad a \longmapsto a^p$$

is surjective. In particular every finite field is perfect.

Proof. In characteristic zero a nonzero irreducible polynomial cannot have zero derivative, so the criterion proves separability. In characteristic p, an inseparable irreducible polynomial has derivative zero and is of the form $g(x^p)$. Surjectivity of Frobenius makes each coefficient a p-th power, so this is a p-th power polynomial, impossible for a nonconstant irreducible polynomial. Conversely, if a has no p-th root, then $x^p - a$ has derivative zero and an irreducible factor which is inseparable. Frobenius is injective on every field and hence bijective on a finite field. □

Before turning this count into a definition, notice the strategy: build the field one algebraic generator at a time. At each step an embedding has only as many possible extensions as the minimal polynomial has roots. The Tower Law then turns this local bound into a global one.

Theorem 11.7 (Embedding-count inequality)**.** For finite E/F,

$$\#\operatorname{Hom}_F(E, \overline{F}) \leq [E : F],$$

Proof. Choose generators

$$F = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_r = E, \qquad F_i = F_{i-1}(\alpha_i).$$

An embedding $F_{i-1} \to \overline{F}$ extends across α_i only by sending it to a root of the transported minimal polynomial. By the root-extension lemma, there are at most $[F_i : F_{i-1}]$ choices. Successively extending embeddings therefore gives

$$\#\operatorname{Hom}_F(E, \overline{F}) \leq \prod_i [F_i : F_{i-1}] = [E : F].$$

Equality at every step holds exactly when each transported minimal polynomial has its full number of distinct roots. We analyze that equality after introducing separable degree. □

Definition 11.8. For a finite extension, the number

$$[E : F]_s = \#\operatorname{Hom}_F(E, \overline{F})$$

is its **separable degree**. Thus

$$[E : F]_s \leq [E : F],$$

by Theorem 11.7.

The ordinary degree is multiplicative in a tower by the Tower Law. The following result says that the count of genuinely distinct embeddings has the same multiplicative behavior. Its proof is a refined version of the preceding counting argument: first choose an embedding of the middle field, then count its possible continuations.

Proposition 11.9 (Multiplicativity of separable degree)**.** If

$$F \subseteq K \subseteq E$$

is a tower of finite extensions, then

$$[E : F]_s = [E : K]_s[K : F]_s.$$

Proof. Choose a tower of simple extensions

$$K = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_r = E, \qquad K_i = K_{i-1}(\alpha_i),$$

and let d_i be the number of distinct roots of the minimal polynomial of α_i over K_{i-1}. The root-extension lemma shows that an embedding of K_{i-1} has exactly d_i extensions to K_i. Indeed, the transported minimal polynomial is obtained by applying the embedding to its coefficients. It has the same degree and its derivative is the transported derivative, so its gcd with its derivative has the same degree; hence it has exactly the same number d_i of distinct roots.

It follows that every F-embedding $\sigma : K \to \overline{F}$ has

$$\prod_{i=1}^{r} d_i = [E : K]_s$$

extensions to E. The equality with $[E : K]_s$ is obtained by applying the same step-by-step count beginning with the identity embedding of K. Counting first the possible restrictions to K, then their extensions, proves the formula. □

Theorem 11.10 (Separable degree criterion)**.** For a finite extension E/F,

$$[E : F]_s = [E : F]$$

if and only if E/F is separable.

Proof. Choose a simple generating tower as in the proof of Proposition 11.9. The embedding count there gives

$$[E:F]_s = \prod_i d_i, \qquad [E:F] = \prod_i [K_i : K_{i-1}].$$

If E/F is separable, the minimal polynomial of α_i over K_{i-1} divides its separable minimal polynomial over F, so it is separable. Thus $d_i = [K_i : K_{i-1}]$ for every i, proving equality.

Conversely, equality of the two products forces equality in every factor, so each α_i is separable over K_{i-1}. To see that every $\gamma \in E$ is separable over F, set $L = F(\gamma)$. By Proposition 11.9,

$$[E:F]_s = [E:L]_s[L:F]_s.$$

Both factors are bounded above by their ordinary degrees, while their product is $[E:F] = [E:L][L:F]$. Hence

$$[L:F]_s = [L:F].$$

For the simple extension $L = F(\gamma)$, this says precisely that the minimal polynomial of γ has its full number of distinct roots. Thus γ is separable. □

Proposition 11.11 (Separability in towers)**.** For a finite tower

$$F \subseteq K \subseteq E,$$

E/F is separable if and only if both E/K and K/F are separable. The same statement holds for arbitrary algebraic towers.

Proof. If E/F is separable, elements of K are already separable over F, and a minimal polynomial over K divides the corresponding separable minimal polynomial over F; hence both subextensions are separable. Conversely, if both are separable, the multiplicativity formula and Theorem 11.10 give

$$[E:F]_s = [E:K][K:F] = [E:F].$$

Thus E/F is separable. For arbitrary algebraic towers, apply the finite statement as follows. The forward direction is immediate for K/F, since $K \subseteq E$; and, for $\alpha \in E$, the minimal polynomial over K divides the separable minimal polynomial over F, proving that E/K is separable.

Conversely, suppose K/F and E/K are separable, and choose $\alpha \in E$. The minimal polynomial

$$m_{\alpha,K}(x)$$

has only finitely many coefficients. Let K_0 be the subfield of K generated over F by those coefficients. Then K_0/F is finite; it is separable because every element of the subfield $K_0 \subseteq K$ is already separable over F. The polynomial $m_{\alpha,K}$ lies in $K_0[x]$ and is separable because E/K is separable. Hence the minimal polynomial of α over K_0, which divides $m_{\alpha,K}$, is separable. Thus $K_0(\alpha)/K_0$ is finite separable. The finite tower statement, applied to

$$F \subseteq K_0 \subseteq K_0(\alpha),$$

shows that α is separable over F. Since α was arbitrary, E/F is separable. □

Lemma 11.12 (Cyclicity lemma)**.** Every finite subgroup of the multiplicative group of a field is cyclic.

Proof. Let G be such a subgroup and let m be its exponent. Every element of G is a root of $x^m - 1$, so $|G| \leq m$. Conversely, by the structure theorem for finite abelian groups, G has an element of order equal to its exponent m, hence $m \leq |G|$. Therefore $m = |G|$, and an element of order m generates G. □

The primitive element theorem is not merely a convenient notation theorem. It reduces a finite separable extension to a single minimal polynomial, so that its embeddings and conjugates can be studied through one element. Over an infinite field, the proof chooses a linear combination $\alpha + c\beta$ avoiding the finitely many values of c that would make two embeddings coincide; over a finite field, cyclicity of the multiplicative group supplies a generator instead.

Theorem 11.13 (Primitive element theorem)**.** Every finite separable extension is simple.

Proof. First suppose F is infinite and $E = F(\alpha, \beta)$. List the $[E : F]$ F-embeddings of E into an algebraic closure as σ_i, which exist by Theorem 11.10. For each pair of distinct images, the equality

$$\sigma_i(\alpha) + c\sigma_i(\beta) = \sigma_j(\alpha) + c\sigma_j(\beta)$$

excludes at most one value of $c \in F$. Choose c outside this finite set and put $\gamma = \alpha + c\beta$. The conjugates of γ are then $[E : F]$ distinct values. They are values of distinct restrictions of the σ_i to $F(\gamma)$: if two restrictions agreed, they would give the same displayed value. Hence

$$\# \operatorname{Hom}_F(F(\gamma), \overline{F}) \geq [E : F].$$

The embedding-count theorem gives the reverse bound by $[F(\gamma) : F]$, while the Tower Law gives $[F(\gamma) : F] \le [E : F]$. Thus equality holds throughout. Since $F(\gamma) \subseteq E$, the Tower Law gives $E = F(\gamma)$. Induction reduces finitely many generators to this case.

If F is finite, then $E^\times$ is a finite subgroup of a field and is cyclic by Lemma 11.12. A generator γ of $E^\times$ has $E = F(\gamma)$. □

Example 11.14 (A primitive element)**.** The extension

$$\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$$

is finite and separable. The proof above predicts a primitive element; here one may take $\gamma = \sqrt{2} + \sqrt{3}$. Since

$$\gamma^2 = 5 + 2\sqrt{6},$$

we recover $\sqrt{6}$, and then

$$\sqrt{2} = \frac{\gamma^2 - 1}{2\gamma}, \qquad \sqrt{3} = \gamma - \sqrt{2}.$$

Thus $\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2} + \sqrt{3})$.

11.3 Normal Extensions and Galois Extensions

Definition 11.15. An algebraic extension E/F is **normal** if every irreducible polynomial in $F[x]$ having one root in E splits over E. It is **Galois** if it is algebraic, normal, and separable.

The guiding slogan is:

normal means that no conjugate root escapes.

Thus normality is a condition on the **whole** field, whereas separability is a condition on the distinctness of roots. The normality criteria below translate this root-theoretic statement into the embedding language developed above.

Theorem 11.16 (Normality criteria)**.** For finite E/F, the following are equivalent:

1. E/F is normal;
2. every F-embedding $E \to \overline{F}$ has image E;

3. E is a splitting field over F of a collection of polynomials.

Proof. Assume first that E/F is normal. If $\sigma : E \to \overline{F}$ is an F-embedding and $\alpha \in E$, then $\sigma(\alpha)$ is a conjugate root of $m_{\alpha,F}$. Normality makes this root belong to E; hence $\sigma(E) \subseteq E$. Both fields have degree $[E : F]$ over F, so $\sigma(E) = E$.

Conversely, suppose every embedding has image E. If an irreducible $f \in F[x]$ has a root $\alpha \in E$, every root β of f gives an embedding $F(\alpha) \to \overline{F}$ by Chapter 10. Extend it to E by Theorem 10.24. Its image is E, so $\beta \in E$. Thus f splits in E, and E/F is normal.

Finally, suppose E is a splitting field over F. Every F-embedding of E fixes the defining coefficients and permutes the corresponding roots. Its image is consequently contained in E; equality follows from degrees, and the already proved implication shows that E/F is normal. If E/F is finite normal, choose generators $\alpha_1, \ldots, \alpha_r$. The splitting field in E of the finitely many minimal polynomials $m_{\alpha_i,F}$ contains every generator and is therefore E. $\square$

Example 11.17 (Normality and separability are independent)**.** $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is separable because characteristic-zero fields are perfect, but it is not normal: the two nonreal roots

$$\omega\sqrt[3]{2}, \qquad \omega^2\sqrt[3]{2}$$

of $x^3 - 2$ are absent. Its normal closure is obtained by adjoining a primitive cube root of unity ω. On the other hand, put

$$E = \mathbb{F}_p(t^{1/p}), \qquad F = \mathbb{F}_p(t).$$

For every $\alpha \in E$, the Frobenius identity gives

$$\alpha^p \in F.$$

Thus the minimal polynomial of α divides

$$x^p - \alpha^p = (x - \alpha)^p$$

in an algebraic closure and has only one distinct root. Every irreducible polynomial over F having a root in E consequently splits in E as a power of a linear factor, so E/F is normal. It is not separable, because the generator $t^{1/p}$ has derivative-zero minimal polynomial $x^p - t$.

The following three familiar extensions summarize the distinctions. The middle example has all of its distinct conjugates but not all of them inside the displayed field; the last loses embeddings because its minimal polynomial has a repeated root.

extension	degree	F-embeddings	separable	normal	Galois
$\mathbb{Q}(\sqrt{2})/\mathbb{Q}$	2	2	yes	yes	yes
$\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$	3	3	yes	no	no
$\mathbb{F}_p(t^{1/p})/\mathbb{F}_p(t)$	p	1	no	yes	no

Proposition 11.18 (Finite Galois criteria)**.** A finite extension is Galois precisely when it is the splitting field of a separable polynomial, equivalently when

$$|\operatorname{Aut}_F(E)| = [E : F].$$

Proof. A splitting field is normal by Theorem 11.16. When its defining polynomial is separable, the tower criterion gives separability. Conversely, if E/F is Galois, choose generators

$$E = F(\alpha_1, \dots, \alpha_r).$$

Normality makes every $m_{\alpha_i,F}$ split in E, and separability makes each of these polynomials separable. The product of the distinct polynomials among them is separable, and its splitting field contains every generator and is contained in E; hence it is E.

If E/F is normal and separable, Theorem 11.10 gives

$$\# \operatorname{Hom}_F(E, \overline{F}) = [E : F].$$

Theorem 11.16 says every such embedding has image E, so all of them are automorphisms and

$$|\operatorname{Aut}_F(E)| = [E : F].$$

Conversely, if this automorphism count holds, then

$$[E : F] = |\operatorname{Aut}_F(E)| \leq \# \operatorname{Hom}_F(E, \overline{F}) \leq [E : F].$$

Equality holds throughout. Theorem 11.10 gives separability, and every F-embedding is already one of the automorphisms. Its image is therefore E, so Theorem 11.16 gives normality. Thus E/F is Galois. $\square$

Theorem (Finite Galois closures)**.** If E/F is finite separable, then there is a finite Galois extension L/F containing E.

Proof. Choose generators

$$E = F(\alpha_1, \dots, \alpha_r),$$

and let f_i be the minimal polynomial of α_i over F. Each f_i is separable because E/F is separable. Let L be the splitting field over F of the product of the distinct f_i's. A splitting field of finitely many polynomials is finite, and L/F is normal. The product is separable, so its splitting field is separable. Thus L/F is finite Galois, and it contains every α_i, hence contains E. □

Such an L is called a finite Galois closure (or normal Galois closure) of E/F. Uniqueness is not needed here.

11.4 Finite Fields

Theorem 11.19 (Finite fields)**.** For each prime power $q = p^r$, there is, up to isomorphism, one field $\mathbb{F}_q$. It is the splitting field of

$$x^q - x$$

over $\mathbb{F}_p$, and $\mathbb{F}_q^\times$ is cyclic of order $q-1$. Its subfields are exactly $\mathbb{F}_{p^d}$ for $d \mid r$.

Proof. Let $q = p^r$. In an algebraic closure of $\mathbb{F}_p$, the roots of

$$x^q - x$$

form a field: in characteristic p,

$$(a+b)^q = a^q + b^q, \qquad (ab)^q = a^q b^q,$$

and nonzero roots remain roots on inversion. The derivative is -1, so there are q distinct roots. This gives a field with q elements.

Conversely, if K has q elements, then $K^\times$ has order $q-1$, so $a^{q-1} = 1$ for every nonzero a, and $a^q = a$ for all $a \in K$. Thus K is exactly the root set of $x^q - x$, proving uniqueness up to isomorphism and showing that it is a splitting field. Cyclicity of $\mathbb{F}_q^\times$ is Lemma 11.12.

For subfields, start with an arbitrary field

$$\mathbb{F}_p \subseteq K \subseteq \mathbb{F}_{p^r}$$

and put $d = [K : \mathbb{F}_p]$. Then $|K| = p^d$, so every element of K satisfies $x^{p^d} - x = 0$. Thus K is the unique field with p^d elements in the chosen algebraic closure, namely $\mathbb{F}_{p^d}$, and the Tower Law gives $d \mid r$. Conversely, if $d \mid r$, every root of

$$x^{p^d} - x$$

lies in $\mathbb{F}_{p^r}$, because

$$x^{p^d} - x \mid x^{p^r} - x$$

when $d \mid r$. To justify the divisibility, write $r = dk$. If $a^{p^d} = a$, iterating this equality k times gives

$$a^{p^r} = a.$$

Thus every root of $x^{p^d} - x$ is a root of $x^{p^r} - x$. The former polynomial has derivative -1, hence has no repeated roots, and this root containment proves the divisibility. Its p^d roots form the required subfield. This proves both the existence of each listed subfield and the exhaustion of all subfields. □

Example 11.20 (The field with four elements)**.** The polynomial

$$x^2 + x + 1$$

has no root in $\mathbb{F}_2$, so

$$\mathbb{F}_4 \cong \mathbb{F}_2[\alpha]/(\alpha^2 + \alpha + 1).$$

Its elements are

$$0, \quad 1, \quad \alpha, \quad \alpha + 1,$$

and the relation $\alpha^2 = \alpha + 1$ makes multiplication concrete:

$$(\alpha + 1)^2 = \alpha^2 + 1 = \alpha.$$

The Frobenius automorphism fixes $0, 1$ and exchanges the two nontrivial elements:

$$\alpha^2 = \alpha + 1, \qquad (\alpha + 1)^2 = \alpha.$$

Example 11.21 (The field with eight elements)**.** By Example 11.4, the polynomial $x^3 + x + 1$ has no root in $\mathbb{F}_2$ and is irreducible. Thus

$$\mathbb{F}_8 \cong \mathbb{F}_2[\beta]/(\beta^3 + \beta + 1).$$

Every element has a unique form

$$a + b\beta + c\beta^2, \qquad a, b, c \in \mathbb{F}_2,$$

and multiplication is reduced using $\beta^3 = \beta + 1$. For instance,

$$\beta(\beta^2 + 1) = \beta^3 + \beta = 1.$$

Since $\beta \neq 0, 1$ and $\mathbb{F}_8^\times$ has prime order 7, Lagrange's theorem implies that β has multiplicative order 7.

Proposition 11.22 (Frobenius has exact order)**.** For $q = p^r$,

$$\mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q) \cong C_n,$$

generated by Frobenius $x \mapsto x^q$, whose order is exactly n.

Proof. Frobenius fixes $\mathbb{F}_q$, and its n-th power is the identity on $\mathbb{F}_{q^n}$. If its d-th power were the identity for $0 < d < n$, every element would satisfy $x^{q^d} = x$, placing the field among the q^d roots of $x^{q^d} - x$, an impossibility since $q^n > q^d$. Thus its n powers are distinct. They are all $\mathbb{F}_q$-automorphisms, and the embedding count bounds the whole automorphism group by n. □

These cyclic finite Galois groups will become a running example in Chapter 12. There the compatible restrictions among the groups $\mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ assemble into the absolute Galois group of $\mathbb{F}_q$.

Example 11.23 (Subfields of $\mathbb{F}_{p^{12}}$)**.** The divisors of 12 give all subfields:

$$\mathbb{F}_p, \quad \mathbb{F}_{p^2}, \quad \mathbb{F}_{p^3}, \quad \mathbb{F}_{p^4}, \quad \mathbb{F}_{p^6}, \quad \mathbb{F}_{p^{12}}.$$

For example,

$$\mathbb{F}_{p^3} \cap \mathbb{F}_{p^4} = \mathbb{F}_p,$$

since an element in the intersection lies in the unique subfield whose degree divides both 3 and 4.

11.5 Cyclotomic Extensions

Cyclotomic fields are the natural meeting point of roots of unity, splitting fields, and explicit Galois groups. The multiplicative order of a root of unity supplies its arithmetic label, while the way automorphisms change that root will later be read directly from the unit group $(\mathbb{Z}/n\mathbb{Z})^\times$.

Definition 11.24. A **primitive n-th root of unity** is an element of multiplicative order n. The n-th cyclotomic polynomial $\Phi_n(x)$ is the monic polynomial whose roots are the primitive n-th roots of unity.

The finite cyclicity lemma shows that the roots of $x^n - 1$, in characteristic not dividing n, form a cyclic group. Partitioning them by their order gives

$$x^n - 1 = \prod_{d \mid n} \Phi_d(x).$$

Induction in this formula, together with Gauss's Lemma, gives $\Phi_n \in \mathbb{Z}[x]$: after the factors for proper divisors have been constructed as monic integral polynomials, their product divides the monic polynomial $x^n - 1$ in $\mathbb{Q}[x]$, and polynomial division followed by Gauss's Lemma makes the monic quotient integral. The number of primitive roots gives

$$\deg \Phi_n = \varphi(n).$$

For example,

$$\begin{aligned} &\Phi_1 = x - 1, && \Phi_2 = x + 1, && \Phi_3 = x^2 + x + 1, \\ &\Phi_4 = x^2 + 1, && \Phi_5 = x^4 + x^3 + x^2 + x + 1, && \\ &\Phi_6 = x^2 - x + 1, && \Phi_8 = x^4 + 1. && \end{aligned}$$

Theorem 11.25 (Cyclotomic irreducibility)**.** $\Phi_n(x)$ is irreducible over $\mathbb{Q}$.

Proof. Let ζ be primitive and let $g \in \mathbb{Z}[x]$ be its monic irreducible polynomial. Write

$$\Phi_n = gh$$

in $\mathbb{Z}[x]$. Let $p \nmid n$ be prime. Suppose that ζ^p is not a root of g. It is a primitive root and hence a root of h, so $h(\zeta^p) = 0$. Therefore ζ is a root of $h(x^p)$, and minimality of g gives

$$g(x) \mid h(x^p)$$

in $\mathbb{Q}[x]$, hence in $\mathbb{Z}[x]$ by Gauss's Lemma. Reducing modulo p, Frobenius gives

$$\overline{h(x^p)} = \overline{h(x)}^{\,p}.$$

Thus an irreducible factor of $\bar{g}$ also divides $\bar{h}$, so $\overline{\Phi_n} = \bar{g}\bar{h}$ has a repeated irreducible factor. But Φ_n divides $x^n - 1$, whereas

$$(x^n - 1)' = nx^{n-1}$$

is relatively prime to $x^n - 1$ modulo p. This contradicts the derivative criterion. Hence ζ^p is a root of g for every prime $p \nmid n$. Let $a > 0$ with $(a, n) = 1$, and factor

$$a = p_1 \cdots p_s.$$

No p_i divides n. Repeatedly applying the preceding conclusion shows that ζ^a is a root of g. Thus every primitive root is a root of g, and consequently $g = \Phi_n$. ∎

If ζ_n is primitive, irreducibility gives

$$[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n).$$

It is the splitting field of $x^n - 1$, hence is finite Galois. The map

$$(\mathbb{Z}/n\mathbb{Z})^\times \longrightarrow \mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}), \qquad a \longmapsto (\zeta_n \mapsto \zeta_n^a)$$

is an isomorphism. Indeed, ζ_n^a is primitive exactly when $(a, n) = 1$, and then it is a root of the minimal polynomial Φ_n. The root-extension lemma therefore gives a unique $\mathbb{Q}$-embedding sending ζ_n to ζ_n^a; finite dimensionality makes it an automorphism. Composition corresponds to multiplication of exponents, and the map is injective because an automorphism is determined by ζ_n. Conversely every automorphism sends ζ_n to a primitive root, so it has this form. Roots of unity will reappear in the later discussion of radicals.

Example 11.26 (Small cyclotomic Galois groups)**.** For $n = 3$ and $n = 4$,

$$(\mathbb{Z}/n\mathbb{Z})^\times \cong C_2,$$

so both $\mathbb{Q}(\zeta_3)$ and $\mathbb{Q}(\zeta_4) = \mathbb{Q}(i)$ are quadratic Galois extensions. For $n = 5$,

$$(\mathbb{Z}/5\mathbb{Z})^\times \cong C_4,$$

whereas

$$(\mathbb{Z}/8\mathbb{Z})^\times \cong C_2 \oplus C_2.$$

Thus the degree-four fields $\mathbb{Q}(\zeta_5)$ and $\mathbb{Q}(\zeta_8)$ have nonisomorphic Galois groups.

Exercises

Basic

Exercise 11.1 (Basic)**.** Determine whether $x^4 - 2x^2 + 1$ is separable over $\mathbb{Q}$, and explain the answer using the derivative criterion.

Exercise 11.2 (Basic)**.** Use the derivative criterion to decide whether

$$x^p - x \in \mathbb{F}_p[x]$$

is separable.

Exercise 11.3 (Basic)**.** Show that a finite field is perfect directly from the fact that its Frobenius map is bijective.

Core

Exercise 11.4 (Core)**.** Test normality and separability of $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$.

Exercise 11.5 (Core)**.** Let E/F be finite. Explain why

$$[E : F]_s = [E : F]$$

when E/F is generated by a tower of separable simple extensions.

Exercise 11.6 (Core)**.** Find a primitive element for

$$\mathbb{Q}(\sqrt{2}, \sqrt{5})/\mathbb{Q}.$$

Exercise 11.7 (Core)**.** Verify that x^3+x+1 is irreducible over $\mathbb{F}_2$, and use it to list the elements of $\mathbb{F}_8$.

Exercise 11.8 (Core)**.** Determine the subfields of $\mathbb{F}_{p^{12}}$.

Exercise 11.9 (Core)**.** Prove that every finite extension of a finite field is Galois.

Exercise 11.10 (Core)**.** Compute the Frobenius orbit of a nonzero element of $\mathbb{F}_8$ not in $\mathbb{F}_2$.

Further

Exercise 11.11 (Further)**.** Show that

$$\Phi_{10}(x) = x^4 - x^3 + x^2 - x + 1.$$

Exercise 11.12 (Further)**.** Show that $\mathbb{Q}(\zeta_5)$ has Galois group isomorphic to C_4.

Exercise 11.13 (Further)**.** Determine $\mathrm{Gal}(\mathbb{Q}(\zeta_8)/\mathbb{Q})$.

Exercise 11.14 (Further)**.** Prove that the splitting field of

$$x^4 - 2$$

over $\mathbb{Q}$ is normal and separable, and determine its degree.

Looking Ahead

Exercise 11.15 (Challenging)**.** Let E/F be finite and separable. Use the embedding-count theorem to show that E/F is normal exactly when every F-embedding $E \to \overline{F}$ is an automorphism of E.

Exercise 11.16 (Looking Ahead)**.** If E/F is finite Galois, show that

$$F \subseteq E^{\mathrm{Aut}_F(E)}.$$

The reverse inclusion is the starting point of the Galois correspondence.

Chapter 12

Galois Theory, Limits, and Profinite Groups

Galois theory replaces the study of intermediate fields by the study of the symmetries of the larger field. In the finite case this is an exact dictionary. In the infinite case the same dictionary survives once the symmetry group remembers agreement at every finite stage through a topology.

12.1 Galois Groups and Fixed Fields

Definition 12.1. For an extension E/F, its **Galois group** is

$$\mathrm{Gal}(E/F) = \mathrm{Aut}_F(E).$$

If $H \leq \mathrm{Gal}(E/F)$, its **fixed field** is

$$E^H = \{x \in E : \sigma(x) = x \text{ for every } \sigma \in H\}.$$

For an intermediate field $F \subseteq K \subseteq E$, write

$$\mathrm{Gal}(E/K) = \{\sigma \in \mathrm{Gal}(E/F) : \sigma|_K = 1_K\}.$$

The fixed field records the elements on which the chosen symmetries are invisible; it will be the inverse operation to taking an automorphism group.

Lemma 12.2. The set E^H is an intermediate field. Furthermore,

$$K_1 \subseteq K_2 \Longrightarrow \mathrm{Gal}(E/K_2) \leq \mathrm{Gal}(E/K_1), \qquad H_1 \leq H_2 \Longrightarrow E^{H_2} \subseteq E^{H_1}.$$

Proof. Every element of H fixes 0 and 1. If $x, y \in E^H$, then each $\sigma \in H$ fixes $x - y$, xy, and x^{-1} when $x \neq 0$. Thus E^H is a subfield containing F. The two inclusions say, respectively, that fixing more elements imposes more conditions and that asking more automorphisms to fix an element imposes more conditions. $\square$

Example 12.3. Complex conjugation has fixed field $\mathbb{R}$ in $\mathbb{C}$. If d is squarefree, the nonidentity automorphism of $\mathbb{Q}(\sqrt{d})$ sends $\sqrt{d}$ to $-\sqrt{d}$ and fixes precisely $\mathbb{Q}$. In $\mathbb{Q}(\zeta_5)$, the automorphisms are $\zeta_5 \mapsto \zeta_5^a$, $a \in (\mathbb{Z}/5\mathbb{Z})^\times$; its unique order-two subgroup fixes $\mathbb{Q}(\sqrt{5})$.

The following linear-algebra fact is the key input in Artin's theorem. Distinct field homomorphisms cannot satisfy an accidental linear relation as functions.

Lemma 12.4 (Artin independence lemma)**.** *Let $\sigma_1, \dots, \sigma_n : E \to L$ be distinct field homomorphisms. Then they are linearly independent over L as functions $E \to L$.*

Proof. Suppose that a nontrivial relation

$$a_1\sigma_1 + \cdots + a_n\sigma_n = 0$$

exists, and choose one with the fewest nonzero coefficients. Omit zero terms, so every $a_i \neq 0$ and $n \geq 2$. Choose $x \in E$ with $\sigma_1(x) \neq \sigma_n(x)$. Evaluating at xy, then subtracting $\sigma_n(x)$ times the original relation evaluated at y, yields

$$\sum_{i=1}^{n-1} a_i\big(\sigma_i(x) - \sigma_n(x)\big)\sigma_i(y) = 0 \quad \text{for all } y \in E.$$

The coefficient of σ_n has vanished while that of σ_1 is nonzero. This is a shorter nontrivial relation, a contradiction. □

The proof of Artin's theorem has a useful two-sided shape. Independence of the automorphisms first produces enough linearly independent elements to give the lower bound $|G| \leq [E : E^G]$. Orbit polynomials then show that every finitely generated subfield has degree at most $|G|$, producing the reverse bound. The final embedding count identifies the automorphisms. This is the bridge from a concrete group action to the field–subgroup dictionary.

Theorem 12.5 (Artin fixed-field theorem)**.** *If G is a finite group of automorphisms of a field E, then*

$$[E : E^G] = |G|, \qquad \mathrm{Gal}(E/E^G) = G.$$

Proof. Put $F = E^G$ and $n = |G|$. Artin independence implies that there are $x_1, \dots, x_n \in E$ for which the evaluation matrix

$$\big(\sigma_i(x_j)\big)_{1 \leq i,j \leq n}$$

is invertible. Indeed, consider the evaluation vectors

$$v(x) = (\sigma_1(x), \dots, \sigma_n(x)) \in E^n.$$

If these did not span E^n, a nonzero linear functional $(a_1, \dots, a_n)$ would vanish on every $v(x)$, giving

$$a_1\sigma_1 + \cdots + a_n\sigma_n = 0,$$

contrary to Artin independence. Choose n evaluation vectors forming a basis, and use them as the columns of the displayed matrix. The x_j are therefore F-linearly independent, so $n \leq [E : F]$.

For the reverse inequality, let $\alpha \in E$ and form

$$P_\alpha(X) = \prod_{\sigma \in G} (X - \sigma(\alpha)).$$

Every $\tau \in G$ permutes these factors, so the coefficients belong to F. The minimal polynomial of α over F divides P_α, and is separable of degree at most n. Given finitely many elements of E, the field they generate over F is finite separable; the primitive element theorem makes it $F(\alpha)$, hence of degree at most n. Therefore E has no $n + 1$ linearly independent elements over F, and $[E : F] \leq n$.

Finally $G \leq \mathrm{Gal}(E/F)$, while the embedding-count inequality (Theorem 11.7) gives

$$|\,\mathrm{Gal}(E/F)| \leq [E : F] = |G|.$$

Therefore the inclusion is equality. □

Corollary 12.6. If E/F is finite Galois, then

$$E^{\mathrm{Gal}(E/F)} = F.$$

Conversely, if G is a finite subgroup of $\mathrm{Aut}(E)$, then E/E^G is finite Galois with Galois group G.

Proof. The first assertion follows by combining Artin's theorem with the finite Galois criterion (Proposition 11.18). The second follows from the same two results. □

12.2 The Fundamental Theorem of Finite Galois Theory

Theorem 12.7 (Finite Galois correspondence)**.** Let E/F be finite Galois and put $G = \mathrm{Gal}(E/F)$. The maps

$$K \longmapsto \mathrm{Gal}(E/K), \qquad H \longmapsto E^H$$

are inverse inclusion-reversing bijections between intermediate fields $F \subseteq K \subseteq E$ and subgroups $H \leq G$. Moreover,

$$[E : K] = |\,\mathrm{Gal}(E/K)|, \qquad [K : F] = [G : \mathrm{Gal}(E/K)].$$

Proof. For an intermediate K, separability and normality descend from E/F to E/K: a minimal polynomial over K divides the corresponding polynomial over F, so it is separable and splits in E. Thus E/K is finite Galois, and Corollary 12.6 gives

$$E^{\mathrm{Gal}(E/K)} = K.$$

For $H \leq G$, Artin's theorem applied to the finite group H gives

$$\mathrm{Gal}(E/E^H) = H.$$

These identities prove the bijection. The first degree formula is Artin's theorem applied to E/K, and the Tower Law gives

$$[K : F] = \frac{[E : F]}{[E : K]} = \frac{|G|}{|\,\mathrm{Gal}(E/K)|}.$$

$\square$

The reversal is the central visual idea: a larger field has fewer automorphisms that fix it.

$$\begin{array}{ccc} E & & \{1\} \\ \leftarrow\!\!\!\uparrow\!\!\!\rightarrow & & \uparrow \\ K & & H \\ \leftarrow\!\!\!\uparrow\!\!\!\rightarrow & & \uparrow \\ F & & G \end{array}$$

Proposition 12.8 (Normality refinement)**.** For an intermediate field K,

$$K/F \text{ is Galois} \quad \Longleftrightarrow \quad \mathrm{Gal}(E/K) \trianglelefteq G.$$

When these conditions hold, restriction induces

$$G/\,\mathrm{Gal}(E/K) \cong \mathrm{Gal}(K/F).$$

Proof. Write $H = \mathrm{Gal}(E/K)$. First,

$$K/F \text{ is normal} \quad \Longleftrightarrow \quad \sigma(K) = K \text{ for every } \sigma \in G. \tag{12.1}$$

Normality makes every conjugate of an element of K remain in K. Conversely, extend any F-embedding of K to E by Theorem 10.24; normality of E/F makes the extension an automorphism, and stability then proves normality of K/F.

For $\sigma \in G$,

$$\sigma H \sigma^{-1} = \mathrm{Gal}(E/\sigma(K)).$$

The finite correspondence and (12.1) show that H is normal exactly when K/F is normal. Every intermediate extension is separable, so this is equivalent to Galoisness.

When K is stable, restriction

$$\mathrm{res} : G \longrightarrow \mathrm{Gal}(K/F)$$

is well-defined and has kernel H. An F-automorphism of K extends to an F-embedding of E, and normality of E/F makes the extension an automorphism. Thus restriction is onto, and the First Isomorphism Theorem gives the quotient isomorphism. $\square$

Example 12.9 (The full lattice for a cubic splitting field)**.** Put

$$r = \sqrt[3]{2}, \qquad E = \mathbb{Q}(r, \zeta_3).$$

The roots of $x^3 - 2$ are $r, \zeta_3 r, \zeta_3^2 r$, and $\mathrm{Gal}(E/\mathbb{Q}) \cong S_3$ acts by permuting them. Let

$$K_1 = \mathbb{Q}(r), \qquad K_2 = \mathbb{Q}(\zeta_3 r), \qquad K_3 = \mathbb{Q}(\zeta_3^2 r).$$

The intermediate fields have the following lattice; each displayed intermediate field is contained in E, and the arrows are inclusions.

$$\begin{tikzcd}
& & E & \\
K_1 \arrow[urr] & K_2 \arrow[ur] & K_3 \arrow[u] & \mathbb{Q}(\sqrt{-3}) \arrow[ul] \\
& & \mathbb{Q} \arrow[ull] \arrow[ul] \arrow[u] \arrow[ur] &
\end{tikzcd}$$

The order-two subgroup stabilizing the indicated root fixes the corresponding K_i, so the three order-two subgroups correspond to the three cubic subfields. The alternating subgroup A_3 fixes $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\zeta_3)$, and the trivial and full subgroups correspond to E and $\mathbb{Q}$, respectively:

field (degree)	subgroup	order
E (6)	$\{1\}$	1
$K_1,\ K_2,\ K_3$ (3)	three root stabilizers	2
$\mathbb{Q}(\sqrt{-3})$ (2)	A_3	3
$\mathbb{Q}$ (1)	S_3	6

The three order-two subgroups are not normal. Accordingly the three cubic fields are not Galois over $\mathbb{Q}$, whereas $\mathbb{Q}(\sqrt{-3})/\mathbb{Q}$ is Galois. This is a useful contrast between the field lattice itself and the additional normality information carried by the group lattice.

Example 12.10 (A biquadratic contrast)**.** For

$$E = \mathbb{Q}(\sqrt{2}, \sqrt{3}),$$

every $\mathbb{Q}$-automorphism independently changes the signs of $\sqrt{2}$ and $\sqrt{3}$. Hence

$$\mathrm{Gal}(E/\mathbb{Q}) \cong C_2 \oplus C_2.$$

Its three order-two subgroups fix, respectively,

$$\mathbb{Q}(\sqrt{2}), \qquad \mathbb{Q}(\sqrt{3}), \qquad \mathbb{Q}(\sqrt{6}).$$

Unlike the S_3 example, every subgroup is normal because the group is abelian. Thus all three quadratic intermediate fields are Galois over $\mathbb{Q}$. The two examples show that the same-looking degree pattern can have very different normality behavior.

Example 12.11 (A cyclic correspondence)**.** $\mathrm{Gal}(\mathbb{Q}(\zeta_5)/\mathbb{Q}) \cong C_4$. Its unique subgroup of order two fixes $\mathbb{Q}(\sqrt{5})$. All subgroups are normal, so every intermediate field is Galois over $\mathbb{Q}$.

Optional Application: A Galois-Theoretic Proof of the Fundamental Theorem of Algebra

Chapter 8 used the Fundamental Theorem of Algebra to obtain an eigenvalue of a complex operator. We now discharge that dependence using the field and group theory developed so far. Apart from the standard elementary properties of the real numbers, including square roots of nonnegative reals, the only substantive real-analysis input is the Intermediate Value Theorem. The needed facts are developed in **Lecture Notes on Mathematical Analysis** [2]: polynomials are continuous (§5.2), and the Intermediate Value Theorem is Theorem 5.14 (§5.4). Hence every real polynomial of odd degree has a real root. Indeed, if

$$p(x) = a_{2m+1}x^{2m+1} + \cdots, \qquad a_{2m+1} \neq 0,$$

then, for sufficiently large $R > 0$, leading-term behavior makes $p(R)$ and $p(-R)$ have opposite signs. Continuity and the Intermediate Value Theorem therefore give a point $c \in (-R, R)$ with $p(c) = 0$.

Theorem (Fundamental Theorem of Algebra)**.** Every nonconstant polynomial in $\mathbb{C}[x]$ has a root in $\mathbb{C}$. Equivalently, $\mathbb{C}$ is algebraically closed.

Proof. We first show that every finite extension of $\mathbb{R}$ has degree a power of 2. Let $M/\mathbb{R}$ be finite. Characteristic zero makes $M/\mathbb{R}$ separable, so the finite Galois closure theorem supplies a finite Galois extension $L/\mathbb{R}$ containing M. Put $G = \mathrm{Gal}(L/\mathbb{R})$. If an odd prime divided $|G|$, let P be a Sylow 2-subgroup. Then

$$[G : P]$$

is odd and greater than one. By the finite Galois correspondence, $L^P/\mathbb{R}$ has this odd degree. The primitive element theorem writes $L^P = \mathbb{R}(\beta)$, so the minimal polynomial of β has odd degree. It has a root in $\mathbb{R}$ by the stated external fact, which contradicts irreducibility unless the degree is one. Thus G is a 2-group. Since

$$[M : \mathbb{R}] \mid [L : \mathbb{R}] = |G|,$$

the claimed power-of-two conclusion follows.

Fix an algebraic closure $\overline{\mathbb{R}}$ and an element $i \in \overline{\mathbb{R}}$ with $i^2 = -1$, so that $\mathbb{C} = \mathbb{R}(i)$. Every nontrivial quadratic extension of $\mathbb{R}$ is this field. Indeed, if $\mathbb{R}(\alpha)/\mathbb{R}$ is quadratic, the minimal polynomial of α is an irreducible quadratic with negative discriminant. Completing the square therefore produces an element of the extension whose square is -1: after division by the leading coefficient, a polynomial $x^2 + bx + c$ gives

$$\left(\alpha + \frac{b}{2}\right)^2 = \frac{b^2 - 4c}{4} < 0,$$

and rescaling by the positive real square root of its negative gives a square root of -1. It is i or $-i$, so the extension is $\mathbb{R}(i) = \mathbb{C}$.

We next show directly that every element of $\mathbb{C}$ has a square root. For $z = a + bi$, put

$$r = \sqrt{a^2 + b^2}.$$

If $r + a > 0$, set

$$x = \sqrt{\frac{r+a}{2}}, \qquad y = \frac{b}{2x}.$$

Then $x^2 - y^2 = a$ and $2xy = b$, so $(x+iy)^2 = z$. If $r+a = 0$, then $b = 0$ and $a \le 0$, and $(i\sqrt{-a})^2 = z$. Consequently the quadratic formula shows that every quadratic polynomial over $\mathbb{C}$ has a root. Hence $\mathbb{C}$ has no nontrivial quadratic extension.

We claim that no finite Galois extension $L/\mathbb{R}$ has degree greater than 2. Otherwise its group G is a 2-group of order greater than two. Every nontrivial finite 2-group has a normal subgroup of index two: for order two this is clear, and in general the center theorem and Cauchy's theorem supply a central subgroup of order two; induction in the quotient then gives the asserted subgroup. Choose a normal subgroup $H \trianglelefteq G$ of index two. The finite correspondence makes

$$L^H/\mathbb{R}$$

a quadratic extension, hence $L^H = \mathbb{C}$. Now $H = \mathrm{Gal}(L/\mathbb{C})$ is a nontrivial 2-group, so it has a subgroup J of index two. The finite correspondence for $L/\mathbb{C}$ then makes

$$L^J/\mathbb{C}$$

a quadratic extension, a contradiction. Therefore $[L : \mathbb{R}] \le 2$.

Finally, let $\alpha \in \overline{\mathbb{R}}$. The finite extension $\mathbb{R}(\alpha)/\mathbb{R}$ is separable, so the finite Galois closure theorem gives a finite Galois extension containing it, whose degree is at most two by the preceding argument. The Tower Law therefore gives $[\mathbb{R}(\alpha) : \mathbb{R}] \le 2$, so it is either $\mathbb{R}$ or, in the quadratic case, $\mathbb{C}$. Thus every α belongs to $\mathbb{C}$, and $\overline{\mathbb{R}} = \mathbb{C}$. Hence $\mathbb{C}$ is algebraically closed. □

Remark. Chapter 8 met the Fundamental Theorem of Algebra through complex linear algebra, while its classical proof belongs to complex analysis. The present proof instead reveals field extensions, finite 2-groups, and the Galois correspondence behind the same theorem.

12.3 Radicals and Classical Constructions

Definition 12.12. A **radical extension** is an extension contained in a tower

$$F = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_r,$$

where $K_i = K_{i-1}(\alpha_i)$ and $\alpha_i^{n_i} \in K_{i-1}$ for some $n_i > 0$. A polynomial is **solvable by radicals** if its splitting field is contained in a radical extension of its coefficient field.

Roots of unity govern the conjugates of an n-th root, which explains the following honest part of the radicals–groups connection.

Proposition 12.13 (A root-of-unity radical direction)**.** Let

$$F = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_r$$

be a tower such that every K_i/F is finite Galois and

$$K_i = K_{i-1}(\alpha_i), \qquad \alpha_i^{n_i} \in K_{i-1}.$$

Assume that K_{i-1} contains all n_i-th roots of unity. Then

$$\mathrm{Gal}(K_r/F)$$

is solvable.

Proof. If $\alpha_i = 0$, then $K_i = K_{i-1}$, so the relevant Galois group is trivial. Otherwise define

$$\chi_i : \mathrm{Gal}(K_i/K_{i-1}) \longrightarrow \mu_{n_i}, \qquad \chi_i(\sigma) = \frac{\sigma(\alpha_i)}{\alpha_i}.$$

The equation $\alpha_i^{n_i} \in K_{i-1}$ shows that $\chi_i(\sigma)$ is an n_i-th root of unity. Since $\mu_{n_i} \subseteq K_{i-1}$, every σ fixes it pointwise; consequently

$$\chi_i(\sigma\tau) = \frac{\sigma\tau(\alpha_i)}{\alpha_i} = \frac{\sigma\big(\chi_i(\tau)\alpha_i\big)}{\alpha_i} = \chi_i(\tau)\chi_i(\sigma).$$

The target is abelian, so this is the homomorphism law. Moreover, $\chi_i(\sigma) = 1$ means that σ fixes α_i; since $K_i = K_{i-1}(\alpha_i)$, it is then the identity. Thus χ_i is injective. Therefore $\mathrm{Gal}(K_i/K_{i-1})$ is isomorphic to a subgroup of the abelian group μ_{n_i}, and hence is abelian.

Put

$$G_i = \mathrm{Gal}(K_r/K_i).$$

The hypothesis that K_i/F is Galois makes G_i normal in $G_0 = \mathrm{Gal}(K_r/F)$. The finite normality refinement (Proposition 12.8) gives

$$G_{i-1}/G_i \cong \mathrm{Gal}(K_i/K_{i-1}).$$

Consequently

$$G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_r = 1$$

is a normal series with abelian factors. Thus G_0 is solvable. $\square$

Remark 12.14. The classical theorem in characteristic zero says that a polynomial is solvable by radicals if and only if its Galois group is solvable. The converse requires Kummer theory, which is outside these notes. The proposition above captures the group-theoretic mechanism after a radical tower has been placed inside a suitable Galois tower; it does not prove the converse. Accordingly, we do not claim a proof here of the general quintic theorem.

Definition 12.15. A real number is **constructible over** $\mathbb{Q}$ if it belongs to a tower of subfields of $\mathbb{R}$

$$\mathbb{Q} = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_r \subseteq \mathbb{R}$$

whose successive degrees are at most two. This is the field-theoretic form of straightedge-and-compass construction: line intersections use field operations and circle intersections require a quadratic equation.

The algebraic definition records a necessary consequence of the ordinary geometric construction rules. Indeed, the intersection of two lines has coordinates obtained by field operations from the preceding coordinates. After eliminating one variable, the intersection of a line and a circle, or of two circles, is obtained by solving a quadratic equation. Hence every ordinary straightedge-and-compass construction produces coordinates in a tower whose successive degrees are at most two. This one direction is all that is needed for the impossibility results below.

Proposition 12.16 (Degree obstruction)**.** If α is constructible over $\mathbb{Q}$, then

$$[\mathbb{Q}(\alpha) : \mathbb{Q}]$$

divides a power of 2.

Proof. The construction operations give a tower with successive degrees at most two. The Tower Law makes the degree of its top field a power of two, and the degree of the subfield $\mathbb{Q}(\alpha)$ divides that degree. $\square$

Corollary 12.17. Doubling the cube and trisecting a general angle are impossible with straightedge and compass.

Proof. $\sqrt[3]{2}$ has irreducible minimal polynomial $X^3 - 2$. For angle trisection, a $60°$ angle is constructible. If arbitrary angle trisection were possible, then $20°$ would be constructible. But $\cos 20°$ satisfies

$$8X^3 - 6X - 1 = 0,$$

which is irreducible over $\mathbb{Q}$ by the rational-root test. Both degrees are 3, so Proposition 12.16 applies. Squaring the circle uses the transcendence of π, outside the present scope. □

12.4 Direct and Inverse Limits

Infinite Galois theory is assembled from finite stages. Direct limits join objects that grow; inverse limits record compatible information at all finite quotients. The common indexing language is designed so that a finite number of comparisons can always be made at one later stage.

Definition 12.18. A partially ordered set I is **directed** if it is nonempty and every pair i, j has an upper bound k. Examples are $\mathbb{N}$, finite subsets of a set ordered by inclusion, and finite intermediate extensions ordered by inclusion. Repeatedly applying the pair condition gives a common upper bound for every finite collection. This is exactly what lets a direct limit turn a finite sum of representatives into one representative later on.

Definition 12.19. A **direct system** of R-modules over I is a family $(M_i)_{i\in I}$ and maps

$$\phi_i^j : M_i \longrightarrow M_j \qquad (i \le j)$$

with $\phi_i^i = 1_{M_i}$ and

$$\phi_j^k \phi_i^j = \phi_i^k \qquad (i \le j \le k).$$

For $i \le j \le k$, this says that the following triangle commutes:

$$\begin{tikzcd} M_i \arrow[rr, "\phi_i^k"] \arrow[dr, "\phi_i^j"'] & & M_k \\ & M_j \arrow[ur, "\phi_j^k"'] & \end{tikzcd}$$

Thus passing from i to k does not depend on whether one pauses at j.

Definition 12.20. A **direct limit** of a direct system is an R-module $\varinjlim_i M_i$, together with maps

$$\iota_i : M_i \longrightarrow \varinjlim_i M_i,$$

such that $\iota_j \phi_i^j = \iota_i$ for $i \le j$, and satisfying the following universal property: whenever maps $f_i : M_i \to N$ satisfy

$$f_j \phi_i^j = f_i,$$

there is a unique map

$$u : \varinjlim_i M_i \longrightarrow N$$

with $u\iota_i = f_i$ for every i.

$$\begin{tikzcd}
M_i \arrow[rr, "\phi_i^j"] \arrow[dr, "\iota_i"'] \arrow[ddr, bend right, "f_i"'] & & M_j \arrow[dl, "\iota_j"] \arrow[ddl, bend left, "f_j"] \\
& \varinjlim_i M_i \arrow[d, dashed, "u"] & \\
& N &
\end{tikzcd}$$

Proposition 12.21 (Uniqueness of direct limits)**.** Two direct limits of the same direct system are uniquely isomorphic by an isomorphism commuting with every canonical map.

Proof. Let (M, ι_i) and (M', ι'_i) be direct limits. The universal property of M, applied to the compatible family (ι'_i), gives a unique $u : M \to M'$ with $u\iota_i = \iota'_i$. Similarly there is a unique $v : M' \to M$ with $v\iota'_i = \iota_i$. Both vu and 1_M have the same composites with every ι_i, so uniqueness gives $vu = 1_M$. Likewise $uv = 1_{M'}$. $\square$

Proposition 12.22 (Construction of direct limits)**.** Every direct system of R-modules has a direct limit. It is

$$\varinjlim_{i \in I} M_i = \left(\bigoplus_{i \in I} M_i \right) \Big/ D,$$

where D is generated by

$$e_i(m) - e_j(\phi_i^j(m)) \qquad (i \le j,\ m \in M_i).$$

Proof. Let $\iota_i(m)$ be the class of $e_i(m)$. The displayed relations give $\iota_j\phi_i^j = \iota_i$. A compatible family $f_i : M_i \to N$ defines

$$\sum_i e_i(m_i) \longmapsto \sum_i f_i(m_i).$$

The sums are finite because the source is a direct sum, and each defining relation maps to

$$f_i(m) - f_j(\phi_i^j(m)) = 0.$$

Hence the map descends to a map u from the quotient. It satisfies $u\iota_i = f_i$. Since every quotient class is a finite sum of classes from the summands, these identities also force uniqueness. $\square$

Proposition 12.23 (Maps induced on direct limits)**.** Suppose (M_i, ϕ_i^j) and (N_i, ψ_i^j) are direct systems and maps $\rho_i : M_i \to N_i$ satisfy

$$\rho_j\phi_i^j = \psi_i^j\rho_i \qquad (i \le j).$$

Then there is a unique map

$$\rho : \varinjlim_i M_i \longrightarrow \varinjlim_i N_i$$

with $\rho\iota_i = \kappa_i\rho_i$, where κ_i are the canonical maps for the second limit.

Proof. The maps $\kappa_i\rho_i$ are compatible, since

$$\kappa_j\rho_j\phi_i^j = \kappa_j\psi_i^j\rho_i = \kappa_i\rho_i.$$

The universal property of $\varinjlim_i M_i$ gives exactly the asserted map. $\square$

Proposition 12.24 (Element criteria)**.** Every element of a direct limit has the form $\iota_i(m)$ for some $i \in I$ and $m \in M_i$. In addition,

$$\iota_i(m) = 0 \quad \Longleftrightarrow \quad \phi_i^j(m) = 0 \text{ for some } j \ge i,$$

and

$$\iota_i(m) = \iota_j(n) \quad \Longleftrightarrow \quad \phi_i^k(m) = \phi_j^k(n) \text{ for some } k \ge i, j.$$

Proof. A quotient element is represented by

$$\sum_{\ell=1}^{r} e_{i_\ell}(m_\ell).$$

Choose k above every i_ℓ. Compatibility gives

$$\sum_{\ell=1}^{r} \iota_{i_\ell}(m_\ell) = \iota_k\left(\sum_{\ell=1}^{r} \phi_{i_\ell}^{k}(m_\ell)\right),$$

which proves the first assertion. The reverse implications in the displayed criteria are compatibility. For the forward implications, the asserted zero or equality in the quotient is a finite sum of defining relations. Choose an index above all of the finitely many stages occurring in this expression. Mapping every summand to that stage cancels the relations and gives the desired zero or equality. □

Example 12.25 (A directed union written as fractions)**.** Consider the direct system indexed by $n \geq 0$

$$\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \xrightarrow{\times 2} \cdots .$$

Its direct limit is naturally

$$\mathbb{Z}[1/2] = \left\{ \frac{m}{2^n} : m \in \mathbb{Z},\ n \geq 0 \right\}.$$

Indeed, send the integer m at stage n to $m/2^n$. The transition map sends m to $2m$, and

$$\frac{2m}{2^{n+1}} = \frac{m}{2^n},$$

so these maps are compatible. They induce a map from the direct limit to $\mathbb{Z}[1/2]$, and it is onto by definition. If two representatives m at stage n and a at stage r give the same fraction, then after multiplying their equality by a sufficiently large power of 2, their images agree at a common later stage. Proposition 12.24 therefore makes the induced map one-to-one. This example turns the abstract identification of forward representatives into the familiar rule for equivalent fractions.

Corollary 12.26 (Directed unions)**.** If $M_i \subseteq M_j$ for $i \leq j$ and the transition maps are inclusions, then

$$\varinjlim_{i} M_i \cong \bigcup_i M_i.$$

Proof. The inclusions into the union are compatible. The induced map is onto, and the zero criterion in Proposition 12.24 makes it one-to-one. □

Theorem 12.27 (Exactness of directed direct limits)**.** Suppose that

$$A_i \xrightarrow{u_i} B_i \xrightarrow{v_i} C_i$$

is exact for every i, and all transition maps commute with these maps. Then

$$\varinjlim_i A_i \longrightarrow \varinjlim_i B_i \longrightarrow \varinjlim_i C_i$$

is exact. Thus directed direct limits preserve injections and surjections when these occur stagewise.

Proof. The composite is zero at every stage. Conversely, suppose that $\iota_i(b)$ maps to zero. The zero criterion gives $j \geq i$ with

$$v_j(\phi_i^j(b)) = 0.$$

At stage j, exactness supplies $a \in A_j$ with $u_j(a) = \phi_i^j(b)$. Therefore $\iota_i(b)$ is the image of $\iota_j(a)$. Applying this to sequences beginning or ending in zero gives the final statement. $\square$

Definition 12.28. An **inverse system** of R-modules over the directed set I is a family $(N_i)_{i \in I}$ and maps

$$\psi_i^j : N_j \longrightarrow N_i \qquad (i \leq j)$$

such that $\psi_i^i = 1_{N_i}$ and

$$\psi_i^j \psi_j^k = \psi_i^k \qquad (i \leq j \leq k).$$

For $i \leq j \leq k$, the corresponding reversed triangle commutes:

$$\begin{tikzcd} & N_j \arrow[dr, "\psi_i^j"] & \\ N_k \arrow[ur, "\psi_j^k"] \arrow[rr, "\psi_i^k"'] & & N_i \end{tikzcd}$$

Thus information at a later stage restricts consistently to every earlier stage. Unlike a direct limit, an inverse limit will retain only collections of data that agree under all these restrictions.

Definition 12.29. An **inverse limit** of an inverse system is an R-module

$$\varprojlim_i N_i,$$

together with maps $\pi_i : \varprojlim_i N_i \to N_i$ satisfying

$$\psi_i^j \pi_j = \pi_i \qquad (i \leq j),$$

with the following universal property. Whenever maps $g_i : X \to N_i$ satisfy $\psi_i^j g_j = g_i$, there is a unique map

$$u : X \longrightarrow \varprojlim_i N_i$$

for which $\pi_i u = g_i$ for every i.

$$\begin{tikzcd}
 & X \arrow[d, dashed, "u"] \arrow[ddl, bend right, "g_i"'] \arrow[ddr, bend left, "g_j"] & \\
 & \varprojlim_i N_i \arrow[dl, "\pi_i"'] \arrow[dr, "\pi_j"] & \\
N_i & & N_j \arrow[ll, "\psi_i^j"]
\end{tikzcd}$$

Proposition 12.30 (Uniqueness of inverse limits)**.** Two inverse limits of the same inverse system are uniquely isomorphic by an isomorphism commuting with all projections.

Proof. Let (N, π_i) and (N', π_i') be inverse limits. The universal property of N', applied to the compatible family (π_i), produces a unique $u : N \to N'$ with $\pi_i' u = \pi_i$. Similarly there is a unique $v : N' \to N$ with $\pi_i v = \pi_i'$. The maps vu and 1_N have the same composites with every π_i; uniqueness gives $vu = 1_N$. The same argument gives $uv = 1_{N'}$. $\square$

Proposition 12.31 (Construction of inverse limits)**.** Every inverse system of R-modules has an inverse limit. It is

$$\varprojlim_{i \in I} N_i = \left\{ (n_i) \in \prod_{i \in I} N_i : \psi_i^j (n_j) = n_i \text{ whenever } i \leq j \right\}.$$

Proof. The displayed set is a submodule of the product: the compatibility equations are preserved under addition and scalar multiplication. Its coordinate projections are compatible. A compatible family $g_i : X \to N_i$ gives the map

$$x \longmapsto (g_i(x))_{i \in I}$$

to this submodule, because the assumed equations say precisely that this tuple is compatible. Its i-th coordinate is $g_i(x)$, and a map to a submodule of a product is determined by its coordinates. This proves the universal property. Thus a direct limit is a quotient of a direct sum, whereas an inverse limit is a compatible submodule of a direct product. □

Proposition 12.32 (Maps induced on inverse limits)**.** Suppose (M_i, ϕ_i^j) and (N_i, ψ_i^j) are inverse systems and maps $\rho_i : M_i \to N_i$ satisfy

$$\psi_i^j \rho_j = \rho_i \phi_i^j \qquad (i \le j).$$

Then there is a unique map

$$\rho : \varprojlim_i M_i \longrightarrow \varprojlim_i N_i$$

whose i-th coordinate is ρ_i applied to the i-th coordinate.

Proof. For $(m_i)_i \in \varprojlim_i M_i$, the compatibility calculation

$$\psi_i^j(\rho_j(m_j)) = \rho_i(\phi_i^j(m_j)) = \rho_i(m_i)$$

shows that $(\rho_i(m_i))_i$ is a compatible tuple. The asserted formula therefore defines ρ; uniqueness follows from the coordinate projections. □

Example 12.33 (The 2-adic and profinite integers)**.** The inverse limit of the reductions

$$\mathbb{Z}/2^{n+1}\mathbb{Z} \longrightarrow \mathbb{Z}/2^n\mathbb{Z}$$

is $\mathbb{Z}_2$. An element is a compatible sequence

$$(a_n \bmod 2^n)_{n \ge 1};$$

the ordinary integer 5, for instance, gives $(1 \bmod 2, 1 \bmod 4, 5 \bmod 8, 5 \bmod 16, \ldots)$. Ordered by divisibility,

$$\widehat{\mathbb{Z}} = \varprojlim_{n \ge 1} \mathbb{Z}/n\mathbb{Z}.$$

The former retains only powers of 2; the latter retains every modulus. $\mathbb{Z}_2$ also contains compatible sequences not coming from a nonnegative ordinary integer in any fixed finite stage. For example,

$$(-1 \bmod 2,\ -1 \bmod 4,\ -1 \bmod 8,\ -1 \bmod 16, \ldots)$$

is compatible and represents -1, while the proposed first two entries

$$(0 \bmod 2,\ 1 \bmod 4)$$

cannot occur in a compatible sequence, because 1 mod 4 reduces to 1 mod 2, not to 0 mod 2. Compatibility is therefore a real restriction, not merely a way of listing residues.

Proposition 12.34 (Left exactness of inverse limits)**.** Inverse limits preserve kernels and hence are left exact, but need not preserve surjectivity at the right end.

Proof. In the product description, a compatible family lies in the kernel of an induced map exactly when every coordinate lies in the corresponding kernel. The same compatibility equations remain, so the kernel is the inverse limit of the kernels. Compatible lifts of a compatible family need not exist, which is why right exactness is not claimed. $\square$

Example 12.35 (Stagewise surjections need not lift compatibly)**.** For every $n \geq 1$, let $X_n = \mathbb{Z}$ with identity transition maps, let

$$Y_n = \mathbb{Z}/2^n\mathbb{Z}$$

with the reduction maps, and let $q_n : X_n \to Y_n$ be the quotient map. Each q_n is surjective, and the maps commute with the transitions. Yet

$$\varprojlim_n X_n \cong \mathbb{Z}, \qquad \varprojlim_n Y_n \cong \mathbb{Z}_2,$$

and the induced map $\mathbb{Z} \to \mathbb{Z}_2$ is not surjective. For example, the compatible sequence

$$(1 \bmod 2,\ 1 \bmod 4,\ 5 \bmod 8,\ 5 \bmod 16,\ 21 \bmod 32, \ldots)$$

is not the image of an ordinary integer. Its 2-adic binary expansion has infinitely many zeros and ones. A nonnegative ordinary integer has only finitely many nonzero binary digits, whereas a negative ordinary integer has a 2-adic binary expansion that is eventually all ones. Thus compatible stagewise lifts need not be selectable simultaneously.

Remark 12.36 (Looking Ahead: the Mittag–Leffler condition). The preceding proposition identifies the difficulty with inverse limits: one can take compatible kernels coordinatewise, but compatible choices of preimages need not exist. For an inverse system (M_i, ψ_i^j), the **Mittag–Leffler condition** says that for every fixed i, there is a $j \geq i$ such that

$$\mathrm{im}(\psi_i^k) = \mathrm{im}(\psi_i^j) \qquad \text{for every } k \geq j.$$

In other words, the images inside the fixed earlier module M_i, as seen from later and later stages, eventually stabilize. Surjective transition maps satisfy the condition because all these images are M_i. It also holds when each M_i is finite as a set, since a descending family of submodules of a fixed finite module stabilizes; for finite-dimensional vector spaces, one may instead use the eventually constant nonincreasing dimensions of the images.

Under standard hypotheses, a Mittag–Leffler condition on the kernel system in an inverse system of short exact sequences restores surjectivity at the right-hand end after taking inverse limits. It is not needed for the present Galois application: restriction maps between finite Galois stages are already surjective. The derived obstruction to exactness is usually denoted $\varprojlim^1$; its systematic treatment belongs to later homological algebra.

12.5 Topological Preliminaries

In finite Galois theory the group alone remembers all the relevant information. For an infinite extension, however, one must also remember when automorphisms agree on larger and larger finite Galois subextensions. Topology is the concise language for this finite-stage approximation: in the Krull topology, two automorphisms are close when they agree on a sufficiently large finite Galois field. We develop only the small amount of topology needed to make this statement precise.

12.5.1 Basic Point-Set Topology

Definition 12.37. A **topology** on a set X is a collection τ of subsets of X such that

$$\varnothing,\ X \in \tau;$$

an arbitrary union of members of τ is in τ; and a finite intersection of members of τ is in τ. The members of τ are the **open sets**. A set is **closed** if its complement is open, and a **neighborhood** of $x \in X$ is a set containing an open set that contains x.

The discrete topology, in which every subset is open, is the important example here: every finite Galois group will be given this topology. At the opposite extreme, the indiscrete topology has only $\varnothing$ and X open. The usual topology on $\mathbb{R}$ is familiar intuition, but no analysis on $\mathbb{R}$ is needed below.

Definition 12.38. A collection $\mathcal{B}$ of open sets is a **basis** if every open set is a union of members of $\mathcal{B}$. A collection of neighborhoods of x is a **neighborhood basis at** x if every neighborhood of x contains one of its members. A map $f : X \to Y$ is **continuous** if $f^{-1}(U)$ is open whenever U is open. A bijective continuous map whose inverse is continuous is a **homeomorphism**.

For $A \subseteq X$, its **closure** $\overline{A}$ is the intersection of all closed subsets of X containing A. We say that A is **dense** when $\overline{A} = X$. For example, a nonempty subset of a finite discrete space is closed, so it is dense exactly when it is the whole space.

If $Y \subseteq X$, the **subspace topology** on Y declares

$$U \subseteq Y \text{ open} \quad \Longleftrightarrow \quad U = Y \cap V \text{ for some open } V \subseteq X.$$

For a family of spaces, the product topology on

$$\prod_{i \in I} X_i$$

has as a basis the sets which restrict only finitely many coordinates to open sets and leave every remaining coordinate unrestricted. Thus a basic neighborhood records finite information. When every X_i is finite discrete, it may simply prescribe the values of finitely many coordinates. For example, in

$$\prod_{n \geq 1} \mathbb{Z}/2^n\mathbb{Z},$$

the condition

$$a_1 = 1 \bmod 2, \qquad a_2 = 3 \bmod 4$$

with all higher coordinates unrestricted is a basic open set of the full product. Compatibility is imposed only after one passes to the inverse-limit subspace. Inverse limits will be compatible subspaces of precisely such products.

Definition 12.39. A space is **Hausdorff** if distinct points have disjoint neighborhoods. It is **compact** if every open cover has a finite subcover. A space is **connected** if it cannot be written as the union of two disjoint nonempty open sets, and **totally disconnected** if every connected subset has at most one point.

Finite discrete spaces are Hausdorff, compact, and totally disconnected: singletons separate points and are open, while an open cover of a finite set has an evident finite subcover.

Theorem 12.40 (Tychonoff's theorem (external))**.** An arbitrary product of compact spaces is compact in the product topology.

Remark 12.41 (Foundational scope)*.* In standard set theory with the Axiom of Choice, Tychonoff's theorem gives compactness for arbitrary products of compact spaces. In fact, over ZF, this full arbitrary-product form is equivalent to the Axiom of Choice. The present notes use it only as an external input for products of finite discrete spaces; this is consistent with the earlier use of Zorn's Lemma in the construction of algebraic closures.

12.5.2 Topological Groups

Definition 12.42. A **topological group** is a group G with a topology for which

$$G \times G \longrightarrow G, \qquad (x, y) \longmapsto xy,$$

and

$$G \longrightarrow G, \qquad x \longmapsto x^{-1},$$

are continuous. A homomorphism of topological groups is **continuous** when it is continuous as a map of spaces; a **topological-group isomorphism** is a group isomorphism that is also a homeomorphism.

Proposition 12.43. For $g \in G$, left and right translation,

$$L_g(x) = gx, \qquad R_g(x) = xg,$$

are homeomorphisms of a topological group. Consequently, the neighborhoods of the identity determine all neighborhoods: if U is a neighborhood of 1, then gU is a neighborhood of g.

Proof. The map $x \mapsto (g, x)$ is continuous, and left translation is its composite with multiplication. Similarly, $x \mapsto (x, g)$ followed by multiplication is right translation. Their inverse translations are $L_{g^{-1}}$ and $R_{g^{-1}}$, which are continuous by the same argument. The final assertion follows by applying L_g to a neighborhood of 1. $\square$

Proposition 12.44. If $\phi : G \to H$ is a continuous homomorphism and H is discrete, then $\ker \phi$ is open. If H is finite, this kernel has finite index. Further, every subgroup with its subspace topology and every product of topological groups with coordinatewise operations are topological groups.

Proof. Since $\{1\}$ is open in a discrete group,

$$\ker \phi = \phi^{-1}(\{1\})$$

is open. The First Isomorphism Theorem identifies $G/\ker\phi$ with the finite group $\operatorname{im}\phi$. The remaining assertions follow by restricting the continuous group operations, respectively by checking continuity coordinatewise on basic product opens. $\square$

12.5.3 Profinite Groups

Definition 12.45. A **profinite group** is an inverse limit of finite discrete groups, with the subspace topology inherited from

$$\prod_{i\in I} G_i.$$

The topology says that two elements are close when they have the same image at many finite stages. The next proposition makes this intuition precise in the particularly useful form of a neighborhood basis.

One may think of a profinite group as an infinite symmetry group observed through compatible finite stages. No single stage sees the entire element, but a compatible answer at every stage does. The prototype is $\widehat{\mathbb{Z}}$: an element gives a residue class modulo every positive integer, with all reductions agreeing. This viewpoint is exactly suited to an infinite algebraic extension, whose individual elements and automorphisms can be tested on finite subextensions.

Proposition 12.46. An inverse limit of finite discrete groups is a closed topological subgroup of their product. Consequently every profinite group is compact, Hausdorff, and totally disconnected.

Proof. The product is a topological group and is compact by Tychonoff's theorem. It is Hausdorff: distinct points differ in one coordinate, and disjoint open neighborhoods in that finite discrete factor pull back to disjoint product neighborhoods. It is totally disconnected for the same clopen-cylinder reason: if $x \neq y$, choose a coordinate where they differ; fixing that coordinate to x_i gives a clopen set containing x but not y, so no connected subset can contain both points. For $i \leq j$, define

$$\rho_i^j : \prod_k G_k \longrightarrow G_i \times G_i, \qquad (g_k)_k \longmapsto (\psi_i^j(g_j), g_i).$$

This map is continuous. The compatibility condition $\psi_i^j(g_j) = g_i$ is the inverse image under ρ_i^j of the diagonal, which is closed because the finite discrete target is Hausdorff. Intersecting these closed conditions over all $i \leq j$ gives the inverse limit. A closed subspace of a compact space is compact, since an open cover of it extends by its open complement to an open cover of the ambient space. Hausdorffness and total disconnectedness pass to subspaces: separating open sets and clopen cylinders may simply be intersected with the subspace. □

Proposition 12.47 (The projection-kernel basis)**.** Let

$$G = \varprojlim_{i \in I} G_i$$

be a profinite group, indexed by a directed set. The subgroups

$$\ker(\pi_i) \qquad (i \in I)$$

form a neighborhood basis at the identity. They are open normal subgroups, and every open subgroup has finite index and is closed.

Proof. In the product topology, a basic neighborhood of the identity restricts only finitely many coordinates, say those indexed by $i_1, \ldots, i_r$. Choose k above all these indices. Compatibility implies

$$\ker(\pi_k) \subseteq \bigcap_{\ell=1}^{r} \ker(\pi_{i_\ell}),$$

so one projection kernel lies inside the given neighborhood. Conversely, each $\ker(\pi_i)$ is open because it is the inverse image of the open singleton $\{1\} \subseteq G_i$. It is normal, and its quotient is a subgroup of the finite group G_i. An open subgroup contains one such kernel, hence has finite index. Its complement is a finite union of its open cosets, so it is closed. □

Example 12.48 (The topology of $\mathbb{Z}_p$)**.** For a prime p, define

$$\mathbb{Z}_p = \varprojlim_{n \geq 1} \mathbb{Z}/p^n\mathbb{Z}$$

using the usual reduction maps. The projection

$$\mathbb{Z}_p \longrightarrow \mathbb{Z}/p^n\mathbb{Z}$$

has kernel customarily denoted $p^n\mathbb{Z}_p$. Thus

$$p^n\mathbb{Z}_p \qquad (n \geq 1)$$

is a neighborhood basis at 0. In particular, congruence modulo a high power of p is the topological notion of being close in $\mathbb{Z}_p$. This is the additive prototype for the Krull topology below.

12.6 Infinite Galois Theory

Let E/F be algebraic Galois. Each element of E lies in a finite Galois subextension: take the splitting field in E of its separable minimal polynomial. If K/F and L/F are finite Galois subextensions, their **compositum** KL is the smallest subfield of E containing both. It is finite over F, separable because it lies in the separable extension E/F, and normal because K and L are splitting fields over F of finitely many separable polynomials, while KL is the splitting field of the product of those polynomials. Thus KL/F is again finite Galois and contains both fields. These subextensions are therefore directed by compositum, and

$$E = \bigcup_{K/F \text{ finite Galois}} K.$$

For the running finite-field example, the finite stages inside $\mathbb{F}_q^{\text{sep}}$ are the fields $\mathbb{F}_{q^n}$, ordered by divisibility of n. Restriction from a larger finite field to a smaller one is the concrete model for every inverse-limit transition map used below.

Lemma 12.49 (Extension to an algebraic closure)**.** Let L/F be algebraic inside a fixed algebraic closure $\overline{F}$. Every F-embedding

$$\sigma : L \longrightarrow \overline{F}$$

extends to an element of $\mathrm{Aut}_F(\overline{F})$.

Proof. The embedding-extension theorem (Theorem 10.24) extends σ to an F-embedding

$$\widetilde{\sigma} : \overline{F} \longrightarrow \overline{F}.$$

Its image is an algebraically closed subfield. Since $\overline{F}/F$ is algebraic, $\overline{F}/\widetilde{\sigma}(\overline{F})$ is algebraic. An algebraically closed field has no proper algebraic extension: if B/A is algebraic and A is algebraically closed, then the minimal polynomial over A of any $\beta \in B$ has a root in A, so irreducibility forces it to be linear. Hence $\beta \in A$. Thus $\widetilde{\sigma}$ is onto, as required. □

Lemma 12.50 (Extension through a normal algebraic extension)**.** If E/F is algebraic and normal and L/F is an algebraic subextension of E/F, every F-embedding $L \to \overline{F}$ that maps L into E extends to an F-automorphism of E. In particular, every F-automorphism of a finite subextension L/F extends to an F-automorphism of E.

Proof. Extend the given embedding to $\widetilde{\sigma} \in \mathrm{Aut}_F(\overline{F})$ by Lemma 12.49. If $x \in E$, then $\widetilde{\sigma}(x)$ is another root of the minimal polynomial of x over F. Normality says that all such roots lie in E, so $\widetilde{\sigma}(E) \subseteq E$. Applying the same argument to $\widetilde{\sigma}^{-1}$ gives the reverse inclusion. Hence $\widetilde{\sigma}|_E$ is the desired automorphism. □

Theorem 12.51 (Infinite Galois groups as inverse limits)**.** For a Galois extension E/F, restriction induces an isomorphism

$$\mathrm{Gal}(E/F) \cong \varprojlim_{K/F \text{ finite Galois}} \mathrm{Gal}(K/F).$$

Proof. For $K \subseteq L$, restriction $\mathrm{Gal}(L/F) \to \mathrm{Gal}(K/F)$ is well-defined and surjective by extension through the finite normal extension L/F, and the composition law is immediate.

An automorphism of E gives compatible restrictions. It is determined by them because the finite Galois fields cover E. Conversely, for a compatible family $(\sigma_K)_K$, define $\sigma(x) = \sigma_K(x)$ at any finite Galois stage containing x. A common larger stage proves this is well-defined and respects addition and multiplication. It fixes F, and the compatible inverse family defines its inverse. Thus it is an automorphism of E. □

Definition 12.52. The **Krull topology** on $\mathrm{Gal}(E/F)$ has basic neighborhoods of the identity

$$\mathrm{Gal}(E/K),$$

where K/F is finite Galois. Their cosets are the other basic open sets.

This topology is forced by the finite-stage viewpoint: two automorphisms σ, τ are close when they agree on a large finite Galois subextension K/F. Equivalently,

$$\tau^{-1}\sigma \in \mathrm{Gal}(E/K).$$

Thus the displayed stabilizers are exactly the natural identity neighborhoods, and their translates are all other basic neighborhoods. They do form a basis: for finite Galois K_1/F and K_2/F, the directedness argument above gives a finite Galois compositum K_1K_2/F, and

$$\mathrm{Gal}(E/K_1) \cap \mathrm{Gal}(E/K_2) = \mathrm{Gal}(E/K_1K_2).$$

Under Theorem 12.51, these subgroups are exactly the kernels of finite-stage projections. Thus the Krull topology is the inverse-limit topology, the displayed isomorphism is an isomorphism of topological groups, and $\mathrm{Gal}(E/F)$ is profinite. Concretely, in the finite-field case a basic neighborhood of the identity consists of the automorphisms that fix a prescribed $\mathbb{F}_{q^n}$. The topology therefore says that two automorphisms are close when their Frobenius powers agree on a sufficiently large finite field.

The infinite correspondence follows the same finite-stage logic as the finite theorem, with one new feature: closedness is what guarantees that a subgroup is recovered from all of its finite-stage images. The proof first recovers an intermediate field from its stabilizer, then uses the finite correspondence at every finite Galois stage, and finally translates finiteness and normality into openness and normality of the subgroup. In the finite case every subgroup is automatically closed. In the infinite case a nonclosed subgroup can omit automorphisms that its elements approximate at every finite stage, so it cannot be recovered from its finite images.

Theorem 12.53 (Infinite Galois correspondence)**.** For a Galois extension E/F, the assignments

$$K \longmapsto \mathrm{Gal}(E/K), \qquad H \longmapsto E^H$$

are inverse inclusion-reversing bijections between intermediate fields and closed subgroups of $\mathrm{Gal}(E/F)$. Further,

$$K/F \text{ finite} \quad \Longleftrightarrow \quad \mathrm{Gal}(E/K) \text{ open},$$

and

$$K/F \text{ Galois} \quad \Longleftrightarrow \quad \mathrm{Gal}(E/K) \text{ normal}.$$

Consequently, finite Galois intermediate extensions correspond to open normal subgroups.

Proof. For $\alpha \in E$, choose a finite Galois subextension L/F containing α. Its point stabilizer contains the open subgroup $\mathrm{Gal}(E/L)$, hence is itself open. It has finite index and is closed by Proposition 12.47. Therefore, for an intermediate K,

$$\mathrm{Gal}(E/K) = \bigcap_{\alpha \in K} \mathrm{Stab}(\alpha)$$

is closed. We have

$$E^{\mathrm{Gal}(E/K)} = K. \tag{12.2}$$

Indeed, if $\alpha \notin K$, its minimal polynomial over K has degree larger than one. The extension E/K is normal and separable, so that polynomial has in E a second, distinct root β. The map $K(\alpha) \to K(\beta)$ fixing K and taking α to β extends, by the preceding lemma applied over K, to a K-automorphism of E. It moves α, proving (12.2).

Let H be closed, and let H_L be its image in $\mathrm{Gal}(L/F)$ for a finite Galois L/F. The field L^{H_L} lies in E^H. If σ fixes E^H, then $\sigma|_L$ fixes L^{H_L}, so the finite correspondence gives $\sigma|_L \in H_L$. Thus the image of σ lies in the image of H at every finite stage; equivalently, every basic open neighborhood of σ meets H. Hence σ lies in the closure of H, and so in H. Therefore $\mathrm{Gal}(E/E^H) = H$, completing the correspondence.

If K/F is finite, then it is separable; by the finite Galois closure theorem and normality of E/F, choose a finite Galois extension L/F in E containing K. Then $\mathrm{Gal}(E/L)$ is open and is contained in $\mathrm{Gal}(E/K)$. Conversely, openness supplies a finite Galois L/F with $\mathrm{Gal}(E/L)$ contained in $\mathrm{Gal}(E/K)$. Taking fixed fields in (12.2) gives $K \subseteq L$. For the normality assertion, write $G = \mathrm{Gal}(E/F)$ and $H = \mathrm{Gal}(E/K)$. First observe that

$$\sigma H \sigma^{-1} = \mathrm{Gal}(E/\sigma(K)) \qquad (\sigma \in G).$$

Indeed, an element on the left fixes $\sigma(K)$, and the converse follows by conjugating back. If K/F is Galois, each $\sigma \in G$ preserves K, so this formula gives $\sigma H \sigma^{-1} = H$. Conversely, if H is normal, the formula and (12.2) give

$$\sigma(K) = E^{\mathrm{Gal}(E/\sigma(K))} = E^H = K \qquad (\sigma \in G).$$

To prove that K/F is normal, let $\tau : K \to \overline{F}$ be an F-embedding. By Lemma 12.49, it extends to an automorphism $\widetilde{\tau}$ of $\overline{F}$ fixing F. Since E/F is normal, $\widetilde{\tau}(E) = E$, so its restriction belongs to G. The displayed stability of K therefore yields $\tau(K) = K$. Thus K/F is normal; it is separable because it is an intermediate extension of the separable extension E/F. Hence it is Galois. □

Definition 12.54. Inside a fixed algebraic closure, F^{sep} is the **separable closure** of F, and

$$G_F = \mathrm{Gal}(F^{\mathrm{sep}}/F)$$

is its **absolute Galois group**.

Theorem 12.55 (The absolute Galois group of a finite field)**.** Let $\mathbb{F}_q^{\mathrm{sep}}$ be a separable closure of $\mathbb{F}_q$. Then

$$G_{\mathbb{F}_q} \cong \varprojlim_{n\geq 1} \mathbb{Z}/n\mathbb{Z} = \widehat{\mathbb{Z}}$$

as topological groups. Under this identification, Frobenius

$$\mathrm{Fr}_q : x \longmapsto x^q$$

corresponds to $(1 \bmod n)_{n\geq 1}$ and is a dense topological generator.

Proof. Every element of $\mathbb{F}_q^{\mathrm{sep}}$ has finite degree over $\mathbb{F}_q$, and hence lies in some $\mathbb{F}_{q^n}$. The finite extensions $\mathbb{F}_{q^n}$ are directed by divisibility, so Theorem 12.51 gives

$$G_{\mathbb{F}_q} \cong \varprojlim_{n\geq 1} \mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q).$$

By Proposition 11.22, the n-th group is cyclic of order n, generated by Fr_q. Identify it with $\mathbb{Z}/n\mathbb{Z}$ by sending the class of a to Fr_q^a. If $n \mid m$, restriction from $\mathbb{F}_{q^m}$ to $\mathbb{F}_{q^n}$ sends Fr_q^a to Fr_q^a, which is exactly the reduction map

$$\mathbb{Z}/m\mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z}.$$

The resulting inverse limit is $\widehat{\mathbb{Z}}$. Frobenius maps to the compatible family $(1 \bmod n)_n$. Its cyclic subgroup maps onto every finite quotient $\mathbb{Z}/n\mathbb{Z}$, so it meets every basic open coset; the projection-kernel basis in Proposition 12.47 therefore shows that it is dense. ☐

Here dense generator is deliberately topological language. Frobenius does not generate $\widehat{\mathbb{Z}}$ as an abstract cyclic group; rather, the ordinary cyclic subgroup generated by Frobenius is dense in $\widehat{\mathbb{Z}}$.

Under this identification,

$$\mathrm{Gal}(\mathbb{F}_q^{\mathrm{sep}}/\mathbb{F}_{q^n})$$

corresponds to

$$\ker(\widehat{\mathbb{Z}} \longrightarrow \mathbb{Z}/n\mathbb{Z}) = n\widehat{\mathbb{Z}},$$

the unique open subgroup of index n; its uniqueness also follows from the infinite Galois correspondence together with the uniqueness of the degree-n extension $\mathbb{F}_{q^n}/\mathbb{F}_q$. Thus the finite fields, Frobenius, inverse limits, and the open-subgroup part of the infinite Galois correspondence meet in one explicit example.

12.7 Further Topics in Algebra

This final section is orientation rather than prerequisite material. Its aim is to show how the four viewpoints developed in these notes become the common language of later algebra. The universal properties of free objects, quotients, tensor products, and limits encountered throughout these notes are organized systematically by category theory, whose formal development belongs to later study.

12.7.1 Groups, Actions, and Linear Algebra

At its most elementary, a representation of a group G is a homomorphism

$$G \longrightarrow \mathrm{GL}(V).$$

Thus representation theory combines group theory with linear algebra. Actions from Part I are actions on sets; representations are the especially structured case in which the set is a vector space. Invariant subspaces, irreducible representations, characters, and induction and restriction organize this richer form of symmetry.

Normal subgroups, quotients, semidirect products, and exact sequences lead to the study of group extensions. Group cohomology later organizes extension and obstruction questions. Groups also arise outside algebra: fundamental groups, covering spaces, and deck transformations make group actions central in algebraic topology. Matrix groups, Lie groups, and Lie algebras are a further meeting point of group theory and linear algebra.

12.7.2 Rings, Local Structure, and Geometry

Part II is the entry point to commutative algebra. Ideals, localization, polynomial rings, prime and maximal ideals, and modules lead onward to Noetherian rings, primary decomposition, integral dependence, Krull dimension, completions, and local algebra. Localization deserves special emphasis: it is algebraic zooming-in near a prime ideal.

Affine algebraic geometry studies spaces encoded by commutative rings. Informally, $\mathrm{Spec}(R)$ has prime ideals as points; localization describes what is visible near a point, residue fields record its scalar data, and modules supply algebraic analogues of geometric linear data. Scheme theory systematically combines rings, localization, modules, tensor products, and field extensions.

12.7.3 Modules, Exactness, and Multilinear Algebra

Exact sequences, projective and injective modules, flatness, Hom, tensor products, and direct and inverse limits form the starting vocabulary of homological algebra. Projective and injective resolutions, Tor, Ext, derived functors, and spectral sequences are later tools built from this vocabulary. The Mittag–Leffler remark and the symbol $\varprojlim^1$ are early signals of that theory.

Tensor products, dual spaces, exterior powers, and alternating forms also lead to differential geometry: they organize differential forms, orientation, volume forms, and tangent and cotangent spaces. No manifold theory is needed to recognize this continuation.

12.7.4 Fields, Arithmetic, and Galois Symmetry

Finite extensions of $\mathbb{Q}$ lead to number fields. Their rings of algebraic integers, ideals in Dedekind domains, factorization of prime ideals, ramification, and decomposition and inertia groups combine the themes of rings, localization, field extensions, and Galois groups. The present notes also deliberately stop before the full solvability-by-radicals theorem; Kummer theory is a natural next step when suitable roots of unity are present.

Absolute Galois groups act on algebraic objects, and Galois cohomology later organizes these actions in field arithmetic, descent, and number theory. Continuous homomorphisms

$$G_F \longrightarrow \mathrm{GL}_n(R)$$

for suitable topological coefficient rings are Galois representations, a major meeting point of Galois theory, representation theory, and topology. Algebraic geometry has a parallel profinite fundamental group: finite étale covers play the role of finite covering spaces. Differential Galois theory and covering-space Galois theory offer further variants of the same symmetry principle.

12.7.5 A Synthesis

$$\begin{aligned} \text{Groups} &\longrightarrow \text{symmetry and actions,} \\ \text{Rings} &\longrightarrow \text{arithmetic and local structure,} \\ \text{Modules} &\longrightarrow \text{linearized algebra and exactness,} \\ \text{Fields} &\longrightarrow \text{algebraic equations and Galois symmetry.} \end{aligned}$$

Later subjects arise by combining these viewpoints: representation theory is groups plus linear algebra; homological algebra is modules plus exact sequences and Hom/tensor; algebraic geometry is commutative rings plus localization, modules, and field extensions; algebraic number theory is rings plus fields and Galois theory; and arithmetic geometry joins algebraic geometry, number theory, and Galois representations. These are conceptual slogans, not formal identities. The recurring methods of the notes—quotient, localization, universal property, action, kernel and image, exact sequence, tensor construction, passage to finite stages, and symmetry—continue to reappear throughout advanced mathematics.

Exercises

Basic

Exercise 12.1 (Basic)**.** Prove directly that the fixed set of field automorphisms is a field.

Exercise 12.2 (Basic)**.** Verify Artin independence for the two embeddings of $\mathbb{Q}(\sqrt{2})$ in $\mathbb{C}$.

Exercise 12.3 (Basic)**.** Determine the correspondence for $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$.

Exercise 12.4 (Basic)**.** Compute the direct limit of $\mathbb{Z} \xrightarrow{\times 3} \mathbb{Z} \xrightarrow{\times 3} \cdots$.

Exercise 12.5 (Basic)**.** Describe the compatible sequence in $\mathbb{Z}_2$ associated with each of $-1, 0, 1, 2$.

Exercise 12.6 (Basic)**.** Prove that every discrete space is Hausdorff, and describe a basic open set in a product of finite discrete spaces.

Core

Exercise 12.7 (Core)**.** Reprove Lemma 12.4 using any chosen pair of distinct embeddings.

Exercise 12.8 (Core)**.** List all subgroup-field pairs in the splitting field of $X^3 - 2$, including their degrees.

Exercise 12.9 (Core)**.** Work out the correspondence for $\mathbb{Q}(\zeta_5)/\mathbb{Q}$, including normality.

Exercise 12.10 (Core)**.** Prove the equality criterion in Proposition 12.24 from the quotient construction.

Exercise 12.11 (Core)**.** Use the element criteria to prove exactness for a specified directed system of short exact sequences.

Exercise 12.12 (Core)**.** Give an inverse system for which surjectivity at each stage does not give surjectivity on inverse limits.

Exercise 12.13 (Core)**.** Show that projection kernels form a neighborhood basis in a profinite group.

Exercise 12.14 (Core)**.** Prove that the kernel of a continuous homomorphism to a discrete group is open, and use translations to describe its cosets as open sets.

Exercise 12.15 (Core)**.** Verify directly the gluing step in Theorem 12.51.

Exercise 12.16 (Core)**.** Give a direct alternative proof that $\mathrm{Gal}(E/K)$ is closed in the Krull topology.

Exercise 12.17 (Core)**.** Show that an open subgroup of a profinite group has finite index.

Further

Exercise 12.18 (Further)**.** Prove the root-of-unity radical proposition in full for a single radical step.

Exercise 12.19 (Further)**.** Use the degree obstruction to show that a regular heptagon is not straightedge-and-compass constructible.

Exercise 12.20 (Further)**.** Show that $\widehat{\mathbb{Z}}$ maps onto $\mathbb{Z}/n\mathbb{Z}$ for every $n > 0$.

Exercise 12.21 (Further)**.** Prove surjectivity of restriction from an infinite Galois group to a finite Galois stage.

Exercise 12.22 (Further)**.** Let E/F be Galois and let K/F be finite. Choose a finite Galois extension L/F in E containing K. Use the finite correspondence in L/F to decide when $\mathrm{Gal}(E/K)$ is normal in $\mathrm{Gal}(E/F)$.

Looking Ahead

Exercise 12.23 (Looking Ahead)**.** For $n \mid m$, verify that the restriction map

$$\mathrm{Gal}(\mathbb{F}_{q^m}/\mathbb{F}_q) \longrightarrow \mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$$

corresponds, under the Frobenius generators, to reduction

$$\mathbb{Z}/m\mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z}.$$

Exercise 12.24 (Looking Ahead)**.** Explain why finite separable subextensions of F^{sep}/F correspond to open subgroups of G_F.

Bibliography

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